STRUCTURAL CALCULATIONS · SAMPLE EXCERPT

Crane Runway Girder Design

Cedar Works · New 5-ton runway in the existing 5-bay building

1450 Foundry Road, Riverton, Ohio (fictitious)

PURPOSE

Sample calculation package

Shows the format and level of detail of a runway calculation; not a design for construction.

Calculation	DEMO-FOCUS-005 · Rev 0 · 2026-10-13 · Issued for review
Prepared by	MSB
Building code	2021 IBC as adopted by the fictitious demonstration jurisdiction · AISC 360-16 / ASCE 7-16
Crane	Demo crane CR-5 · 5 US tons: 2 wheels per runway; per wheel 7.5 kip maximum, 1 kip side thrust
Runway girder	W14X61 + C15X50, Fy 50 ksi; 5 simple spans of 25'-0"
Result	Demonstration analysis completed for assumed design criteria; governing D/C 0.83 (Tapered haunch): Column stiffener fillets, throat fatigue (A-3.1 item 8.2)
Pages	35: cover, contents, scope, calculation excerpt and drawing sheets S-00 to S-03

Engineer's seal and signature

NOT SEALED · SAMPLE

Fictitious demonstration project: the building, crane, supplier data and site are fictitious. Not a sealed design; not for construction.

Contents

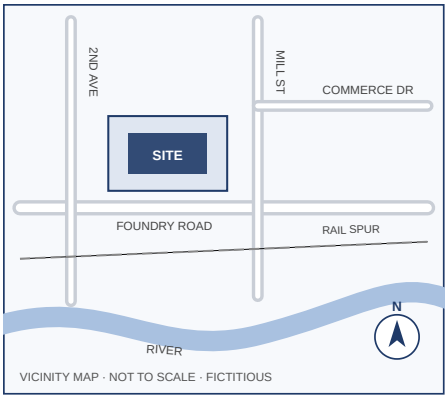
Sections of the full calculation, with their page in this sample. The other sections, Appendix A (every computed result with its equation and reference) and the DXF drawings are available on request.

Calculation		
	Design summary and governing checks	4
1 Design basis and inputs		
1.1	Codes and design basis	5
1.2	Notation	5
1.3	Materials and geometry	6
1.4	Crane data and design criteria	6
2 Load development		
2.1	Crane wheel loads and impact	7
2.2	Side thrust, traction and bumper forces	8
2.3	Dead loads, load cases and combinations	8
2.4	Fatigue loading	9
3 Analysis		
3.1	Analysis method and verification	10
3.2	Governing design actions	10
3.3	Support reactions by load type	11
3.4	Support, haunch and restraint forces	12
3.5	Forces delivered to the building	13
4 Member checks		
4.1	Runway girder, capped section properties and stability	14
4.2	Runway girder, strength and stability	15
4.3	Runway girder serviceability	17
4.4	Runway girder fatigue	17
4.5	Tapered haunch, inputs and actions	18
4.6	Tapered haunch, member checks	19
4.7	Existing column	20
5 Connection checks		
5.1	Cap channel welds and end development	21
5.2	Bolted end bearings	22
5.3	Tapered haunch welds, fatigue and column local effects	23
5.4	Cap tieback	26
5.5	Rail, clips, joints and anchorage	full calc
5.6	Runway end stops	full calc
6 Detailing, fabrication and applicability		
6.1	Girder arrangement	full calc
6.2	Detailing checklist	full calc
6.3	Bearing and end connection details	full calc
6.4	Cap tieback and rail joint details	full calc
6.5	Fabrication, inspection and erection notes	full calc
6.6	Applicability and project coordination	full calc
6.7	Items by others and forces delivered	28
Drawings		
S-00	General notes, criteria and materials	30
S-01	General arrangement	31
S-02	Connections, cap channel, supports and brackets	32
S-03	Cap tieback and end stops	33
Appendices		
A	Appendix A · every computed result (also as CSV)	full calc
B	Appendix B · references	full calc

Scope of work

PROJECT LOCATION

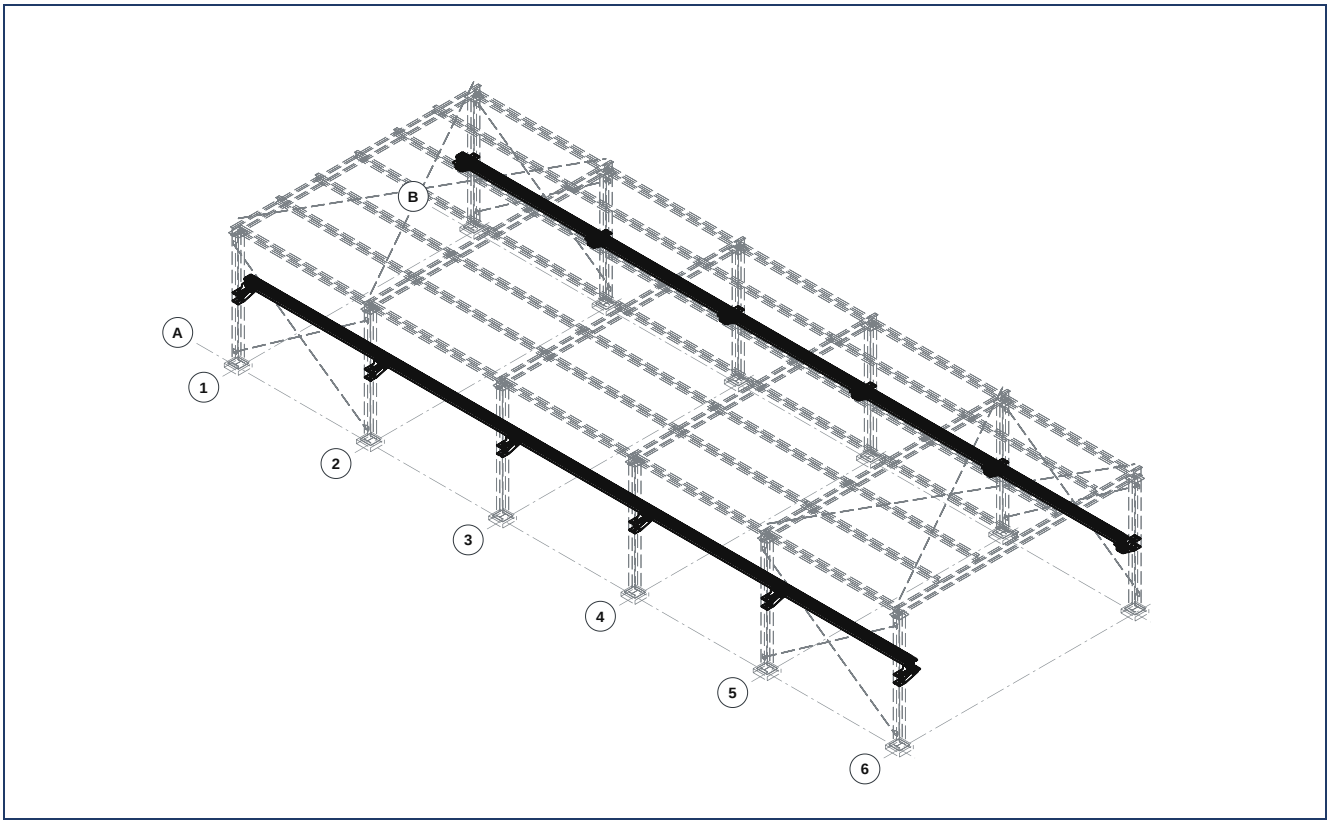
Cedar Works
1450 Foundry Road
Riverton, Ohio
Fictitious site



One crane runway, two identical girder lines (grid B opposite hand to grid A), each of five simply supported spans of 25'-0", 125'-0" overall, for Demo crane CR-5 (5 US tons). This calculation designs and checks the W14X61 + C15X50 capped runway girders; the Demo R-6 crane rail with its rail clips, joints and anchorage; the bolted end bearings and bearing stiffeners; the built-up tapered I haunches welded to the existing columns, the column stiffeners and the column local effects (AISC 360-16 J10); the cap tieback angles bolted to the channel cap and to the column; the girder-mounted end stops at both ends of both runways; and the existing columns for the added crane loads.

Codes: AISC 360-16 · AIST TR-13 · AISC DG7 · ASCE 7-16 · AWS D1.1/D1.1M:2020; owner deflection criteria. Editions and use: section 1.1.

Building and runway



New work (this calculation): girders, rail, haunches, tiebacks, bearings, end stops ------ Existing building (survey)

Isometric view of the existing 5-bay building with the new runway on grid lines A and B, from sheet S-01 (not to scale).

Cedar Works · New 5-ton runway in the existing 5-bay building

DEMO-FOCUS-005 · 1450 Foundry Road, Riverton, Ohio (fictitious) · Crane runway girder calculation · Rev 0, 2026-10-13

Prepared by	MSB	**Checked by**	_____
Engineer of record	_____	**Firm**	_____
License no.	_____	**Jurisdiction**	_____
Adopted code	2021 IBC as adopted by the fictitious demonstration jurisdiction · AISC 360-16 / ASCE 7-16		
Issue status	**DEMONSTRATION - NOT FOR CONSTRUCTION**		

ENGINEER'S SEAL AND SIGNATURE

NOT SEALED

Rev	Date	Description	By
0	2026-10-13	Issued for review	MSB

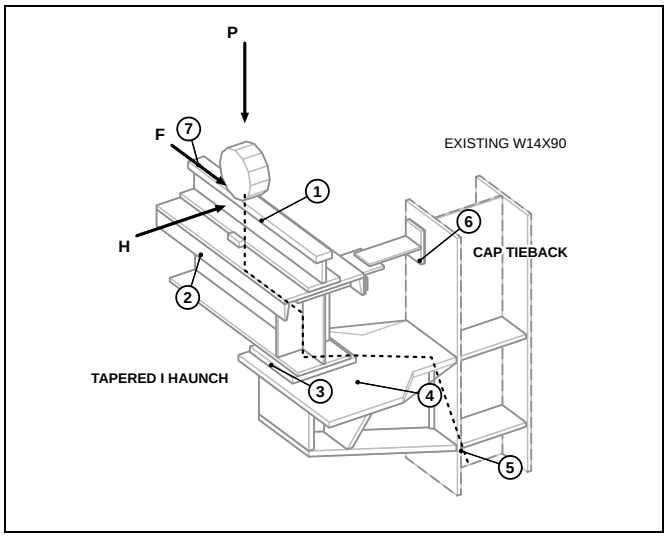
DEMONSTRATION, FICTITIOUS INPUTS. Fictitious loads, materials and owner criteria; not a construction or fabrication package.

Demonstration analysis completed for assumed design criteria. Governing D/C = 0.83 (Tapered haunch): Column stiffener fillets, throat fatigue (A-3.1 item 8.2).

Items by others and the forces delivered to them: section 6.7.

Scope: one crane runway, two identical girder lines (grid B opposite hand to grid A), each of five simply supported spans of 25'-0", 125'-0" overall, for Demo crane CR-5 (5 US tons). This calculation designs and checks the W14X61 + C15X50 capped runway girders; the Demo R-6 crane rail with its rail clips, joints and anchorage; the bolted end bearings and bearing stiffeners; the built-up tapered I haunches welded to the existing columns, the column stiffeners and the column local effects (AISC 360-16 J10); the cap tieback angles bolted to the channel cap and to the column; the girder-mounted end stops at both ends of both runways; and the existing columns for the added crane loads.

Design summary



Item	Specified
Crane	Demo crane CR-5 · 5 US tons: 2 wheels per runway; per wheel 7.5 kip maximum, 1 kip side thrust
Design basis	LRFD, AIST TR-13 / AISC 360-16 / ASCE 7-16; class B, 400,000 fatigue cycles
Girder	W14X61 + C15X50, Fy 50 ksi; 5 simple spans of 25'-0"
Rail	Demo R-6 crane rail; weld-on rail clips per manufacturer, base welds 5/16" fillets; joint bars 2-PL 1 1/4" × 3 3/4", 7/8" bolts
Bearing	Bearing PL 1" × 1'-0" × 1'-0"; stiffeners 2-PL 1" × 4 1/2", 5/16" fillets; 4-3/4" A325 bolts each end (locating slip-critical, sliding sleeved in slots)
Haunch	Tapered I haunch 2'-4" long, 1'-5" to 0'-9" deep: web PL 3/8", top flange PL 1" × 2'-1 1/2", bottom flange PL 1" × 1'-2", wing stiffeners 2-PL 1"; flanges CJP and web 1/4" fillets to the column
Column stiffeners	2 pairs PL 3/4" × 6 3/4", full depth, one pair at each haunch flange; 5/8" fillets to the flanges and 5/8" to the web, both faces; field welded inside the existing column
Cap tieback	L4X4X1/2 × 6 1/2" cap angle with 2-5/8" A325 bolts in sleeved horizontal slots; tie PL 3/8" × 4 1/4" × 9 3/4" with 2-5/8" A325 slip-critical bolts; end PL 5/8" × 4 1/4" × 6", 5/16" fillets both faces; 2-5/8" A325 bolts through shims to the column flange (field-drilled)
Column	Existing W14X90, Fy 50 ksi, 26'-0" base to top support
Longitudinal	Longitudinal force to the building bracing (by others)
End stops	Bolted stop: base PL 3/4" × 9" × 1'-0", face PL 3/4", stiffeners PL 1/2", 3/4" A325 bolts
Governing LRFD actions	M_u = 134 kip-ft; V_u = 26.3 kip; ΣR_u = 29.0 kip per grid; vertical deflection 0.22 in (one crane, static)

Figure 1. Typical support, isometric (near girder omitted). (1) rail; (2) girder; (3) bearing; (4) tapered haunch with column stiffeners; (5) existing column; (6) cap tieback; (7) longitudinal force F to the crane-level strut; (8) end stops (runway ends, not shown). P wheel load, H side thrust, F longitudinal force.

Governing check of each element along the load path

Element	Governing check	Equation	D/C	Result
(1) **Rail, clips and joints**	Rail clip base welds to the cap channel web AISC DG7 §11.2, §14.1; AISC 360-16 J2.4; AIST TR-13 §5.16	$f_w = \frac{1}{2t_e l_w}\sqrt{V^2 + \left(T + \frac{6M}{l_w}\right)^2} \le \phi(0.6F_{EXX})$ where l_w length of each of the two base welds (across the rail); V lateral force on the clip; T uplift on the clip; M moment about the weld centroid (V at mid-height of the rail base, or T at the clip nose)	0.21	OK

Element	Governing check	Equation	D/C	Result
② **Runway girder strength**	H1 interaction, bottom flange AISC 360-16 H1; AISC DG7 §11.2, §14.1	$\frac{P_r}{2P_c} + \frac{M_{rx}}{M_{cx}} + \frac{M_{ry}}{M_{cy}} \le 1$ (H1-1b)	0.44	OK
Runway girder deflection	Vertical crane-load deflection AIST TR-13 §5.8.7; Owner criteria	$\Delta_v \le L/1000 = 0.30$ in where L span	0.74	OK
③ **Bolted end bearings**	Sliding end, sleeves under bolt pretension AIST TR-13 §5.14	$1.5T_b \le \phi F_y A_g;\ \phi = 1.0$ (sleeve)	0.76	OK
④ **Tapered haunch**	Column stiffener fillets, throat fatigue (A-3.1 item 8.2) AISC 360-16 Table A-3.1	$f_{sr} = \frac{\Delta P}{2(0.707w)l} \le F_{SR} = 100(1.5/n_{SR})^{0.167} \ge 8$ ksi	0.83	OK
⑤ **Existing column (conditional)**	Existing column, torsion from the longitudinal force offset (TR-13 §5.9.2) AIST TR-13 §5.9.2; AISC 360-16 H3	$\frac{P_r}{P_c} + \frac{8}{9}\left(\frac{M_{rx}}{M_{cx}} + \frac{M_{ry}}{M_{cy}} + \frac{\sigma_{ws}}{\phi F_y}\right) \le 1;\ \tau_t + \tau_{ws} \le \phi 0.6F_y$	0.28	OK
⑥ **Cap tieback**	Angle slots, travel each way along the runway Fisher & Van de Pas (EJ 2002); AISC DG7 §11.2	$\theta h_b + \alpha L \Delta T + t_{set} \le u$ where L distance from the locating support; ΔT temperature change; h_b height of the cap bolts above the bearing; u slot travel each way; θ end rotation	0.80	OK
⑦ ***Longitudinal system***	*Crane-level strut, collector, brace connections and braced bay: building engineer of record (section 6.7). Forces delivered: section 3.5.*		—	***By others***
⑧ **End stops**	Girder top flange, thickness for no prying (ratio t_{min}/t) AISC Manual Part 9; AIST TR-13 §3.6.3, §3.10	$t_{min} = \sqrt{\frac{4Tb'}{\phi p F_u}}$ where T tension per front bolt; b' bolt line to the face of the heel, less half the bolt diameter; p tributary length per bolt	0.68	OK

137 design checks evaluated, and 356 more screened below the fatigue threshold, 9 N/A; 82 detailing rules (10 with small margin, section 6.2). All results: Appendix A.

1 DESIGN BASIS AND INPUTS

1.1 Codes and design basis

Basis: AIST Technical Report 13 (2021); AISC 360-16; AISC Design Guide 7, third edition with February 2023 errata. ASCE 7-16 strength combinations (§2.3) are used for the columns, with the crane loads as live load; the building frame beyond these members is outside this calculation.

Codes, standards and guides

Document	Edition	Used for
AISC 360-16	2016	Member and connection strength (Chapters B to K), fatigue (Appendix 3), stability bracing (Appendix 6)
AIST TR-13	2021	Crane loads (impact, side thrust, traction, bumper), runway load combinations, girder serviceability, fatigue duty and runway detailing
AISC DG7	Third edition	Crane runway girder practice: capped sections, tiebacks, bearings, fatigue details
Owner criteria	Project	Deflection limits (section 1.3)
ASCE 7-16	2016	Crane loads §4.9 where they exceed AIST; load combinations for the building elements (Chapter 2); seismic
AWS D1.1/D1.1M:2020	2020	Welding, cyclically loaded connections (Clause 4 Part C), welding to the existing steel (Clause 11)

Other references: Appendix B. Items by others: section 6.7. All results: Appendix A and the check-results file (CSV, US units) issued with this revision.

1.2 Notation

Crane runway symbols. Other symbols follow AISC 360-16, the AISC Manual and AISC DG9; symbols used in one check only are defined under that check.

Crane loads (AIST TR-13, AISC DG7)			
P_{max}	Maximum static wheel load from the supplier schedule, without impact	C_v	Vertical crane load from the lifted load (C_{vm} multiple cranes, C_{vs} single crane, $C_{vs,min}$ least lifted load)
P_{min}	Minimum wheel load on the opposite runway	C_i	Vertical impact, $I \cdot P_{max}$
P_d	Maximum load on the driven wheels	I	Vertical impact factor
Q	Rated capacity	C_{ss}	Side thrust on one runway (ΣC_{ss}: whole crane)
T	Trolley weight	C_{ls}	Longitudinal (tractive) force
B	Bridge weight	C_{bs}	Bumper (crane stop) force
W	Whole crane, B + T + Q	W_E	Bridge and trolley weight effective in the bumper

C_d	Crane dead load: wheel loads of the empty crane (C_{dm} multiple cranes, C_{ds} single crane)	η	Bumper efficiency
Capped girder, flange bending and rail			
A_c	Area of the cap channel	M_f	Flange lateral bending moment (side thrust and torsion)
Q_c	First moment of the cap channel about the neutral axis	M_{cf}	Available flexural strength of one flange about its own axis
t_{wc}	Web thickness of the cap channel	S_f	Elastic section modulus of one flange about its own axis
a'	Terminal development length of the cap welds (as F13.3)	f_{un}	Total normal stress, factored loads, including warping
r_w	Available strength of one cap weld per unit length	f_{uv}	Total shear stress, factored loads, including torsion
q_w	Shear flow on one cap weld	Δ_r	Lateral movement of the rail head
h_r	Rail height	Δ_o	Lateral deflection at the shear center
Serviceability and fit			
Δ_v	Vertical deflection, one crane, static wheel loads	t_{set}	Setting tolerance (TR-13 §5.18)
Δ_h	Lateral deflection of the top flange, one crane	t_{fab}	Fabrication tolerance on hole and part location (AISC Code of Standard Practice §6.4)
Fatigue (unfactored duty-bin loads)			
f_{max}	Largest stress under the unfactored fatigue loads	$f_{sr,eq}$	Equivalent constant-amplitude stress range at $n_{SR} = \Sigma n_i$ (Miner's rule)
f_{min}	Smallest stress under the unfactored fatigue loads	k_c	Share of the bracket moment $V \cdot e_c$ that bends the receiving column at the bracket
$f_{sr,i}$	Stress range of duty bin i; $F_{SR,i}$ is the allowable range at its n_i cycles		

1.3 Materials and geometry

Materials

Element	Specification	Properties
Runway girder W14X61 + C15X50	ASTM A992	Fy 50 ksi, Fu 65 ksi, E 29,000 ksi
Cap channel	ASTM A572 Grade 50 channel	Fy 50 ksi, Fu 65 ksi
Plates, brackets, stiffeners, angles	ASTM A572 Gr. 50	Fy 50 ksi, Fu 65 ksi
Crane rail, Demo R-6	ASTM A759	Fy 60 ksi, Fu 90 ksi
High-strength bolts	ASTM F3125 Gr. A325 (Group A)	Pretensioned (RCSC Table 8.1)
Nuts and washers	ASTM A563 Gr. DH heavy hex nuts; ASTM F436 hardened washers	Plate washers as plates above
Bolt sleeves	Machined from ASTM A108 cold-finished carbon steel bar (Grade 1018 or 1045) to the OD, bore and length detailed	Fy 50 ksi min. by mill test report
Weld electrode	AWS A5.1 / A5.5 E70XX	F_{EXX} 70.0 ksi
Existing column W14X90 (mill certificates)	ASTM A572 Grade 50	Fy 50 ksi, Fu 65 ksi

Geometry and girder data

Input	Value	Input	Value
Member	W14X61 + C15X50	Method / system	LRFD / simple spans
Spans	5 @ 25'-0"	Fy / Fu	50 ksi / 65 ksi
Lateral restraint spacing, top / bottom	25'-0" / 25'-0"	Unbraced length	25'-0"
Girder / rail weight	0.11 kip/ft / 0.038 kip/ft	Additional dead load	0 kip/ft
Rail head above flange	6 in	Rail eccentricity	1/4 in
AIST building class	B (400,000 cycles; section 1.4)	Rail rotation limit	None specified (reported only)
Deflection limits (one crane, static)	vertical L/1000 = 0.30 in (AIST Class B governs); lateral L/400	Rail gauge	40'-0 1/2"
Camber / level tolerance	0 in / 1/8 in		

Specified components: the design summary at the start of the calculation. Girder section properties: section 4.1.

1.4 Crane data and design criteria

Demo crane CR-5 · 5 US tons

Item	Value
Load source	Fictitious supplier schedule DEMO-CS-05: 7.5 kip max / 3.5 kip empty per wheel on the near runway, 3.7 kip minimum on the far runway; bridge speed 100 ft/min; bumper stroke 4 in, efficiency 0.5; end-truck bumper face 15 in beyond the end-wheel centerline.
Service / building class	Fictitious CMAA C crane; AIST Class B building (400,000 cycles counted)

Item	Value
Rated capacity Q / trolley T / bridge B	10 kip / 2.5 kip / 10 kip
AIST crane type and control	Maintenance, pendant
Wheels on this runway	2 at 10'-0"
Travel of the first wheel	1'-11" to 113'-1" from grid 1, ending at stop contact
Stop contact, left end (bumper fully compressed)	End wheel 1'-11" from grid 1 = stop face 1'-0" + bumper overhang 1'-3" − stroke 0'-4"
Stop contact, right end (bumper fully compressed)	End wheel 1'-11" from grid 6 = stop face 1'-0" + bumper overhang 1'-3" − stroke 0'-4"
Supplied impact / longitudinal force	25% / 2 kip
Driven-wheel load on this runway ΣP_d / side-thrust share	7.5 kip / 0.5
Bumper force	10 kip

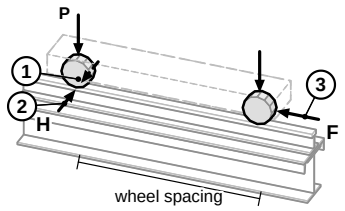
Side-thrust share to this runway: s = 0.5 (Fictitious supplier distribution: one-half of crane side thrust assigned to this runway).

Design assumptions and confirmed inputs

Item	Basis and source	Value
AIST building classification	Class B: TR-13 Table 1.1 classifies the building by its number of load cycles, and no basis limiting the count to full-load repetitions is stated, so all 400,000 crane load cycles are counted (Class B, 100,000 to 500,000 cycles), where the owner's 30,000 full-load repetitions alone would give Class C.	400,000
One fatigue cycle count throughout	Fatigue checks, duty bins and the stated duty sources all use N = 400,000 crane passages: one stress cycle per passage at the girder, its connections and supports; at the wheel-local details each wheel group passing is a cycle (AISC DG7 and AIST TR-13 cycle counting), n = N × wheel groups: rail clip base welds 800,000 (2 groups), cap welds under the wheels 800,000 (2 groups), rail joint bars and bolts 800,000 (2 groups).	—
Maximum load on one runway	This runway's 2 maximum static wheel loads sum to 15 kip, against at most W − 7.4 kip = 15.1 kip with the other runway carrying at least 7.4 kip (2 × the minimum wheel load 3.7 kip).	15.0 kip
Lifted-load share on one runway	Loaded minus empty wheel loads on this runway: 15 kip − 7 kip = 8 kip, the share of the lifted load that reaches this runway; it cannot exceed the rated load 10 kip.	8.00 kip
Wheel loads sum to the crane weight	Maximum wheel loads 15 kip on this runway plus 2 × 3.7 kip = 7.4 kip on the other runway give 22.4 kip, 0.1 kip from the crane weight W = 22.5 kip (tolerance 3% for rounding of the supplier schedule).	0.10 kip
Existing column, survey and existing load effects	Section: the surveyed receiving-column plates (d, bf, tf, tw), without rolled-fillet credit.	—
Existing column, longitudinal force path and seat moment (weak axis)	The crane longitudinal force leaves the column through a crane-level strut on the girder line, at least 10 times the column's own stiffness there ($k_s \geq 22.7$ kip/in): the column keeps at most 1/11 of the force, checked as weak-axis bending and torsion.	—

2 LOAD DEVELOPMENT

2.1 Crane wheel loads and impact



Wheel loads and reversible thrust

Detail: Demo crane CR-5 · 5 US tons: 2 wheels per runway; per wheel 7.5 kip maximum, 1 kip side thrust

Load path: (1) Wheel loads P with impact → (2) Side thrust H, either direction → (3) Traction F along the runway

ISOMETRIC · NOT TO SCALE

Demo crane CR-5 · 5 US tons

Wheel loads on this runway (supplier schedule, static). C_d = empty-crane wheel load; $C_v = P_{max} - C_d$; $C_i = I \cdot P_{max}$, one crane only.

Wheel	Offset	Max static P_{max}	Empty C_d	Lifted C_v	Impact C_i	Vertical $P_{max}(1+I)$	Side thrust C_{ss}
1	0 in	7.50 kip	3.50 kip	4.00 kip	1.88 kip	9.38 kip	1.00 kip
2	120 in	7.50 kip	3.50 kip	4.00 kip	1.88 kip	9.38 kip	1.00 kip
Σ		**15.0 kip**	7.00 kip	8.00 kip	3.75 kip	**18.8 kip**	**2.00 kip**

$$I = \max(\text{supplier},\ \text{TR-13},\ \text{ASCE 7}) = \max(25\%,\ 10\%,\ 10\%) = 25\%$$

(AIST TR-13 Table 3.2; ASCE 7-16 §4.9.3; pendant control). $P_{max}(1+I)$ = 9.38 kip.

Each crane position is analyzed with the crane loaded, empty and absent.

2.2 Side thrust, traction and bumper forces

Demo crane CR-5 · 5 US tons

Whole-crane side thrust, the greater of AIST TR-13 Table 3.2 and ASCE 7-16 §4.9.4, and the traction:

$$\Sigma C_{ss} = 0.10(Q + T + B) = 0.10(10.0\,\text{kip} + 2.50\,\text{kip} + 10.0\,\text{kip}) = 2.25\,\text{kip} \quad \text{(TR-13)}$$

$$\Sigma C_{ss} = 0.20(Q + T) = 0.20(10.0\,\text{kip} + 2.50\,\text{kip}) = 2.50\,\text{kip} \quad \text{(ASCE 7)}$$

$$C_{ls} = \max(0.20\Sigma P_d,\ 0.10\Sigma P_{max}) = \max(0.20 \times 7.50\,\text{kip},\ 0.10 \times 15.0\,\text{kip}) = 1.50\,\text{kip} \quad \text{(TR-13, ASCE 7)}$$

The supplier assigns s = 0.5 of the side thrust to this runway. A smaller share would need a two-runway stiffness analysis.

Crane design loads: code minimum against the supplier value

Load	AIST TR-13 minimum	ASCE 7-16 minimum	Supplier	Governing value · source	Per wheel
Vertical impact I	10% (Table 3.2, pendant)	10% (§4.9.3)	25%	**25%** · Supplier	$C_i = I \cdot P_{max}$ = 1.88 kip at each of 2 wheels
Side thrust C_{ss}, this runway	0.5 × 2.25 kip = 1.13 kip	0.5 × 2.50 kip = 1.25 kip (§4.9.4)	2.00 kip	**2.00 kip** · Supplier	1.00 kip at each of 2 wheels, at the rail head, either direction
Traction C_{ls}, this runway	$0.20\Sigma P_d$ = 1.50 kip	$0.10\Sigma P_{max}$ = 1.50 kip (§4.9.5)	2.00 kip	**2.00 kip** · Supplier	1.00 kip at each of 2 wheels, at the rail head; to the locating bearing
Bumper C_{bs}	§3.6.3: energy at 100% travel speed, checked with the end stops	—	10.0 kip	**10.0 kip** · Supplier	at the end stop, crane stop combinations only

Side thrust and traction act at the top of rail, either direction, one crane at a time. Side thrust goes through the top flange to the cap tiebacks; traction goes either way to the girder under the driven wheels (the rail seat) or, once the floating rail string has crept against an end stop, through that stop into the end bay toward the stop, wherever the crane is. No rail-seat friction is relied on: the girders, locating bearings and supports are checked for both paths, each with the full traction (alternative cases), and the end stops for the larger bumper force. The bumper force acts on the end stops in the crane stop combinations only.

Bumper

Item	Basis and source	Value
Bumper, deceleration at 50% speed	Peak deceleration at 50% of the rated speed 2.08 ft/s² against 16 ft/s² (TR-13 commentary to §3.6.3).	2.08 ft/s²
Bumper, peak force absorbing the full energy at 100% speed	Energy at 100% speed $W_{Ev}^2/2g$ = 0.31 kip-ft (W_E = 7.25 kip), absorbed over the 4 in stroke at efficiency 0.5: peak force 1.88 kip against the supplier bumper force 10 kip, for which the runway, end stops and their load path are designed.	1.88 kip
Bumper, peak force within the stop design strength	Peak bumper force 1.88 kip against the girder-mounted stop design strength 21.9 kip: the 10 kip design bumper force over the largest strength D/C of the stop checks (a no-prying thickness ratio is squared to a strength ratio).	1.88 kip

2.3 Dead loads, load cases and combinations

Dead / live load	Value	Load	Value
Girder self-weight	0.11 kip/ft	Rail	0.038 kip/ft
Rail clips and joint bars	0.0054 kip/ft	Additional dead load	0 kip/ft
Total dead load D	**0.15 kip/ft**		
Runway live load L	0 kip/ft	Rail eccentricity / height	0.25 in / 6.00 in

Load cases. D dead load; C_d crane dead (empty) wheel loads; C_v lifted load; C_i impact; C_{ss} side thrust; C_{ls} traction; C_{bs} bumper; L runway live load. Subscript m: every crane on the runway (multiple-crane case); s: a single crane. One crane runs on this runway, so the multiple-crane terms are that crane: $C_{dm} = C_{ds}$, $C_{vm} = C_{vs}$. Runway live load L = 0 kip/ft.

New steel dead load, one runway line

Item	Carried by	Number	Each	Total
Girder (W and cap channel)	girder (uniform)	1	13.9 kip	13.9 kip
Rail	girder (uniform)	1	4.79 kip	4.79 kip
Rail clips (pairs)	girder (uniform)	68	0.008 kip	0.54 kip
Rail joint bars (pairs)	girder (uniform)	3	0.033 kip	0.1 kip
Bearing plates and stiffeners (each girder end)	seat, at its grid	10	0.073 kip	0.73 kip
End stops	seat, at its grid	2	0.086 kip	0.17 kip
Cap tiebacks (each girder end)	column (support steel)	10	0.016 kip	0.16 kip
Tapered haunch	column (support steel)	6	0.43 kip	2.57 kip

Item	Carried by	Number	Each	Total
Column stiffeners (four plates)	column (support steel)	6	0.072 kip	0.43 kip
Bolts, washers, sleeves and welds (5% of the connection steel)	with the items	—	—	0.24 kip
One runway line / both				**23.6 kip / 47.2 kip**

The girder carries its own weight, the rail and the clips and joint bars (D above). The bearing plates and stiffeners and the end stops add to the dead reaction on the seat at their grid, and the support steel goes straight into the column: both are in the dead reactions D of section 3.3, the column loads and the seismic weight.

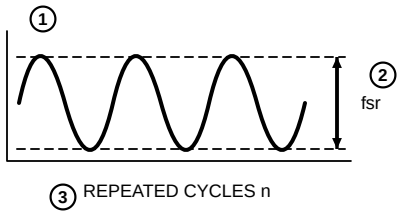
Basis. The LRFD crane load combinations of AIST TR-13 §3.10, also given in AISC Design Guide 7, 3rd ed. (2019), with combinations 2b (p. 55) and 2c (p. 56) as corrected by the DG7 revisions and errata of February 16, 2023. **Impact and side thrust are in different combinations:** 2b carries the side thrust without impact, $1.6(C_{vm} + C_{ss} + C_{ls})$, and 2c the impact without side thrust, $1.6(C_{vs} + C_i + C_{ls})$, for one crane. Roof live, snow, rain, wind and seismic loads do not act on the girder, so each ASCE 7-16 combination reduces to the runway terms shown. **ASCE 7 reading.** ASCE 7-16 §4.9 makes every crane load, the crane weight C_d included, a live load, where TR-13 factors C_d as dead load: A2b and A2c repeat 2b and 2c with C_d and L factored 1.6. The girder, bearings, supports and connections are checked for the envelope of both readings, so the larger governs every check. Every crane position and state (loaded, empty, absent) is evaluated in each combination, with side thrust in both directions.

Combination	TR-13 case	ASCE 7-16 basis	Equation	D	C_d	C_v	C_i	C_{ss}	C_{ls}	C_{bs}	L
LRFD 1a	1a	§2.3.1 (1)	$1.4D + 1.4C_{dm}$	1.4	1.4	—	—	—	—	—	—
LRFD 2a	2a	§2.3.1 (2)	$1.2(D + C_{dm}) + 1.6L + C_{vm} + C_{ss} + C_{ls}$	1.2	1.2	1	—	1	1	—	1.6
LRFD 2b	2b (DG7 errata 2023)	§2.3.1 (2)	$1.2(D + C_{dm}) + 1.6(C_{vm} + C_{ss} + C_{ls}) + L$	1.2	1.2	1.6	—	1.6	1.6	—	1
LRFD 2c	2c (DG7 errata 2023); one crane	§2.3.1 (2)	$1.2(D + C_{ds}) + 1.6(C_{vs} + C_i + C_{ls}) + L$	1.2	1.2	1.6	1.6	—	1.6	—	1
LRFD 7a	7a	§2.3.1 (5), §2.3.6 (7)	$0.9D + C_{dm}$	0.9	1	—	—	—	—	—	—
LRFD 8	8; one crane	— (AIST)	$1.2(D + C_{ds}) + C_{vs} + C_{bs}$	1.2	1.2	1	—	—	—	1	—
LRFD 9	9; one crane	— (AIST)	$0.9(D + C_{ds}) + 1.6C_{vs,min} + 1.6C_{ss}$	0.9	0.9	1.6	—	1.6	—	—	—
LRFD 10 (traction)	10 (traction); one crane	— (AIST)	$0.9(D + C_{ds}) + 1.6C_{ls}$	0.9	0.9	—	—	—	1.6	—	—
LRFD 10 (bumper)	10 (bumper); one crane	— (AIST)	$0.9(D + C_{ds}) + C_{bs}$	0.9	0.9	—	—	—	—	1	—
LRFD A2b	2b with C_d as L	§2.3.1 (2), §4.9	$1.2D + 1.6(C_{dm} + C_{vm} + C_{ss} + C_{ls} + L)$	1.2	1.6	1.6	—	1.6	1.6	—	1.6
LRFD A2c	2c with C_d as L; one crane	§2.3.1 (2), §4.9	$1.2D + 1.6(C_{ds} + C_{vs} + C_i + C_{ls} + L)$	1.2	1.6	1.6	1.6	—	1.6	—	1.6

Not listed, bounded by the cases shown: LRFD 1, 2, 6 and 7 (no crane on the runway); LRFD 3a (bounded by 2a: 1.0L ≤ 1.6L; no roof live, snow or rain load on the girder); LRFD 4a (bounded by 2a: no wind on the girder; 2a adds the side thrust and traction).

The existing column is checked with ASCE 7-16 §2.3.1 combinations 1–5 and §2.3.6 combinations 6–7, all crane loads as live load L, and again with the crane dead load C_d factored as dead load (the TR-13 rows; section 4.7). Fatigue uses unfactored loads (section 2.4).

2.4 Fatigue loading



Stress range at a cyclic detail

Detail: Class B, 400,000 cycles

Steps: (1) Each wheel passage gives a stress range fsr at the detail → (2) Detail category: AISC 360 Appendix 3 → (3) Cycles ni in each duty bin; Miner equivalent range fsr,eq at nSR

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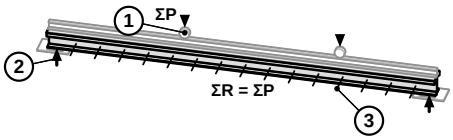
Fatigue loads are unfactored, without impact: C_{ds} + (lift fraction)·C_{vs} + 0.5C_{ss} of one crane (AIST TR-13 §3.10.2.3), side thrust reversing. Design life: 400,000 cycles (Class B, section 1.4), loaded and empty passes included.

Duty bin	Lift fraction	Cycles	Vertical per wheel	Lateral per wheel
Rated lifts	1.00	60,000	7.50 kip	0.50 kip
Routine half-load lifts	0.50	260,000	5.50 kip	0.50 kip
Light lifts	0.25	80,000	4.50 kip	0.50 kip
Total		**400,000**		

The 60,000 rated-lift cycles are the owner's 30,000 full-load repetitions at 2 stress cycles each. Each cycle takes the full range of its bin, so empty passes are counted at no less than the Light lifts range (4.50 kip per wheel). Miner's rule gives the equivalent range at 400,000 cycles, without endurance-limit credit. The largest bin range is also checked at all cycles. Stress ranges by detail: section 4.4.

3 ANALYSIS

3.1 Analysis method and verification



Equilibrium and refinement

Detail: W14X61 + C15X50, Fy 50 ksi; 5 simple spans of 25'-0"

Steps: (1) Wheel loads ΣP → (2) Reactions ΣR: equilibrium → (3) Refine crane travel and mesh until converged

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Vertical. The simply supported girder (5 spans of 25'-0") is a finite-element beam with nodes at the supports, every wheel, restraint and fatigue-detail station. Each crane is moved along its travel (1'-11" to 113'-1" from grid 1) in steps, with additional positions placing a wheel over each support and critical station; the step is halved until the governing results change by less than the travel tolerance. Influence-line superposition gives the envelope of every action for each combination of section 2.3: 9,408 design cases. The crane stop combinations (LRFD 8, LRFD 10 (bumper)) act only with the crane at stop contact (section 1.4): its bumper fully compressed against the stop and every wheel on the rail, with the concurrent wheel loads, bumper force, girder axial force and bending, bearing reactions and support actions all taken from that position.

Lateral and torsion. Side thrust at the rail head and the rail eccentricity load the girder in torsion with warping (thin-walled open section, AISC DG9): flange lateral bending and warping with the top and bottom flange restraints as finite springs, free warping at the supports (restraints with play at their secant stiffness in the strength cases, section 5.4). Second-order effects use AISC 360-16 C2 / Appendix 8 amplification with 0.8 of the elastic stiffness for strength.

> **Verification:** the largest equilibrium residual is 0.8% of its tolerance; halving the travel step, station spacing and warping mesh changes the governing results by less than 1% (largest 0.2%). Values: Appendix A.

Component models. The cap tieback is a DG7 top-flange tie connected directly to the column, with its bolted plies idealized as T-stubs and their stiffnesses combined in series. The existing column is a single member under the ASCE 7 combinations of the design method (§2.3 LRFD, §2.4 ASD), checked to AISC 360-16 E3, E4, E7, F2, F3, F6, G2 and H1 with B1 from Appendix 8.

3.2 Governing design actions

Factored LRFD envelope maxima of the girder. Crane position: the first wheel, from the grid named; x is the section where the action is largest, from grid 1 (the axial force is constant along the bay).

Action	Value	Combination	Crane position (first wheel)	At x
Major-axis moment M_u	**134 kip-ft**	LRFD A2c	10'-3" past grid 1	10.3 ft
Shear V_u	**26.3 kip**	LRFD A2c	15'-0" past grid 4	100.0 ft
Grid reaction ΣR_u (both girder ends)	**29.0 kip**	LRFD A2c	25'-0" past grid 4	100.0 ft
Axial force P_u (traction / bumper)	**10.0 kip**	LRFD 8	1'-11" past grid 1	—
Top flange lateral moment M_f	**21.2 kip-ft**	LRFD A2b	5'-0" past grid 5	115.0 ft
Bottom flange lateral moment	**8.41 kip-ft**	LRFD A2b	5'-0" past grid 5	115.0 ft
H1 interaction U, top flange (positive moment, M_{cx} 448 kip-ft)	**0.34**	LRFD A2b	10'-3" past grid 1	10.3 ft
H1 interaction U, bottom flange (positive moment, M_{cx} 486 kip-ft)	**0.44**	LRFD A2b	5'-0" past grid 1	15.0 ft
Uplift: none (least reaction)	0 kip	—	—	—
Torsion (shear-center torque)	1.93 kip-ft	—	—	—

Lateral-torsional results (torsion with warping)

Result	Value	Combination and crane position	At x
Normal stress, flange tip (with warping)	**19.4 ksi**	LRFD A2b, crane at 10'-0" past grid 3	60.0 ft
Shear stress (with St. Venant and warping)	**8.46 ksi**	LRFD A2c, crane at 0'-1" past grid 5	100.0 ft
Elastic critical load multiplier	**4.60**	LRFD A2c, crane at 7'-10" past grid 5	—
Rail-head lateral displacement, one crane	**0.15 in**	one crane, static wheel loads, crane at 7'-6" past grid 3	62.5 ft
Twist, one crane	**0.011 rad**	one crane, static wheel loads, crane at 7'-6" past grid 3	62.5 ft

Envelope by combination

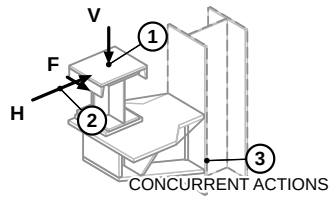
Combination	Mx	Shear	Reaction	Axial	H1 D/C
LRFD 1a $1.4D + 1.4C_{dm}$	55.6 kip-ft	10.5 kip	13.5 kip	0 kip	0.13
LRFD 2a $1.2(D + C_{dm}) + 1.6L + C_{vm} + C_{ss} + C_{ls}$	79.6 kip-ft	15.4 kip	18.1 kip	2.00 kip	0.30
LRFD 2b $1.2(D + C_{dm}) + 1.6(C_{vm} + C_{ss} + C_{ls}) + L$	98.8 kip-ft	19.3 kip	22.0 kip	3.20 kip	0.41
LRFD 2c $1.2(D + C_{ds}) + 1.6(C_{vs} + C_i + C_{ls}) + L$	123 kip-ft	24.1 kip	26.8 kip	3.20 kip	0.31
LRFD 7a $0.9D + C_{dm}$	38.5 kip-ft	7.34 kip	9.22 kip	0 kip	0.09
LRFD 8 $1.2(D + C_{ds}) + C_{vs} + C_{bs}$	73.8 kip-ft	14.2 kip	15.3 kip	10.0 kip	0.19
LRFD 9 $0.9(D + C_{ds}) + 1.6C_{vs,min} + 1.6C_{ss}$	35.7 kip-ft	6.78 kip	8.66 kip	0 kip	0.25
LRFD 10 (traction) $0.9(D + C_{ds}) + 1.6C_{ls}$	35.7 kip-ft	6.78 kip	8.88 kip	3.20 kip	0.09
LRFD 10 (bumper) $0.9(D + C_{ds}) + C_{bs}$	33.7 kip-ft	6.30 kip	7.33 kip	10.0 kip	0.09
LRFD A2b $1.2D + 1.6(C_{dm} + C_{vm} + C_{ss} + C_{ls} + L)$	110 kip-ft	21.5 kip	24.2 kip	3.20 kip	0.44
LRFD A2c $1.2D + 1.6(C_{ds} + C_{vs} + C_i + C_{ls} + L)$	134 kip-ft	26.3 kip	29.0 kip	3.20 kip	0.34

Maximum of each action within each combination (not concurrent), except H1, which uses concurrent actions at the more highly stressed flange with M_{cx} by the sign of the moment at the station (positive 448 kip-ft, negative 174 kip-ft for the flange in compression; the flange in tension takes its yielding strength). Combinations not listed are bounded by those shown (section 2.3).

Deflection (one crane, static, no impact) and end rotation

Result	Value	Result	Value
Vertical deflection	0.22 in	Top-flange lateral deflection	0.067 in
Vertical deflection by bay	0.22 in, every bay	Lateral deflection by bay	0.067 in, every bay
Girder end rotation, factored	0.0054 rad	Service / dead-load rotation	0.0024 rad / 0.00048 rad

3.3 Support reactions by load type



Forces transferred to the building

Detail: Existing W14X90, Fy 50 ksi, 26'-0" base to top support

Load path: (1) Girder reaction V at the support → (2) Concurrent lateral H and longitudinal F → (3) Column and building carry them together

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Unfactored reactions at each grid for checking the building, downward positive. C_d, C_v and C_i are from the crane position that maximizes each at the grid; C_i at 25%. C_{ss} is the largest single-crane side thrust at the grid, at the rail head. Under ASCE 7 §4.9 every crane load is live load L. D is all the new steel at the grid (section 2.3): the girders, rails and attachments, the bearing hardware and end stops on the seat, and the support steel D_s that goes straight into the column.

Grid	D	of which D_s	L	Crane empty C_d	Lifted C_v	Impact C_i	Side thrust C_{ss}	Traction couple ±	Min. crane reaction
1	2.64 kip	0.54 kip	0 kip	5.06 kip	5.79 kip	2.71 kip	1.45 kip	0.14 kip	0 kip
2	4.58 kip	0.56 kip	0 kip	5.60 kip	6.40 kip	3.00 kip	1.60 kip	0.14 kip	0 kip
3	4.58 kip	0.56 kip	0 kip	5.60 kip	6.40 kip	3.00 kip	1.60 kip	0.14 kip	0 kip
4	4.58 kip	0.56 kip	0 kip	5.60 kip	6.40 kip	3.00 kip	1.60 kip	0.14 kip	0 kip
5	4.58 kip	0.56 kip	0 kip	5.60 kip	6.40 kip	3.00 kip	1.60 kip	0.14 kip	0 kip
6	2.64 kip	0.54 kip	0 kip	5.06 kip	5.79 kip	2.71 kip	1.45 kip	0.14 kip	0 kip

Traction C_{ls} = 2.00 kip acts at the rail head. Its height above the bearing adds $\pm C_{ls} \cdot e/L$ to the reactions at the ends of the loaded bay (column “Traction couple”); the force itself goes to the crane-level strut at the locating support.

3.4 Support, haunch and restraint forces

Factored forces at each grid for the haunch, bearing and cap tieback design.

Haunch design forces, factored LRFD envelope

Grid	Case	Combination and crane position	ΣR_u	Left R_u	Right R_u	$M_{y,seat}$	F	H_{tie}	H_{seat}
1	Max V and M	LRFD A2c, crane at 1'-11" past grid 1, traction on bay 1	24.4 kip	—	24.4 kip	7.33 kip-ft	−3.20 kip	±0.43 kip	∓0.43 kip
	Min V	LRFD 10 (traction), crane empty at grid 2, traction on bay 1	1.67 kip	—	1.67 kip	0.50 kip-ft	+3.20 kip	±0.01 kip	∓0.01 kip
	Max F	LRFD 8, crane at 1'-11" past grid 1, stop force on bay 1	15.3 kip	—	15.3 kip	4.58 kip-ft	−10.0 kip	±0.23 kip	∓0.23 kip
	Max H	LRFD A2b, crane at 1'-11" past grid 1, traction on bay 1	20.1 kip	—	20.1 kip	6.03 kip-ft	−3.20 kip	±3.78 kip	∓1.47 kip
2	Max V, M and F	LRFD A2c, crane at grid 2, traction on bay 2	29.0 kip	2.41 kip	26.6 kip	20.4 kip-ft	−3.20 kip	±0.47 kip	∓0.47 kip
	Max V and M	LRFD A2c, crane at 15'-0" past grid 1, traction on bay 1 (a)	29.0 kip	26.6 kip	2.41 kip	−20.4 kip-ft	—	±0.45 kip	∓0.45 kip
	Min V and end V	LRFD 10 (traction), crane empty at grid 3, traction on bay 2	3.40 kip	1.81 kip	1.59 kip	−0.19 kip-ft	+3.20 kip	±0.02 kip	∓0.02 kip
	Max F	LRFD A2c, crane at 15'-0" past grid 1, traction on bay 2 (a)	28.6 kip	26.4 kip	2.19 kip	−20.4 kip-ft	+3.20 kip	±0.45 kip	∓0.45 kip
	Max H	LRFD A2b, crane at grid 2, traction on bay 2	24.2 kip	2.41 kip	21.8 kip	16.3 kip-ft	−3.20 kip	±4.09 kip	∓1.53 kip
3	Max V, M and F	LRFD A2c, crane at grid 3, traction on bay 3	29.0 kip	2.41 kip	26.6 kip	20.4 kip-ft	−3.20 kip	±0.47 kip	∓0.47 kip
	Max V and M	LRFD A2c, crane at 15'-0" past grid 2, traction on bay 2 (a)	29.0 kip	26.6 kip	2.41 kip	−20.4 kip-ft	—	±0.45 kip	∓0.45 kip
	Min V and end V	LRFD 10 (traction), crane empty at grid 4, traction on bay 3	3.40 kip	1.81 kip	1.59 kip	−0.19 kip-ft	+3.20 kip	±0.02 kip	∓0.02 kip
	Max F	LRFD A2c, crane at 15'-0" past grid 2, traction on bay 3 (a)	28.6 kip	26.4 kip	2.19 kip	−20.4 kip-ft	+3.20 kip	±0.45 kip	∓0.45 kip
	Max H	LRFD A2b, crane at grid 3, traction on bay 3	24.2 kip	2.41 kip	21.8 kip	16.3 kip-ft	−3.20 kip	±4.09 kip	∓1.53 kip
4	Max V, M and F	LRFD A2c, crane at grid 4, traction on bay 4	29.0 kip	2.41 kip	26.6 kip	20.4 kip-ft	−3.20 kip	±0.47 kip	∓0.47 kip
	Max V and M	LRFD A2c, crane at 15'-0" past grid 3, traction on bay 3 (a)	29.0 kip	26.6 kip	2.41 kip	−20.4 kip-ft	—	±0.45 kip	∓0.45 kip
	Min V and end V	LRFD 10 (traction), crane empty at grid 5, traction on bay 4	3.40 kip	1.81 kip	1.59 kip	−0.19 kip-ft	+3.20 kip	±0.02 kip	∓0.02 kip
	Max F	LRFD A2c, crane at 15'-0" past grid 3, traction on bay 4 (a)	28.6 kip	26.4 kip	2.19 kip	−20.4 kip-ft	+3.20 kip	±0.45 kip	∓0.45 kip
	Max H	LRFD A2b, crane at grid 4, traction on bay 4	24.2 kip	2.41 kip	21.8 kip	16.3 kip-ft	−3.20 kip	±4.09 kip	∓1.53 kip
5	Max V and M	LRFD A2c, crane at grid 5, traction on bay 5	29.0 kip	2.41 kip	26.6 kip	20.4 kip-ft	−3.20 kip	±0.47 kip	∓0.47 kip
	Max V and M	LRFD A2c, crane at 15'-0" past grid 4, traction on bay 4 (a)	29.0 kip	26.6 kip	2.41 kip	−20.4 kip-ft	—	±0.45 kip	∓0.45 kip
	Min V and end V	LRFD 10 (traction), crane empty at 15'-0" past grid 3, traction through the rail string to the stop on bay 5	3.40 kip	1.81 kip	1.59 kip	−0.19 kip-ft	+3.20 kip	±0.02 kip	∓0.02 kip
	Max F	LRFD 8, crane at 13'-1" past grid 5, stop force on bay 5	8.47 kip	2.41 kip	6.06 kip	3.07 kip-ft	+10.0 kip	±0.12 kip	∓0.12 kip
	Max H	LRFD A2b, crane at grid 5, traction on bay 5	24.2 kip	2.41 kip	21.8 kip	16.3 kip-ft	−3.20 kip	±4.09 kip	∓1.53 kip
6	Max V and M	LRFD A2c, crane at 13'-1" past grid 5, traction on bay 5	24.4 kip	24.4 kip	—	−7.33 kip-ft	—	±0.42 kip	∓0.42 kip
	Min V	LRFD 10 (traction), crane empty at 15'-0" past grid 4, traction on bay 5	1.67 kip	1.67 kip	—	−0.50 kip-ft	—	±0.01 kip	∓0.01 kip
	Max H	LRFD A2b, crane at 13'-1" past grid 5, traction on bay 5	20.1 kip	20.1 kip	—	−6.03 kip-ft	—	±3.75 kip	∓1.44 kip

Factored LRFD envelope over every crane position and strength combination; each row is one concurrent case.

ΣR_u: grid total. Left / right R_u: girder end reactions including the traction couple; down positive; haunch self-weight excluded. Min V: least total reaction; end V: least reaction of one girder end; negative is uplift.

$M_{y,seat}$: $\Sigma V \cdot e$ about the grid, each reaction at 0.8 of the bearing length from its girder end; + toward the next grid; ± = also reached reversed, with the end reactions exchanged.

F: longitudinal force at the locating bearing, + toward the next grid; a crane stop force acts toward its stop on the end bay only.

H_{tie} / H_{seat}: lateral force of all girder ends at the grid at the top / bottom flange; they reverse together, ∓ = H_{seat} opposite in sign (couple).

“Traction on bay n”: the full traction assigned to that bay when the crane straddles a joint (alternative cases, not added).

“Traction through the rail string to the stop on bay n”: the full traction carried by the floating rail to the end stop on that end bay, the crane anywhere (alternative to

the bay under the crane).

(a) Wheel over the grid assigned to the adjacent girder end.

Governing for the haunch: grids 2–5, LRFD A2c, ΣR_u = 29.0 kip with seat moment ±20.4 kip-ft. Governing for the cap tieback: LRFD A2b, grid 5, H_{tie} = 4.09 kip.

Lateral restraint and local forces

Force	Value	Use
H, factored (all ends at a grid / one end)	4.09 kip / 4.08 kip	LRFD A2b, grid 5; at the tie line, 14.8 in above the W bottom
Appendix 6 brace force $P_{br} = 0.02M_rC_d/h_o + 0.01P_r$	2.40 kip + 0.10 kip = 2.50 kip	AISC 360-16 Appendix 6.3.1b and 6.2.2, summed per 6.4, with M_r = 134 kip-ft, P_r = 10.0 kip, h_o = 13.4 in, C_d = 1
Tie design force F = H + P_{br}, pull and push	**6.58 kip**	Each tie and its column connection (section 5.4); lateral at the tie line in section 3.5
Brace stiffness required / provided	5.68 kip/in / 34.7 kip/in	Appendix 6: $\beta_{br} = (1/\varphi)(10M_rC_d/(L_{br}h_o) + 8P_r/L_{br})$, L_{br} = 300 in, φ = 0.75; provided = cap tieback stiffness (section 5.4), its secant value with the play taken up at P_{br} (262 kip/in after contact)
Tie force range ΔF, fatigue (all ends at a grid / one end)	2.35 kip / 2.35 kip	Pull-to-push range over the duty bins (unfactored)
Longitudinal force at the girder end	10.0 kip	Locating bearing, bolts and longitudinal system
Rail vertical / lateral force per wheel, factored	15.0 kip / 1.60 kip	Rail and rail clips
Rail vertical / lateral force per wheel, fatigue	7.50 kip / 0.50 kip	Rail clip weld and rail joint fatigue

3.5 Forces delivered to the building

Factored forces on each column at the seat and the tie line, as concurrent sets from section 3.4 (also on sheet S-00). For building combinations, use the unfactored reactions by load type in section 3.3, and the bumper force C_{bs} = 10.0 kip at the runway-end grids (section 2.2), which that table does not carry.

Grid	Set	Combination	V	$M_x = V{\cdot}e$	$M_{y,seat}$	H_{tie}	H_{seat}	F
1	Max V	LRFD A2c	24.4 kip	67.2 kip-ft	+7.33 kip-ft	±2.93 kip	∓0.43 kip	−3.20 kip
	Max H	LRFD A2b	20.1 kip	55.3 kip-ft	+6.03 kip-ft	±6.28 kip	∓1.47 kip	−3.20 kip
	Stop	LRFD 8	15.3 kip	42.0 kip-ft	+4.58 kip-ft	±2.72 kip	∓0.23 kip	−10.0 kip
2	Max V and max F	LRFD A2c	29.0 kip	79.9 kip-ft	±20.4 kip-ft	±2.96 kip	∓0.47 kip	−3.20 kip
	Max H	LRFD A2b	24.2 kip	66.7 kip-ft	+16.3 kip-ft	±6.58 kip	∓1.53 kip	−3.20 kip
	Max F	LRFD A2c	28.6 kip	78.7 kip-ft	−20.4 kip-ft	±2.95 kip	∓0.45 kip	+3.20 kip
3	Max V and max F	LRFD A2c	29.0 kip	79.9 kip-ft	±20.4 kip-ft	±2.96 kip	∓0.47 kip	−3.20 kip
	Max H	LRFD A2b	24.2 kip	66.7 kip-ft	+16.3 kip-ft	±6.58 kip	∓1.53 kip	−3.20 kip
	Max F	LRFD A2c	28.6 kip	78.7 kip-ft	−20.4 kip-ft	±2.95 kip	∓0.45 kip	+3.20 kip
4	Max V and max F	LRFD A2c	29.0 kip	79.9 kip-ft	±20.4 kip-ft	±2.96 kip	∓0.47 kip	−3.20 kip
	Max H	LRFD A2b	24.2 kip	66.7 kip-ft	+16.3 kip-ft	±6.58 kip	∓1.53 kip	−3.20 kip
	Max F	LRFD A2c	28.6 kip	78.7 kip-ft	−20.4 kip-ft	±2.95 kip	∓0.45 kip	+3.20 kip
5	Max V	LRFD A2c	29.0 kip	79.9 kip-ft	±20.4 kip-ft	±2.96 kip	∓0.47 kip	−3.20 kip
	Max H	LRFD A2b	24.2 kip	66.7 kip-ft	+16.3 kip-ft	±6.58 kip	∓1.53 kip	−3.20 kip
	Stop	LRFD 8	8.47 kip	23.3 kip-ft	+3.07 kip-ft	±2.61 kip	∓0.12 kip	+10.0 kip
6	Max V	LRFD A2c	24.4 kip	67.2 kip-ft	−7.33 kip-ft	±2.92 kip	∓0.42 kip	—
	Max H	LRFD A2b	20.1 kip	55.3 kip-ft	−6.03 kip-ft	±6.25 kip	∓1.44 kip	—
Strut	Crane-level strut by others, girder line	traction / stop	—	—	—	—	—	±10.0 kip

Factored LRFD. Each row is one concurrent set; do not combine rows.

V: girder reactions on the seat, El. 118'-0", at e = 33.0 in from the column centerline (the far edge of the bearing plate, a conservative reaction position; the girder web is at 27.0 in). $M_x = V{\cdot}e$ (column strong axis).

$M_{y,seat}$: moment of the unbalanced end reactions about the grid (column weak axis).

H_{tie}: top-flange restraint force plus the Appendix 6 brace force (2.50 kip), at the tie line, El. 119'-3 13/16"; design value 6.58 kip.

H_{seat}: bottom-flange restraint force at the bearing bolts, ∓ opposite in sign to H_{tie}.

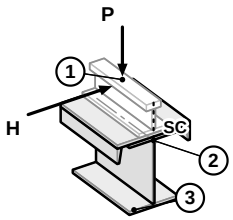
F: longitudinal force at the locating bearing, + toward the next grid. Stop force 10.0 kip at grids 1 and 5 only. Traction reaches the bay under the crane, or through the floating rail string an end bay at its stop.

Strut: the crane-level strut by others on the girder line, 27.0 in from the column centerline, at the seat level of grids 1, 2, 3, 4, 5, axial stiffness k ≥ 22.7 kip/in at each seat: design force F = 10.0 kip (traction 3.20 kip, stop 10.0 kip), carried to the braced bay; the existing column check relies on it.

V includes the bearing hardware on the seat; the haunch and other support steel (0.56 kip per grid, dead) goes straight into the column.

4 MEMBER CHECKS

4.1 Runway girder, capped section properties and stability



Capped section properties

Detail: W14X61 + C15X50, Fy 50 ksi; 5 simple spans of 25'-0"

Load path: (1) Wheel load and side thrust at the rail head → (2) Eccentric to the shear center: torsion and warping → (3) Section resists Mx, My, T

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Properties of the W14X61 + C15X50 capped section and the flexural strength of each compression flange.

Capped section properties

Section property	Value	Section property	Value
A	32.6 in²	Ix / Iy	1,032 in⁴ / 511 in⁴
Sx / Sy	103 in³ / 68.1 in³	J / Cw	4.84 in⁴ / 10,398 in⁶
Zx / Zy	130 in³ / 101 in³	r_t / h_o	4.62 in / 13.4 in
W14X61 depth / flange width	13.9 in / 10 in	W14X61 flange / web thickness	0.65 in / 3/8 in

Capped section design data

Property / input	Value	Property / input	Value
Channel Fy / Fu	50 ksi / 65 ksi	Electrode	70 ksi
Top / bottom Sx	226 in³ / 103 in³	Top lateral It	458 in⁴
Centroid above bottom	10 in	Shear center above bottom	13.1 in
Restraint height from shear center, top / bottom	+1.67 in / −12.8 in	Monosymmetry β_x	−10.7 in
Cap fillet / development	5/16 in / 60 in	Weight	111 plf

Capped section basis

Item	Basis and source	Value
Capped section model	Catalog W shape with a centered, inverted C or MC channel cap; full-length cap with two continuous longitudinal welds.	—
Cap lies wholly within top compression flange	The complete channel and W top flange must lie above the elastic neutral axis for the F4/F5 compression-flange idealization.	—
Channel steel and weld material	ASTM A572 Grade 50 channel, F_y 50 ksi, F_u 65 ksi; weld electrode F_{EXX} 70 ksi.	—
Crane service class for capped girder	CMAA C crane; source: Fictitious supplier confirms CMAA C, 5-US-ton crane; 400,000 wheel stress fluctuations (three duty bins).	—
Continuous cap attachment and bearing contact	Full-length, centered cap; continuous bearing contact and dual longitudinal fillets.	—
Unperforated cap and top flange	No holes in the channel or the beam top flange along the span; only the end stop bolt holes near the girder ends and the cap tieback angle bolt holes in the column-side channel flange within one girder depth of each girder end, outside the governing moment region; those holes are fatigue details (Appendix A).	—
Bearing stiffener top-flange connection	Full-depth fitted stiffeners with complete-joint-penetration weld to the beam top flange; bottom end fitted.	—

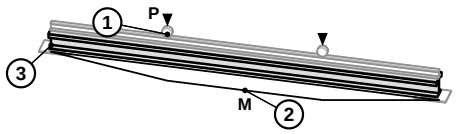
Flexural strength by compression flange (F4)

Compression side	Clause	r_t	L_p	L_r	Nominal M_n
Top flange + channel	F4	4.62 in	122 in	689 in	518 kip-ft
Bottom W flange	F4	2.67 in	70.7 in	241 in	194 kip-ft

Bottom flange shown for reference; simple spans put the cap in compression. C_b = 1 and the lower steel grade. Each flange uses the F4 or F5 equations of its web class (Table B4.1b case 16) with its own r_t, L_p and L_r; with a compact web $R_{pc}M_{yc} = M_p$ (F4-9a) where $I_{yc}/I_y > 0.23$, otherwise R_{pc} = 1.0 and J = 0 (F4-10, F4.2). E4-3 includes the centroid-to-shear-center offset. E7: the least plate effective-width ratio is applied to the whole area.

Properties: median-line open-section model (Ellifritt & Lue 1998), fillets ignored. Cross-section distortion is not modeled.

4.2 Runway girder, strength and stability

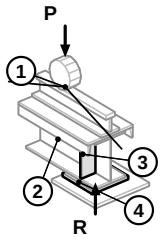


Bending, shear and combined actions

Detail: W14X61 + C15X50, Fy 50 ksi; 5 simple spans of 25'-0"

Load path: (1) Wheel loads P and side thrust H → (2) Girder bending Mx, flange bending and shear (H1, H3 with torsion) → (3) Reactions to the supports

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Wheel patch and bearing zone

Detail: Bearing PL 1" × 1'-0" × 1'-0"; stiffeners 2-PL 1" × 4 1/2", 5/16" fillets; 4-3/4" A325 bolts each end (locating slip-critical, sliding sleeved in slots)

Load path: (1) Wheel P through the rail onto the flange → (2) Web local yielding and crippling over the patch → (3) Reaction R into the bearing stiffeners → (4) Stiffeners and bearing plate on the seat

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Checks: F4 flexure with C_b = 1 and with C_b from the elastic buckling analysis (load at the rail head); G2 shear without tension field; H1 at each flange with its lateral moment; J10 web checks under the wheel; Appendix 6 bracing of the top flange.

Loads: Factored LRFD combinations of section 2.3 at every crane position (section 3.2): M_u = 134 kip-ft (LRFD A2c, crane at 10'-3" past grid 1, x = 10.3 ft); V_u = 26.3 kip (LRFD A2c); grid reaction ΣR_u = 29.0 kip (LRFD A2c); flange lateral moment 21.2 kip-ft (LRFD A2b); axial 10.0 kip (LRFD 8). Normal stress with warping peaks under LRFD A2b at x = 60.0 ft. Factored wheel load P_u = 15.0 kip for the local web checks.

Section classification

Check	Member	Ratio	Limit	Classification
Compression flange compactness AIST TR-13 §5.8; AISC 360-16 B4	Runway girder strength	7.75	9.15	Compact
Web compactness, singly symmetric (Table B4.1b case 16) AIST TR-13 §5.8; AISC 360-16 B4	Runway girder strength	13.9	137	Compact
Tension flange holes / net-section rupture AISC 360-16 F13	Runway girder strength	—	—	N/A
Capped section, singly symmetric proportioning AISC 360-16 F13	Runway girder strength	0.896	0.90	OK

Governing: H1 interaction, bottom flange, LRFD A2b (AISC 360-16 H1; AISC DG7 §11.2, §14.1): D/C = 0.44, OK.

a. Major-axis flexure, C_b = 1 AISC 360-16 F4 · L_b = 25'-0" (bay length); load height in part b **D/C 0.29**

$R_{pc}M_{yc} = M_p = F_y Z_x = 50\,\text{ksi} \times 130\,\text{in}^3 = 540\,\text{kip-ft}$ (compact web, F4-9a)

$F_L = 0.5F_y = 25\,\text{ksi}$ $(S_{xt}/S_{xc} = 103\,\text{in}^3/226\,\text{in}^3 = 0.455,$ F4-6); $F_L S_{xc} = 25\,\text{ksi} \times 226\,\text{in}^3 = 470\,\text{kip-ft}$

$L_p = 1.1 r_t \sqrt{E/F_y} = 122\,\text{in}, \quad L_r = 689\,\text{in}$ $(r_t = 4.62\,\text{in};$ F4-7, F4-8)

$M_{n,LTB} = C_b \left[R_{pc}M_{yc} - (R_{pc}M_{yc} - F_L S_{xc}) \frac{L_b - L_p}{L_r - L_p} \right] = 540\,\text{kip-ft} - (540\,\text{kip-ft} - 470\,\text{kip-ft}) \frac{300\,\text{in} - 122\,\text{in}}{689\,\text{in} - 122\,\text{in}} = 518\,\text{kip-ft}$ (F4-2)

$R_{pt}M_{yt} = 1.263 \times 428\,\text{kip-ft} = 540\,\text{kip-ft} \;\geq\; M_{n,LTB}$ (tension flange yielding does not govern, F4-15)

$\phi_b M_n = 0.90 \times 518\,\text{kip-ft} = 466\,\text{kip-ft} \;\geq\; M_u = 134\,\text{kip-ft}$ (LRFD A2c, crane at 10.3 ft) $D/C = 0.29$

b. Lateral-torsional buckling with the wheel loads at the rail head AISC 360-16 F4 entered at the equivalent unbraced length of the elastic buckling M_{cr} (C2.3, Commentary F1) **D/C 0.30**

Governing case LRFD A2c at x = 35.0 ft: M_u = 134 kip-ft. The buckling analysis of the girder at 0.8EI, with the wheel loads at the rail head 7.48 in above the shear center, gives the load factor λ = 4.74; dividing by 0.8 returns the elastic critical moment. It enters the Chapter F curve at the unbraced length whose uniform-moment (C_b = 1) elastic moment equals M_{cr}; no C_b < 1 factor multiplies the inelastic strength.

$M_{cr} = \lambda M_u / 0.8 = 4.736 \times 134\,\text{kip-ft}/0.8 = 793\,\text{kip-ft}$

$L_b/r_t = 300\,\text{in}/4.62\,\text{in} = 65.0; \qquad \frac{J}{S_{xc}h_o} = 0.00160$

$F_{cr}S_{xc} = \frac{\pi^2 E}{(L_b/r_t)^2} \sqrt{1 + 0.078 \frac{J}{S_{xc}h_o} \left(\frac{L_b}{r_t}\right)^2}\, S_{xc} = \frac{\pi^2 \times 29{,}000\,\text{ksi}}{65.0^2} \sqrt{1 + 0.078 \times 0.00160 \times 65.0^2} \times 226\,\text{in}^3 = 1{,}575\,\text{kip-ft}$ (F4-5, uniform moment)

$F_{cr}(L_e)S_{xc} = M_{cr} \;\Rightarrow\; L_e = 467\,\text{in}, \; L_e/r_t = 101.2 \;\geq L_b$ $(M_{cr}/F_{cr}(L_b)S_{xc} = 0.503,$ moment gradient and load height)

$M_n = R_{pc}M_{yc} - (R_{pc}M_{yc} - F_L S_{xc}) \frac{L_e - L_p}{L_r - L_p} = 540\,\text{kip-ft} - (540\,\text{kip-ft} - 470\,\text{kip-ft}) \frac{467\,\text{in} - 122\,\text{in}}{689\,\text{in} - 122\,\text{in}} = 498\,\text{kip-ft}$ (F4-2 at L_e, $C_b = 1$)

$\phi_b M_n = 0.90 \times 498\,\text{kip-ft} = 448\,\text{kip-ft} \;\geq\; M_u = 134\,\text{kip-ft} \quad D/C = 0.30$

Cross-check, AISC Commentary F1: $C_b = \frac{12.5 M_{max}}{2.5 M_{max} + 3M_A + 4M_B + 3M_C} = \frac{12.5 \times 134\,\text{kip-ft}}{2.5 \times 134\,\text{kip-ft} + 3 \times 85.9\,\text{kip-ft} + 4 \times 127\,\text{kip-ft} + 3 \times 105\,\text{kip-ft}} = 1.184$

With the load-height factor $1.4^{2y/h}$ = 0.54 (y = −13.3 in, the rail head from mid-depth; h = 14.6 in, the overall depth): C_b = 0.64, $\varphi_b M_n$ = 456 kip-ft, D/C = 0.29, against 0.30 from the analysis.

c. Shear AISC 360-16 G2.1, without tension field action — D/C 0.17

$\phi_v V_n = 1.00 \times 0.6 F_y d t_w C_{v1} = 1.00 \times 0.6 \times 50\,\text{ksi} \times 13.9\,\text{in} \times 0.375\,\text{in} \times 1.00 = 156\,\text{kip} \;\geq\; V_u = 26.3\,\text{kip}$ (LRFD A2c, crane at 90.0 ft) $\quad D/C = 0.17$

d. Combined flexure and axial force at each flange AISC 360-16 H1-1b with the flange lateral moment (AISC DG7 §11.2) — D/C 0.44

M_{rx} = $B_1 M_x$ with B_1 = 1.00 (Appendix 8); M_{cx} = the available flexural strength (Chapter F) for the sign of M_{rx} at the station: for the flange in compression with lateral-torsional buckling (with load height where that is lower), for the flange in tension its yielding strength (F2.1 M_p, F4.4 $R_{pt} M_{yt}$), lateral-torsional buckling being a compression-flange limit state; M_f = the flange lateral moment from side thrust and torsion; M_{cf} = the available flexural strength of that flange about its own axis (top flange $\varphi_b M_n$ (F6) = 229 kip-ft, bottom flange $\varphi_b F_y S_f$ = 40.3 kip-ft).

Top flange (LRFD A2b, x = 10.3 ft, positive moment, flange in compression): $\dfrac{P_r}{2P_c} + \dfrac{M_{rx}}{M_{cx}} + \dfrac{M_f}{M_{cf}} = \dfrac{3.2\,\text{kip}}{2 \times 795.6\,\text{kip}} + \dfrac{110.3\,\text{kip-ft}}{447.9\,\text{kip-ft}} + \dfrac{21.25\,\text{kip-ft}}{228.9\,\text{kip-ft}} = 0.341$

Bottom flange (LRFD A2b, x = 15.0 ft, positive moment, flange in tension): $\dfrac{P_r}{2P_c} + \dfrac{M_{rx}}{M_{cx}} + \dfrac{M_f}{M_{cf}} = \dfrac{3.2\,\text{kip}}{2 \times 795.6\,\text{kip}} + \dfrac{110.2\,\text{kip-ft}}{486.2\,\text{kip-ft}} + \dfrac{8.43\,\text{kip-ft}}{40.31\,\text{kip-ft}} = 0.438$

Bottom flange governs: $0.438 \;\leq\; 1.0 \quad D/C = 0.44$

e. Normal stress with warping torsion AISC 360-16 H3.3; AISC DG9 — D/C 0.43

At the flange tips, under the case that maximizes each (section 3.2). b = cap width at the top, b_f at the bottom; W_{no} = 28.9 in² at the cap tips, 64.1 in² at the bottom flange tips; θ and θ″ from the second-order torsion analysis with 0.8 of the elastic stiffness and the restraints at their secant stiffness. Its elastic critical load multiplier is λ = 4.60 (0.8 stiffness), so the twist and lateral bending are amplified about 1/(1 − α/λ) = 1/(1 − 1/4.60) = 1.3 (α = 1).

Top flange, cap tips (LRFD A2b, x = 60.0 ft): $f_{un} = \dfrac{P}{A} + \dfrac{M_x}{S_{xc}} + \dfrac{(M_y + \theta M_x) b}{2 I_y} + E W_{no} \theta''$

$= \dfrac{3.2\,\text{kip}}{32.6\,\text{in}^2} + \dfrac{110\,\text{kip-ft} \times 12\,\text{in/ft}}{226\,\text{in}^3} + \dfrac{(16.3\,\text{kip-ft} + 0\,\text{kip-ft}) \times 12\,\text{in/ft} \times 15\,\text{in}}{2 \times 511\,\text{in}^4} + 2.07\,\text{ksi} \quad (\theta = 0.0309\,\text{rad})$

$= 0.098\,\text{ksi} + 5.84\,\text{ksi} + 2.87\,\text{ksi} + 2.07\,\text{ksi} = 10.9\,\text{ksi} \;\leq\; \phi F_y = 45\,\text{ksi} \quad D/C = 0.24$

Bottom flange tips (LRFD A2b, x = 60.0 ft): $f_{un} = \dfrac{P}{A} + \dfrac{M_x}{S_{xt}} + \dfrac{(M_y + \theta M_x) b}{2 I_y} + E W_{no} \theta''$

$= \dfrac{3.2\,\text{kip}}{32.6\,\text{in}^2} + \dfrac{110\,\text{kip-ft} \times 12\,\text{in/ft}}{103\,\text{in}^3} + \dfrac{(16.3\,\text{kip-ft} + 0\,\text{kip-ft}) \times 12\,\text{in/ft} \times 10\,\text{in}}{2 \times 511\,\text{in}^4} + 4.59\,\text{ksi} \quad (\theta = 0.0309\,\text{rad})$

$= 0.098\,\text{ksi} + 12.8\,\text{ksi} + 1.91\,\text{ksi} + 4.59\,\text{ksi} = 19.4\,\text{ksi} \;\leq\; \phi F_y = 45\,\text{ksi} \quad D/C = 0.43$

f. Web under the wheel AISC 360-16 J10; AIST TR-13 §5.8.6 · P_u = 15.0 kip factored wheel load — D/C 0.17

Web local yielding (J10.2)

$l_b = 2(h_r + t_f + t_{wc}) = 2(6\,\text{in} + 0.645\,\text{in} + 0.716\,\text{in}) = 14.7\,\text{in}$ (TR-13 §5.8.6: the wheel spread through the rail, cap and flange to the web)

$\phi R_n = 1.00 \times F_{yw} t_w l_b = 1.00 \times 50\,\text{ksi} \times 0.375\,\text{in} \times 14.7\,\text{in} = 276\,\text{kip} \;\geq\; P_u = 15\,\text{kip} \quad D/C = 0.05$ (J10.2 stress; no k term)

Web crippling (J10.3)

$l_b = 2(h_r + t_{wc}) = 2(6\,\text{in} + 0.716\,\text{in}) = 13.4\,\text{in}$ on the W flange; $\quad l_b/d = 13.4\,\text{in}/13.9\,\text{in} = 0.966$

$\phi R_n = 0.75 \times 0.40 t_w^2 \left[1 + \left(\dfrac{4 l_b}{d} - 0.2\right)\left(\dfrac{t_w}{t_f}\right)^{1.5}\right] \sqrt{\dfrac{E F_{yw} t_f}{t_w}}$ (J10-5b)

$= 0.75 \times 0.40 \times (0.375\,\text{in})^2 \left[1 + (4 \times 0.966 - 0.2) \times 0.443\right] \sqrt{29{,}000\,\text{ksi} \times 50\,\text{ksi} \times 1.720} = 175\,\text{kip} \;\geq\; P_u = 15\,\text{kip} \quad D/C = 0.09$

Web sidesway buckling (J10.4)

$\dfrac{h/t_w}{L_b/b_f} = \dfrac{11.4\,\text{in}/0.375\,\text{in}}{300\,\text{in}/10\,\text{in}} = 1.015 \leq 1.7; \quad M_u = 134\,\text{kip-ft} < M_y = F_y S_x = 428\,\text{kip-ft}: \; C_r = 960{,}000\,\text{ksi}$

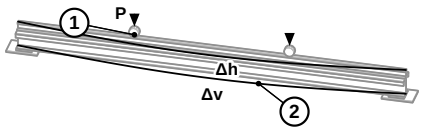
$\phi R_n = 0.85 \times \dfrac{C_r t_w^3 t_f}{h^2}\, 0.4 \left(\dfrac{h/t_w}{L_b/b_f}\right)^3 = 0.85 \times \dfrac{960{,}000\,\text{ksi} \times (0.375\,\text{in})^3 \times 0.645\,\text{in}}{(11.4\,\text{in})^2} \times 0.4 \times 1.015^3 = 89\,\text{kip} \;\geq\; P_u = 15\,\text{kip} \quad D/C = 0.17$

Check · reference	Equation	Value	Limit	D/C	Result
Combined actions at each flange: AISC H1 with the DG7 flange lateral moment, and normal stress with warping (top and bottom flange) Controlling: H1 interaction, bottom flange AISC 360-16 H1; AISC DG7 §11.2, §14.1 Governing case: LRFD A2b	$\frac{P_r}{2P_c} + \frac{M_{rx}}{M_{cx}} + \frac{M_f}{M_{cf}} \leq 1$ (H1-1b)	0.44	1.00	0.44	OK
Web shear including Saint-Venant and warping shear Controlling: Combined shear including Saint-Venant / warping AISC 360-16 H3; Moore & Mueller (EJ 2002)	$f_{uv} = \tau_b + \tau_t + \tau_{ws}$ $\tau_t = 0.8 G t \theta'$, $\tau_{ws} = \frac{0.8 E S_w \theta'''}{t}$ where τ_b shear stress from flexural shear, VQ/(It)	8.46 ksi	27.0 ksi	0.31	OK
Major flexure and lateral-torsional buckling with load height (AISC Chapter F, Cb from the elastic buckling analysis) Controlling: Major flexure, lateral-torsional buckling with load height AISC 360-16 F4, F1; AISC DG7 §11.2, §14.1; Moore & Mueller (EJ 2002)	$M_u \leq \phi_b M_n, \;\; M_n = M_n\big(\max[L_b, L_e]\big)\;(C_b = 1), \quad F_{cr}(L_e) S_{xc} = M_{cr}$ where M_{cr} elastic critical moment from the buckling analysis with the wheels at the rail head; F_{cr} F4-5 (F2-4) elastic stress with Cb = 1; L_e equivalent unbraced length: the length at which the Cb = 1 elastic moment $F_{cr} S_{xc}$ equals M_{cr}, at least Lb	134 kip-ft	448 kip-ft	0.30	OK
Web sidesway buckling AISC 360-16 J10.4(b) Governing case: LRFD A2c	$R_n = \frac{C_r t_w^3 t_f}{h^2} 0.4 \left(\frac{h/t_w}{L_b/b_f}\right)^3$ (J10-7)	15.0 kip	89.0 kip	0.17	OK

Check · reference	Equation	Value	Limit	D/C	Result
Longitudinal girder compression AISC 360-16 E3, E4, E7; AIST TR-13 §3.10 Governing case: LRFD 8	$F_e = \min(F_{ex}, F_{ey}, F_{ez}); \quad P_n = F_{cr} A_e$	10.0 kip	796 kip	0.01	OK

3 other limit states checked, max D/C 0.16 (Appendix A).

4.3 Runway girder serviceability



Deflection and rail movement

Detail: Vertical L/1000, lateral L/400

Steps: (1) One crane, static wheel loads (no impact) → (2) Vertical and rail-head lateral deflection Δ ≤ L / n

ISOMETRIC · NOT TO SCALE

Loads: One crane, static wheel loads without impact or girder dead load, at every position (AIST TR-13, DG7): vertical deflection 0.22 in, top-flange lateral deflection 0.067 in; rail-head displacement with twist 0.15 in (crane at 7'-6" past grid 3), 0.26 in with the restraint play taken up (+0.11 in), the movement the rail-head check uses; twist 0.011 rad (crane at 7'-6" past grid 3).

Governing: Vertical crane-load deflection (AIST TR-13 §5.8.7; Owner criteria): 0.22 in ≤ 0.30 in, D/C = 0.74, OK.

Vertical crane-load deflection:

$\Delta_v \le L/n_v,\ n_v = \max(600,\ 1000,\ 600) = 1000 :\ L/1000 = 300\,\text{in}/1000 = 0.3\,\text{in} \quad (\text{owner } L/600,\ \text{AIST Class B } L/1000,\ \text{CMAA C } L/600)$

$\Delta_v = \dfrac{Pa(3L^2 - 4a^2)}{24EI_x} = \dfrac{7.5\,\text{kip} \times 90\,\text{in} \times [3(300\,\text{in})^2 - 4(90\,\text{in})^2]}{24 \times 29{,}000\,\text{ksi} \times 1{,}032\,\text{in}^4} = 0.2234\,\text{in} \ \le\ 0.3\,\text{in} \quad D/C = 0.74$

$(\text{two static wheels } 120\,\text{in apart, symmetric about midspan, } a = (L - s)/2;\ \text{no impact})$

Lateral bending deflection:

$\Delta_h \le L/n_h,\ n_h = \max(400,\ 400) = 400 :\ L/400 = 300\,\text{in}/400 = 0.75\,\text{in} \quad (\text{owner } L/400,\ \text{AIST TR-13 } L/400)$

$\Delta_h = 0.0671\,\text{in} \ \le\ 0.75\,\text{in} \quad D/C = 0.09 \quad (\text{side thrust on the top flange and cap alone (DG7), one crane})$

Check · reference	Equation	Value	Limit	D/C	Result
Vertical crane-load deflection AIST TR-13 §5.8.7; Owner criteria	$\Delta_v \le L/1000 = 0.30\,\text{in}$ where L span	0.22 in	0.30 in	0.74	OK

2 other limit states checked, max D/C 0.35 (Appendix A).

4.4 Runway girder fatigue

Fatigue of the girder details under the section 2.4 duty bins.

Loads: Unfactored duty-bin loads of section 2.4 (Rated lifts 60,000; Routine half-load lifts 260,000; Light lifts 80,000); stress range = maximum minus minimum signed stress over each bin, at every crane position, including biaxial bending and warping.

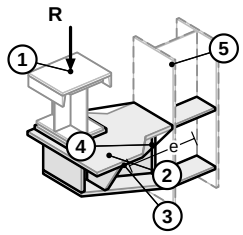
Basis. AISC 360-16 Appendix 3 at n_{SR} = 400,000, the total of the duty bins. D/C = $f_{sr,eq}/F_{SR}$, with $f_{sr,eq} = (\Sigma n_i {f_{sr,i}}^3/n_{SR})^{1/3}$ (Miner's rule, no threshold credit). The largest bin range applied to all cycles, $f_{sr,max}/F_{SR}$, is shown for comparison; where a detail has no Miner ratio (—) it is the design ratio. The design ratio is in bold. Details whose largest range is below F_{TH} are screened (Appendix A); a weld root, Category C″ with R_{FIL} < 1 (Eq. A-3-6), has no threshold.

Peak stress under dead load plus the unfactored cyclic load: f_{max} = 11.4 ksi ≤ 0.66F_y = 33.0 ksi (AISC 360-16 Appendix 3.1), D/C 0.34.

Screened details (21), closest to F_{TH}: Cap weld throat, Category F 6.89 ksi vs 8 ksi; Rail clip base weld 4.89 ksi vs 7 ksi; Top flange at midspan, conservative category D 4.61 ksi vs 7 ksi; Girder detail fatigue stress range 9.89 ksi vs 24 ksi.

No girder detail reaches F_{TH}: every girder detail, at 99 stations on both flange edges, is screened (Appendix A). The evaluated details are at the haunch, the column and the tie (sections 5.3 and 5.4).

4.5 Tapered haunch, inputs and actions



Tapered I haunch

Detail: Tapered I haunch 2'-4" long, 1'-5" to 0'-9" deep: web PL 3/8", top flange PL 1" × 2'-1 1/2", bottom flange PL 1" × 1'-2", wing stiffeners 2-PL 1"; flanges CJP and web 1/4" fillets to the column

Load path: ① Girder reaction R at the bearing → ② Top flange and bearing stiffeners → ③ Tapered web and flanges: M, V and torsion on each section → ④ Flange CJP and web fillets to the column → ⑤ Column flange and web (J10), column stiffeners

ISOMETRIC · NOT TO SCALE

The haunch is welded to the existing column flange and carries both girder bearings on its level top flange; its web lies on the grid line. The unbalanced end reactions twist the haunch, which resists by flange bending with warping fixed at the column. The bearing bolts deliver the longitudinal force and the bottom-flange lateral force to the top flange. Checks are made at 51 vertical sections from the root to the far bearing edge.

Input	Value	Input	Value
Projection L / face to runway center	28 in / 20 in	Root / tip depth, slope	17 in / 9 in, 15.9°
Web	3/8 in	Top flange (seat)	1 in × 25 1/2 in, 14 in at the column, 45° flare from 8 1/4 in
Bottom flange	1 in × 14 in	Bearing stiffeners	2 wing PL 1 in × 12 1/4 in under the seat, tapered to 6 1/2 in on the bottom flange, 5/16 in fillets, at the girder web line
Flanges to column	CJP groove welds: single bevel, 45°, on backing, through 1 1/2 in × 3/4 in weld access holes (J1.6); backing removed, backgouged, 1/4 in reinforcing fillet	Web to column / flanges to web	1/4 in / 5/16 in fillets both sides, the web weld between the access holes
Plates Fy / Fu	50 ksi / 65 ksi	Tip end plate	1 in
Column d / bf	14 in / 14 1/2 in	Column tf / tw	0.71 in / 0.44 in
Column Fy / Fu	50 ksi / 65 ksi	Existing factored column axial	62 kip
Deflection / rotation limits	1/16 in / 0.0020 rad	Column flange unbraced length	120 in
Column stiffeners	2 pairs PL 3/4 in × 6 3/4 in, full depth, 3/4 in clips, field welded	Stiffener fillets to flange / web	5/8 in / 5/8 in both faces

Assumptions and confirmed inputs

Item	Basis and source	Value
Receiving column, surveyed geometry and material	Fictitious survey: existing W14X90 column, ASTM A572 Grade 50 per mill certificates; 10 ft flange unbraced length; 62 kip factored existing gravity axial (1.2D + 1.6S from the building records) for the panel zone.	—
Bracket and tie load paths to the column	A built-up tapered I haunch 28 in long, 17 in deep at the column tapering to 9 in at the tip (bottom flange at 15.9°), field welded to the column flange, carries both girder bearings on its top flange.	—
Haunch bearing stiffeners, J10.8	Provided; the unstiffened haunch web alone carries the 29.6 kip haunch reaction R_h (girder end reactions with the haunch self-weight).	—

Section classification

Check	Member	Ratio	Limit	Classification
Hook bolt applicability AIST TR-13 §5.16	Loads, analysis and detailing	—	—	N/A
Tapered haunch, bottom flange local buckling (B4.1) AISC 360-16 B4.1	Tapered haunch	7.00	12.3	OK
Tapered haunch, bottom flange slope (model limit, DG25) AISC Design Guide 25 §3.3	Tapered haunch	15.9	45.0	OK
Web slenderness without intermediate stiffeners (F13.2, G2.1) AISC 360-16 F13.2, G2.1	Tapered haunch	39.9	232	OK
AIST column bracket, reaction recommendation AIST TR-13 §5.9.2	Tapered haunch	19.0 kip	50.0 kip	OK

Haunch actions at the governing case of each grid

Grid	Combination and crane position	R_h	Seat moment	Root moment	Torque	F	H	T	M_f, top	C cos α	M_f, bottom
1	LRFD A2c, crane at 1'-11" past grid 1, traction on bay 1	25.0 kip	7.33 kip-ft	54.6 kip-ft	5.07 kip-ft	−3.20 kip	0.43 kip	41.2 kip	5.18 kip-ft	40.8 kip	10.9 kip-ft
2	LRFD A2c, crane at grid 2, traction on bay 1	29.6 kip	20.0 kip-ft	64.6 kip-ft	20.0 kip-ft	—	0.47 kip	48.7 kip	31.0 kip-ft	48.2 kip	32.2 kip-ft

Grid	Combination and crane position	R_h	Seat moment	Root moment	Torque	F	H	T	M_f, top	C cos α	M_f, bottom
3	LRFD A2c, crane at grid 3, traction on bay 2	29.6 kip	20.0 kip-ft	64.6 kip-ft	20.0 kip-ft	—	0.47 kip	48.7 kip	31.0 kip-ft	48.2 kip	32.2 kip-ft
4	LRFD A2c, crane at grid 4, traction on bay 3	29.6 kip	20.0 kip-ft	64.6 kip-ft	20.0 kip-ft	—	0.47 kip	48.7 kip	31.0 kip-ft	48.2 kip	32.2 kip-ft
5	LRFD A2c, crane at grid 5, traction on bay 4	29.6 kip	20.0 kip-ft	64.6 kip-ft	20.0 kip-ft	—	0.47 kip	48.7 kip	31.0 kip-ft	48.2 kip	32.2 kip-ft
6	LRFD A2c, crane at 13'-1" past grid 5, traction on bay 5	25.0 kip	−7.33 kip-ft	54.6 kip-ft	−7.33 kip-ft	—	0.42 kip	41.2 kip	11.4 kip-ft	40.8 kip	11.8 kip-ft

R_h = girder end reactions plus haunch self-weight × 1.4, at the far bearing edge. Each flange delivers its force (T or C cos α) and lateral moment M_f to the column, carried by the stiffener pair at that level (section 5.3). Every case of section 3.4 is checked in both lateral directions; the table lists the governing case at each grid.

4.6 Tapered haunch, member checks

Flexure: AISC F4 (F5 if the web is slender) at each vertical section, in stress form (DG25); bottom-flange lateral-torsional buckling with L_b = 2L, C_b = 1. Shear: web alone, no Résal reduction; the vertical component of the sloping flange force is added to the column-face welds. Torsion: flange shears $T/h_o(y)$, warping fixed at the column. The welds to the column act as one elastic group. Sizes: sheet S-03.

Loads: section 3.4 envelope, every concurrent case; governing grid 2, LRFD A2c, ΣR_u = 29.0 kip, $M_{y,seat}$ = 20.4 kip-ft; haunch self-weight × 1.4 (LRFD 1).

Governing: Tapered haunch, top flange local bending under the bearings, LRFD A2c (AIST TR-13 §5.9.2): 4.94 kip ≤ 11.3 kip, D/C = 0.44, OK.

Tapered haunch · top flange local bending under the bearings:

$$m_u = \frac{Pc^2}{2lW} = \frac{26.63\,\text{kip} \times (5.5\,\text{in})^2}{2 \times 6.8\,\text{in} \times 12\,\text{in}} = 4.936\,\text{kip-in/in} \quad (c = W/2 - t_s/2;\ l \text{ the patch spread at } 45^\circ \text{ through the bearing plate})$$

$$\phi M_n = 0.90 \times \frac{F_y t_f^2}{4} = 0.90 \times \frac{50\,\text{ksi} \times (1\,\text{in})^2}{4} = 11.25\,\text{kip} \ \geq\ m_u = 4.936\,\text{kip} \quad D/C = 0.44$$

Bearing stiffeners · column, reactions bearing on the stiffeners (J10.8, J4.4/E3):

$$A = 2b_s t_s + 25 t_w\, t_w = 2 \times 6.5\,\text{in} \times 1\,\text{in} + 9.375\,\text{in} \times 0.375\,\text{in} = 16.52\,\text{in}^2; \quad L_c/r = 6.934\,\text{in}/1.275\,\text{in} = 5.4\ \leq 25\ (\text{J4.4}):\ \phi P_n = 0.90 F_y A = 743.2\,\text{kip}$$

$$\frac{P_r}{P_c} = \frac{26.93\,\text{kip}}{371.6\,\text{kip}} = 0.072\ \leq\ 1 \quad (P_c = \phi P_n/2,\ \text{one stiffener; the reaction bears on it at } x = 10.03\,\text{in}) \quad D/C = 0.07$$

Bearing stiffeners · plate at the web face, flexure and shear (stiffened seat):

$$d_s = h_s - 2c = 9.246\,\text{in} - 2 \times 0.5\,\text{in} = 8.246\,\text{in}; \quad \phi M_n = 0.90 F_y t_s \frac{d_s^2}{4} = 0.90 \times 50\,\text{ksi} \times 1\,\text{in} \times \frac{(8.246\,\text{in})^2}{4} = 63.74\,\text{kip-ft}; \quad \phi V_n = 1.00 \times 0.6 F_y t_s d_s = 247.4\,\text{kip}$$

$$M_u = Pu = P(x - t_w/2) = 26.93\,\text{kip} \times 9.839\,\text{in} = 22.08\,\text{kip-ft}; \quad V_u = P = 26.93\,\text{kip}$$

$$\left(\frac{M_u}{\phi M_n}\right)^2 + \left(\frac{V_u}{\phi V_n}\right)^2 = \left(\frac{22.08\,\text{kip-ft}}{63.74\,\text{kip-ft}}\right)^2 + \left(\frac{26.93\,\text{kip}}{247.4\,\text{kip}}\right)^2 = 0.132\ \leq\ 1 \quad D/C = 0.13$$

Worked: Tapered haunch, web shear along the taper (AISC 360-16 G2.1)

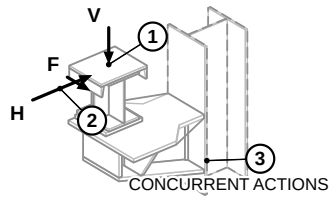
$$A_w = h(y) t_w = 9.571\,\text{in} \times 0.375\,\text{in} = 3.589\,\text{in}^2\ (y = 26\,\text{in}); \quad C_{v1} = 1.00\ (h/t_w = 20.1)$$

$$\phi V_n = 0.90 \times 0.6 F_y A_w C_{v1} = 0.90 \times 0.6 \times 50\,\text{ksi} \times 3.589\,\text{in}^2 \times 1.00 = 96.91\,\text{kip}\ \geq\ V_u = 29.64\,\text{kip} \quad D/C = 0.31$$

Check · reference	Equation	Value	Limit	D/C	Result
Tapered haunch, top flange local bending under the bearings AIST TR-13 §5.9.2 Governing case: LRFD A2c, crane at 15'-0" past grid 1, traction on bay 1, wheel over the grid taken by the adjacent girder end	$m_u = \max[Pc^2/(2lW),\ P(x - x_s)/b_e] \leq \phi F_y t_f^2/4$ where m_u required moment per unit width of the top flange; P bearing force; c flange cantilever from the stiffener face; l bearing patch length; W bearing width; x load position from the web plane; x_s stiffener tip; b_e bearing width spread at 45° to the stiffener tip	4.94 kip	11.3 kip	0.44	OK
Tapered haunch, combined normal stress at the root (H3.3) AISC 360-16 H3.3 Governing case: LRFD A2c, crane at 15'-0" past grid 3, traction on bay 4, wheel over the grid taken by the adjacent girder end	$f_{un} = \frac{P}{A} + \frac{M_x}{S_x} + \frac{M_f}{S_f} \leq \phi F_y$ where P axial force along the haunch; M_f flange bending from torsion and the longitudinal force	17.1 ksi	45.0 ksi	0.38	OK
Bracket, vertical deflection AIST TR-13 §5.9.2 Governing case: one crane, static wheel loads, crane at 15'-0" past grid 3, wheel over the grid taken by the adjacent girder end	$\Delta = \Delta_b + \Delta_s + \theta x$ where Δ_b bending deflection over the taper; Δ_s shear deformation over the taper; θ twist at the bearing; x distance to the outer bearing edge	0.020 in	0.063 in	0.31	OK
Tapered haunch, web shear along the taper (G2.1) AISC 360-16 G2.1 Governing case: LRFD A2c, crane at 15'-1" past grid 1, traction on bay 2	$V_u \leq \phi_v 0.6 F_y A_w C_{v1},\ A_w = h(y) t_w,\ k_v = 5.34$	29.6 kip	96.9 kip	0.31	OK

13 other limit states checked, max D/C 0.38 (Appendix A).

4.7 Existing column



Existing column: crane plus building loads

Detail: Existing W14X90, Fy 50 ksi, 26'-0" base to top support

Load path: (1) Girder reaction V at the support → (2) Concurrent lateral H and longitudinal F → (3) Column and building carry them together

ISOMETRIC · NOT TO SCALE

The existing column under the crane reactions and the existing building loads: axial force and biaxial bending (H1).

Loads: ASCE 7-16 LRFD combinations, crane loads as live load L, repeated with C_d factored as dead load (TR-13 rows); crane reactions by load type (section 3.3). Governing: 1.2D + 1.2Cdm + 1.0W + (Cvm + Ci + Css + Cls + L) + 0.5S, W reversed, at grid 4 (H1 = 0.26).

Item	Value	Item	Value
Section	W14X90 (surveyed)	Fy	50 ksi
Base to top support	312 in	Base to seat	216 in
Rail head above base	238 in	Eccentricity	33 in (far bearing edge)
Strong axis	pinned base, braced top	Weak axis	pinned base, braced top
Lcx / Lcy	312 in / 120 in	Lcz / Lb	120 in / 120 in
Longitudinal force	crane-level strut (by others)	Governing support	grid 4 (x = 75.0 ft)

Existing load effects at grid 4, unfactored (original building calculation sheets 14–17, column lines A and B (the same frames))

Load	P	M_x	M_y	V
D	25.0 kip	15.0 kip-ft	0 kip-ft	1.00 kip
L	0 kip	0 kip-ft	0 kip-ft	0 kip
Lr	12.0 kip	6.00 kip-ft	0 kip-ft	0.50 kip
S	20.0 kip	10.0 kip-ft	0 kip-ft	0.80 kip
R	0 kip	0 kip-ft	0 kip-ft	0 kip
W	−8.00 kip	40.0 kip-ft	0 kip-ft	4.00 kip
E	0 kip	20.0 kip-ft	0 kip-ft	2.00 kip

Governing ASCE 7 LRFD combinations (crane as live load L)

Case	Combination	P_r	M_{rx}	M_{ry}	B_{1x}	H1
TR-13 LRFD 4 (S, W reversed) (side thrust, no impact)	1.2D + 1.2Cdm + 1.0W + (Cvm + Ci + Css + Cls + L) + 0.5S, W reversed	66.8 kip	108 kip-ft	10.5 kip-ft	1.02	0.26
TR-13 LRFD 4 (S, W reversed) (impact, no side thrust)	1.2D + 1.2Cdm + 1.0W + (Cvm + Ci + Css + Cls + L) + 0.5S, W reversed	69.8 kip	107 kip-ft	10.5 kip-ft	1.02	0.26
TR-13 LRFD 4 (Lr, W reversed) (side thrust, no impact)	1.2D + 1.2Cdm + 1.0W + (Cvm + Ci + Css + Cls + L) + 0.5Lr, W reversed	62.8 kip	106 kip-ft	10.5 kip-ft	1.02	0.26

The governing combination and the next two; every combination is in Appendix A. H1 by grid: 1: 0.23, 2–5: 0.26, 6: 0.22. Results above: grid 4.

Conditional result. The column check of this section holds with these restraints, assumed here and confirmed by the building engineer of record (section 6.7): (1) the column top braced across the runway at 312 in (pinned base, braced top); (2) a braced (non-sway) frame, B1 only (frame type not given); (3) bracing of the inner (bracket) flange at 120 in or closer, the length its weak-axis, torsional and lateral-torsional buckling use (the girts at 96 in brace the outer flange only); (4) a crane-level strut on the girder line at each locating column, at least $10k_c$ = 22.7 kip/in along the runway at the seat (the column's own k_c = 2.27 kip/in) and designed for the full longitudinal force; the column keeps at most 1/11 of it, as weak-axis bending and torsion, included above; (5) the runway's own seismic forces within the crane-force design, $C_s \le 0.25$ across and 0.29 along (site seismic data by others; the girder, the bolted end bearings, the cap tiebacks and the haunches are covered to the same limit). Without any of them the column result does not apply.

Braced frame assumed (pinned base, braced top, B1 only). If the frame is a sway frame, repeat with B2 from the frame analysis (by others, section 6.7).

ASCE 7-16 §2.3.1 combinations 1–5 and §2.3.6 combinations 6–7, W and E each way, every crane load as live load L (ASCE 7 §4.9: $L = C_d + C_v + C_i + C_{ss} + C_{ls}\cdot e/L_{bay}$ here, the traction couple over the bay length). The rows marked TR-13 repeat each combination with C_d factored as dead load (AIST TR-13 §3.10); the larger governs. The combinations with E take the existing building seismic effect as entered (M_x = 20 kip-ft); they do not govern (largest H1 = 0.21, LRFD TR-13 6+ (side thrust, no impact)). The runway's own seismic force is by others (runway seismic, section 6.7).

Impact C_i is included in the column load.

L_{cx} = 312 in, base to the braced top. Weak-axis, torsional and lateral-torsional buckling use 120 in, the surveyed unbraced length of the inner flange, which receives the bracket (the girts at 96 in brace the outer flange only).

Drift (AIST TR-13 §5.3): H = 232 in (base to top of girder); limit min(H/400, 2 in) = 0.58 in; displacement taken at the rail head (238 in).

Governing: Existing column, torsion from the longitudinal force offset, TR-13 LRFD 4 (S, W reversed) (side thrust (AIST TR-13 §5.9.2; AISC 360-16 H3): D/C = 0.28, OK.

Terms: Each girder's longitudinal force reaches its locating column at the seat on the girder line, 27 in from the column centerline, while the crane-level strut on the girder line takes it in proportion to its stiffness: with the strut at least 10 k_c = 22.7 kip/in (the column's own stiffness there k_c = (1/k_y + e²/K_θ)⁻¹ = 2.27 kip/in: weak-axis k_y = 22.8 kip/in in series with the twist K_θ/e² = 2.52 kip/in) the column keeps at most 1/11 of it, as weak-axis bending (in H1) and torsion: torque T = 0.41 kip-ft at the seat (TR-13 LRFD 4 (S, W reversed) (side thrust, no impact), support x=900 in). Twist held at the base and at the braced top, warping free (AISC Design Guide 9, exact two-segment solution), a = 104 in: warping stress 0.64 ksi at the flange tips added to H1 as flange lateral bending; flange shear (Saint-Venant plus warping) 1.37 ksi against 27 ksi. Column twist at the seat: 0.0027 rad under the unfactored single-crane traction (torque 0.41 kip-ft, every crane start and stop), moving the girder bearing 0.072 in along the runway and the rail 0.000097 in across it, offset·(1 − cos θ); 0.013 rad under the largest factored torque, 0.36 in along and 0.002 in across.

Worked: Existing column, combined axial and flexure (H1), braced frame assumed (ASCE 7-16 §4.9; AISC 360-16 H1; AIST TR-13 §3.10)

$$P_r/P_c = 66.75\,\text{kip}/973\,\text{kip} = 0.07 < 0.2 \quad (\text{TR-13 LRFD 4 (S, W reversed) (side thrust, no impact)})$$

$$U = \frac{P_r}{2P_c} + \frac{M_{rx}}{M_{cx}} + \frac{M_{ry}}{M_{cy}} = 0.03 + \frac{108.1\,\text{kip-ft}}{563.5\,\text{kip-ft}} + \frac{10.46\,\text{kip-ft}}{271.4\,\text{kip-ft}} = 0.26 \;\le\; 1 \quad D/C = 0.26$$

Check · reference	Equation	Value	Limit	D/C	Result
Existing column, torsion from the longitudinal force offset (TR-13 §5.9.2) AIST TR-13 §5.9.2; AISC 360-16 H3 Governing case: TR-13 LRFD 4 (S, W reversed) (side thrust, no impact)	$\frac{P_r}{P_c} + \frac{8}{9}\left(\frac{M_{rx}}{M_{cx}} + \frac{M_{ry}}{M_{cy}} + \frac{\sigma_{ws}}{\phi F_y}\right) \le 1;\ \tau_t + \tau_{ws} \le \phi 0.6F_y$	0.28	1.00	0.28	OK
Existing column, axial compression ASCE 7-16 §4.9; AIST TR-13 §3.10 Governing case: TR-13 LRFD 3 (S, L) (impact, no side thrust)	$P_n = F_{cr}A_e$ (E3, E4, E7)	83.8 kip	973 kip	0.09	OK
Existing column, runway-level lateral displacement ASCE 7-16 §4.9; AIST TR-13 §5.3	$\Delta \le \min(H/400,\ 2\,\text{in})$ where H height from the column base to the top of the crane girder	0.045 in	0.58 in	0.08	OK
Existing column, strong-axis shear ASCE 7-16 §4.9 Governing case: TR-13 LRFD 4 (S) (side thrust, no impact)	$V_n = 0.6F_yA_wC_{v1}$ (G2)	8.80 kip	166 kip	0.05	OK

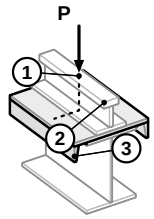
3 other limit states checked, max D/C 0.26 (Appendix A).

Single-column elastic model about each axis. Existing moments are added to the crane peaks regardless of sign. Frame action, base fixity, anchors and foundations are by others (section 6.7).

5 CONNECTION CHECKS

Connections in load-path order: cap channel welds and end development (5.1), bolted end bearings (5.2), tapered haunch welds, fatigue and column local effects (5.3), cap tieback (5.4), rail, clips, joints and anchorage (5.5), runway end stops (5.6).

5.1 Cap channel welds and end development



Cap welds and load transfer

Detail: W14X61 + C15X50, Fy 50 ksi; 5 simple spans of 25'-0"

Load path: (1) Wheel load into the cap channel → (2) Cap-to-flange fillets carry the shear flow VQ/I → (3) Composite section; welds developed at the cap ends

ISOMETRIC · NOT TO SCALE

Shear flow in the cap-to-flange fillets and development at the cap ends.

Loads: Shear flow from the factored girder shear V_u = 26.3 kip (LRFD A2c) with the warping and lateral shear from the torsion analysis, the local wheel force 15.0 kip through the cap, and the channel's share of the axial force 10.0 kip over the end development length; fatigue from the duty bins of section 2.4.

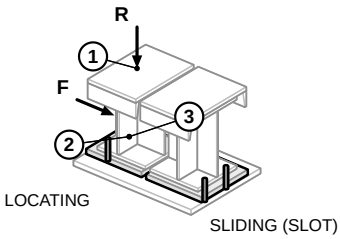
Governing: Cap weld, combined shear flow (AISC DG7 §13.3, §14.1): 1.94 kip/in ≤ 6.96 kip/in, D/C = 0.28, OK.

Check · reference	Equation	Value	Limit	D/C	Result
Cap channel welds (shear flow, connected metal) Controlling: Cap weld, combined shear flow AISC DG7 §13.3, §14.1	$q_w = \frac{V_x Q_x}{2I_x} + \frac{V_y Q_y}{I_y} + ES_w\theta''' + q_r + \frac{P_c A_c}{2A\beta a'}$ where Q_y first moment of half the cap about the vertical axis; q_r shear flow from the wheel loads (TR-13 §5.8.6 spread); A area of the capped section; β end-loaded weld factor, J2.2b	1.94 kip/in	6.96 kip/in	0.28	OK

1 other limit state checked, max D/C 0.11 (Appendix A).

Cap weld base metal (Category B): largest range 4.89 ksi at x = 15.0 ft, below F_{TH} = 16.0 ksi. Cap-end terminations (Category E′) are screened in Appendix A. Weld layout: sheet S-02.

5.2 Bolted end bearings

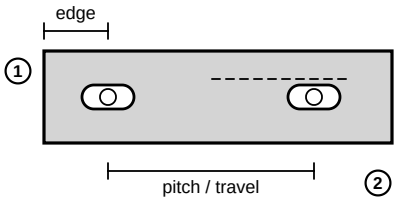


Bolted end bearing

Detail: Bearing PL 1" × 1'-0" × 1'-0"; stiffeners 2-PL 1" × 4 1/2", 5/16" fillets; 4-3/4" A325 bolts each end (locating slip-critical, sliding sleeved in slots)

Load path: (1) Girder reaction R on the bearing plate → (2) Longitudinal force F by slip-critical bolts at the locating end → (3) Uplift and lateral force through the bolts into the seat

ISOMETRIC · NOT TO SCALE



Sliding end: bolt layout and movement allowance

Detail: Bearing PL 1" × 1'-0" × 1'-0"; stiffeners 2-PL 1" × 4 1/2", 5/16" fillets; 4-3/4" A325 bolts each end (locating slip-critical, sliding sleeved in slots)

Steps: (1) Edge distance and pitch of each bolt → (2) Slot length for the movement

SCHEMATIC · NOT TO SCALE

The bolted bearing at each girder end: bearing plate and stiffeners, bolts and sleeves, and the movement allowance at the girder joints.

Loads: ΣR_u = 29.0 kip (LRFD A2c), the full grid reaction, used for one girder end; least reaction 0 kip; longitudinal force at the girder end 10.0 kip (traction or crane stop, section 2.2); bottom-flange lateral force H_{seat} (section 3.4).

Bearing plate and stiffeners

The bearing stiffeners take the reaction from the web to the bearing plate, which spreads it to the seat. They are checked for the full grid reaction ΣR_u = 29.0 kip, which bounds one girder end.

Governing: Bearing stiffener welds, reaction with the lateral flange forces (AISC 360-16 J2.4, J4.2(b)): 39.0 kip ≤ 207 kip, D/C = 0.19, OK.

$M = |T_u| + 0.02 M_r C_d = 1.031\,\text{kip-ft} + 2.679\,\text{kip-ft} = 3.71\,\text{kip-ft}; \quad V = M/h_o = 3.317\,\text{kip}$

$R_{eq} = R + \dfrac{2VQL_w}{I} = 29.04\,\text{kip} + \dfrac{2 \times 3.317\,\text{kip} \times 10.97\,\text{in}^3 \times 9.42\,\text{in}}{68.66\,\text{in}^4} = 39.03\,\text{kip}$

$\phi R_n = \min(262.3\,\text{kip},\ 206.7\,\text{kip},\ 551.1\,\text{kip}) = 206.7\,\text{kip} \quad (\text{web base metal},\ L_w = 9.42\,\text{in}) \ \geq \ R_{eq} = 39.03\,\text{kip} \quad D/C = 0.19$

Worked: Stiffener-to-web load transfer (AISC 360-16 J2.4, J4.2(b))

$L_w = h - 2c = 11.42\,\text{in} - 2 \times 1\,\text{in} = 9.42\,\text{in}$

$\phi R_n = 0.75 \times \min[0.6 F_{EXX}(0.707w)\,4L_w,\ 0.6F_u t_w\,2L_w,\ 0.6F_u t_s\,2L_w] = \min(262.3\,\text{kip},\ 206.7\,\text{kip},\ 551.1\,\text{kip}) = 206.7\,\text{kip} \quad (\text{web base metal}) \ \geq \ R_u = 29.04\,\text{kip} \quad D/C = 0.14$

Worked: Fitted stiffener-end bearing (AISC 360-16 J7)

$A_{pb} = 2(b_s - c)t_s = 2(4.5\,\text{in} - 1\,\text{in}) \times 1\,\text{in} = 7\,\text{in}^2$

$\phi R_n = 0.75 \times 1.8 F_y A_{pb} = 0.75 \times 1.8 \times 50\,\text{ksi} \times 7\,\text{in}^2 = 472.5\,\text{kip} \ \geq \ R_u = 29.04\,\text{kip} \quad D/C = 0.06$

Check · reference	Equation	Value	Limit	D/C	Result
Girder bearing stiffeners (column buckling, fitted bearing, welds) Controlling: Bearing stiffener welds, reaction with the lateral flange forces AISC 360-16 J2.4, J4.2(b)	$R_{eq} = R + \frac{2VL_w t_s b_s (t_w + b_s)}{2I}$ where R_{eq} equivalent reaction on the four web weld lines; R support reaction	39.0 kip	207 kip	0.19	OK
Bearing plate and contact Controlling: Bearing plate flexure beyond the flange tips AISC 360-16 F11	$f_b = \frac{6m}{t_p^2}; \quad m = \frac{f_p a^2}{2}; \quad a = \frac{B - b_f}{2}$ where m moment per unit width; f_p bearing pressure; a cantilever from the stiffener footprint	0.61 ksi	45.0 ksi	0.01	OK

Bolts and sleeves

Each girder end has four bolts through the girder bottom flange, bearing plate and seat, in two rows clear of the stiffeners. Locating end (left end of each bay): standard holes and pretensioned slip-critical bolts for the longitudinal force, the bottom-flange restraint force and uplift. Sliding end: slots in the girder flange under plate washers, with the bolts pretensioned against steel sleeves so the flange can slide; the sleeves carry lateral force in bearing. All runway bolts are pretensioned (AIST TR-13 §5.14; section 6.5). Prying: AISC Manual Part 9.

Governing: Sliding end, sleeves under bolt pretension (AIST TR-13 §5.14): 42.0 kip ≤ 55.2 kip, D/C = 0.76, OK.

$A_g = \dfrac{\pi}{4}(d_o^2 - d_h^2) = \dfrac{\pi}{4}\left[(1.438\,\text{in})^2 - (0.8125\,\text{in})^2\right] = 1.104\,\text{in}^2 \quad (\text{sleeve})$

$1.5T_b = 1.5 \times 28\,\text{kip} = 42\,\text{kip} \ \leq \ \phi F_y A_g = 1.00 \times 50\,\text{ksi} \times 1.104\,\text{in}^2 = 55.22\,\text{kip} \quad D/C = 0.76 \quad (T_b\ \text{Table J3.1})$

Check · reference	Equation	Value	Limit	D/C	Result
End bearing sleeves and plate washers Controlling: Sliding end, sleeves under bolt pretension AIST TR-13 §5.14	$1.5T_b \le \phi F_y A_g;\ \phi = 1.0\ \text{(sleeve)}$	42.0 kip	55.2 kip	0.76	OK

9 other limit states checked, max D/C 0.32 (Appendix A).

Movement allowance

Each bay has independent bearings and separate cap tiebacks to the column. Its left end transfers longitudinal force; its right end slides. Neither end transfers moment across the joint; cap channels stop with their girders (AIST TR-13 §5.8.1). At the runway-end grids each end-bay girder runs 6 in past the grid so that its bearing plate, stiffeners and reaction sit over the haunch web; the reaction acts at 0.8 of the bearing length from the girder end, 3.6 in from the web. The analysis spans grid to grid; the support checks use the bearing positions shown on S-03.

Input	Value
Clear girder gap	1 in
Bearing at a runway-end grid	Centered on the haunch web
Sliding travel, each direction	1/2 in
Temperature rise / fall from erection	54 °F / 54 °F
Setting tolerance at each end	1/8 in

Governing: Sliding end, movement allowance each direction (AISC DG7 §11.2; Owner criteria): 0.31 in ≤ 0.50 in, D/C = 0.61, OK.

Sliding end · movement allowance each direction:

$$\alpha L \Delta T + d\theta + t_{set} = 6.67 \times 10^{-6}\ /^{\circ}\text{F} \times 300\,\text{in} \times 54\,^{\circ}\text{F} + 13.9\,\text{in} \times 0.00536 + 0.125\,\text{in}$$

$$= 0.108\,\text{in} + 0.0745\,\text{in} + 0.125\,\text{in} = 0.3075\,\text{in} \ \le\ u = 0.5\,\text{in} \quad D/C = 0.61$$

Check · reference	Equation	Value	Limit	D/C	Result
Sliding end, movement allowance each direction AISC DG7 §11.2; Owner criteria	$\alpha L \Delta T + d\theta + t_{set} \le u$ where L distance from the locating support; ΔT temperature change; θ end rotation; u slot travel each way	0.31 in	0.50 in	0.61	OK

5.3 Tapered haunch welds, fatigue and column local effects

The haunch welds to the column, the fatigue of the haunch and column details, and the column under the haunch flange forces.

Loads: as section 4.6, with the flange force systems below.

Governing: Column web, shear along the stiffener welds (AISC 360-16 J4.2): 17.8 ksi ≤ 29.3 ksi, D/C = 0.61, OK.

$$\phi F_n = 0.75 \times 0.6 F_u = 0.75 \times 0.6 \times 65\,\text{ksi} = 29.25\,\text{ksi} \ \ge\ f_v = 17.85\,\text{ksi} \quad D/C = 0.61$$

Worked: Web to column, fillet welds (AISC 360-16 J2.4)

$$\phi F_n = 0.75 \times 0.6 F_{EXX} = 0.75 \times 0.6 \times 70\,\text{ksi} = 31.5\,\text{ksi} \ \ge\ f_w = 6.887\,\text{ksi} \quad D/C = 0.22$$

Worked: Column stiffeners at the haunch top flange, plate force (AISC 360-16 J10.8, J4.1, J4.4)

$$g = t_w + b_s = 0.44\,\text{in} + 6.75\,\text{in} = 7.19\,\text{in}; \quad R_{st} = \frac{T}{2} + \frac{M_f}{g} = \frac{48.7\,\text{kip}}{2} + \frac{37.17\,\text{kip-ft} \times 12\,\text{in/ft}}{7.19\,\text{in}} = 86.39\,\text{kip}$$

$$\phi R_n = 0.90 F_y (b_s - c) t_s = 0.90 \times 50\,\text{ksi} \times (6.75\,\text{in} - 0.75\,\text{in}) \times 0.75\,\text{in} = 202.5\,\text{kip} \ \ge\ R_{st} = 86.39\,\text{kip} \quad D/C = 0.43$$

Worked: Column stiffeners at the haunch bottom flange, plate force (AISC 360-16 J10.8, J4.1, J4.4)

$$g = t_w + b_s = 0.44\,\text{in} + 6.75\,\text{in} = 7.19\,\text{in}; \quad R_{st} = \frac{C}{2} + \frac{M_f}{g} = \frac{48.24\,\text{kip}}{2} + \frac{33.11\,\text{kip-ft} \times 12\,\text{in/ft}}{7.19\,\text{in}} = 79.39\,\text{kip}$$

$$\phi R_n = 0.90 F_y (b_s - c) t_s = 0.90 \times 50\,\text{ksi} \times (6.75\,\text{in} - 0.75\,\text{in}) \times 0.75\,\text{in} = 202.5\,\text{kip} \ \ge\ R_{st} = 79.39\,\text{kip} \quad D/C = 0.39$$

Worked: Existing column web, panel shear with axial force (AISC 360-16 J10.6)

$$R_u = \max(T,\ C\cos\alpha) = \max(48.71\,\text{kip},\ 48.24\,\text{kip}) = 48.71\,\text{kip} \quad \text{(net flange force; the } M_f \text{ couples add no panel shear)}$$

$$\frac{\alpha P_r}{P_y} = \frac{91.64\,\text{kip}}{50\,\text{ksi} \times 26.13\,\text{in}^2} = 0.070 \ \Rightarrow\ \min(1,\ 1.4 - 0.070) = 1.000 \quad (P_r \text{ column axial force with the bracket reaction})$$

$$\phi R_n = 0.90 \times 0.6 F_y d_c t_w \times 1.000 = 0.90 \times 0.6 \times 50\,\text{ksi} \times 14\,\text{in} \times 0.44\,\text{in} \times 1.000 = 166.3\,\text{kip} \ \ge\ R_u = 48.71\,\text{kip} \quad D/C = 0.29$$

Check · reference	Equation	Value	Limit	D/C	Result
Column stiffeners at the haunch (strength, welds, web shear) Controlling: Column web, shear along the stiffener welds (J4.2); also Column stiffeners, fillet welds to the flange (J10.8, J2.4) (0.52) AISC 360-16 J4.2	$f_v = \frac{R_{st}}{L_w t_w} \le \min(\phi 0.6 F_y,\ \phi 0.6 F_u)$ where R_{st} force on one column stiffener plate	17.8 ksi	29.3 ksi	0.61	OK
Haunch-to-column welds (flange CJPs, web fillets, connected metal) Controlling: Top flange to column, CJP groove weld AISC 360-16 J2.4 Governing case: LRFD A2c, crane at 15'-0" past grid 3, traction on bay 4, wheel over the grid taken by the adjacent girder end	$f \le \phi F_y;\ \ f_v = 1.5 V_f / A_f \le \phi 0.6 F_y$ where f flange normal stress; V_f lateral shear in the flange; A_f flange area	0.38	1.00	0.38	OK

Fatigue

Unfactored duty-bin loads without impact. Ranges are maximum minus minimum of the signed response over each bin, from zero with no crane present; the warping stress reverses as the crane crosses the grid. AISC Appendix 3 categories: flanges at the column welds, item 5.4 (CJP, C) or 5.6 (fillets, toe C, root C″ with R_{FIL}); the compression bottom flange at its weld toe with its full range (residual tension at the toe); flange-to-web fillets, item 3.1 (B); stiffener toes, 5.7 (C); flare corner, 1.3 (C); column flange, 7.2 (E′); fillet throats, 8.2 (F, largest range at all cycles).

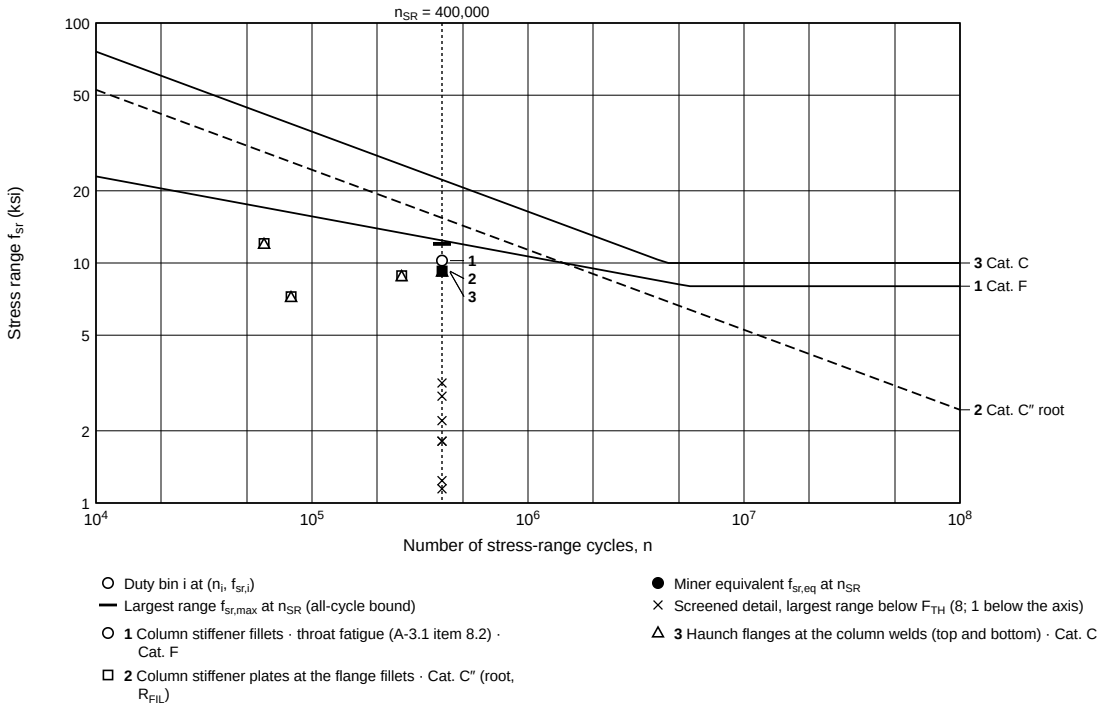
Stress ranges at the haunch details by duty bin

Duty bin	Cycles	Top flange (C)	Bottom flange (C)	Flange-web base (B)	Bearing stiffener toe (C)	Flare corner (C)	Column flange (E′)	Web fillet (F)	Flange-web fillet (F)	Bearing stiffener fillet (F)	Column stiffener force range, top / bottom level
Rated lifts	60,000	11.5 ksi	12.0 ksi	1.24 ksi	0.53 ksi	1.15 ksi	1.81 ksi	2.79 ksi	3.16 ksi	2.21 ksi	52.5 kip / 54.3 kip
Routine half-load lifts	260,000	8.47 ksi	8.77 ksi	0.92 ksi	0.40 ksi	0.84 ksi	1.33 ksi	2.05 ksi	2.32 ksi	1.62 ksi	38.6 kip / 39.8 kip
Light lifts	80,000	6.94 ksi	7.18 ksi	0.75 ksi	0.33 ksi	0.69 ksi	1.09 ksi	1.68 ksi	1.90 ksi	1.32 ksi	31.7 kip / 32.6 kip

Fatigue basis: section 4.4.

Detail and location	Cat.	F_{TH}	F_{SR} at n_{SR}	$f_{sr,max}$	$f_{sr,eq}$	$\Sigma n_i/N_i$	Miner $f_{sr,eq}/F_{SR}$	All-cycle $f_{sr,max}/F_{SR}$	Result
① **Column stiffener fillets, throat fatigue (A-3.1 item 8.2)** Tapered haunch	F	8 ksi	12.4 ksi	10.2 ksi	—	—	—	**0.83**	**OK**
② **Column stiffener plates at the flange fillets** Tapered haunch	C″ (root, R_{FIL})	none	15.4 ksi	12.1 ksi	9.25 ksi	0.22	**0.60**	0.78	**OK**
③ **Haunch flanges at the column welds (top and bottom)** Tapered haunch	C	10 ksi	22.2 ksi	12 ksi	9.16 ksi	0.07	**0.41**	0.54	**OK**

Screened details (8), closest to F_{TH}: Existing column flange at the haunch weld toes 1.81 ksi vs 4.5 ksi; Flange-to-web fillets, throat 3.16 ksi vs 8 ksi; Web-to-column fillets, throat 2.79 ksi vs 8 ksi; Bearing stiffener fillets, throat 2.21 ksi vs 8 ksi.



Fatigue design curves and duty bins

Design curves (AISC 360-16 Appendix 3) of the evaluated details, numbered as in the table, each through F_{SR} at n_{SR} down to F_{TH} (none for a C″ weld root). Open markers: duty-bin ranges (n_i, $f_{sr,i}$). Filled: $f_{sr,eq}$ at n_{SR}. Bar: $f_{sr,max}$ at n_{SR}. ×: screened details at their largest range. Points below their curve pass.

Column local effects

Each haunch flange delivers an axial force (T = M/h_o + P/2 at the top, C cos α at the bottom) and a lateral moment M_f from the haunch torque. Unstiffened, the existing column would take each as one concentrated force R_u = T + 3M_f/b, twice the more loaded half flange: top, T = 48.7 kip with M_f = 37.2 kip-ft: R_u = T + 3M_f/b = 48.7 kip + 3 × 37.2 kip-ft × 12 in/ft/14.0 in = 144 kip; bottom, C = 48.2 kip with M_f = 33.1 kip-ft: R_u = C + 3M_f/b = 48.2 kip + 3 × 33.1 kip-ft × 12 in/ft/14.0 in = 133 kip (LRFD A2c (a)).

Level	Unstiffened column (basis for the stiffener decision)	R_u (same at the level)	φR_n	D/C	Result
Top flange	J10.1 flange local bending	144 kip	142 kip	1.02	Exceeded
Top flange	J10.2 web local yielding	144 kip	166 kip	0.87	OK
Bottom flange	J10.2 web local yielding	133 kip	167 kip	0.80	OK
Bottom flange	J10.3 web crippling	133 kip	197 kip	0.68	OK
Bottom flange	J10.4 web sidesway buckling	133 kip	—	—	N/A *

* J10.4 does not apply: (h/t_w)/(L_b/b_f) > 1.7, J10.4(b)(2).

Column stiffeners required. Unstiffened, flange local bending (J10.1) is exceeded at the top flange (D/C 1.02), and the haunch needs M_f resisted at the column. Full-depth stiffener pairs (2 pairs PL 3/4" × 6 3/4", full depth, one pair at each haunch flange; 5/8" fillets to the flanges and 5/8" to the web, both faces; field welded inside the existing column) carry each flange's full force system with no credit for the unstiffened column: R_{st} = T/2 ± M_f/g, g = t_w + b_s = 7.19 in, giving 86.4 kip at the top (D/C 0.43) and 79.4 kip at the bottom (D/C 0.39). The pairs are checked for J4.1 yielding, J4.4 buckling with the J10.8 web strip, J2.4 welds and J4.2 web shear. Full depth, beyond the half depth J10.8(3) requires, takes the couple to the far flange; J10.5 and J10.7 do not apply to this continuous column. Panel shear J10.6: 48.7 kip, D/C 0.29; the stiffeners add no panel strength. Field welding controls: section 6.5.

Worked: unstiffened J10 at the haunch flanges (basis for the stiffener decision)

Top flange (LRFD A2c, crane at 15’-0” past grid 3, traction on bay 4, wheel over the grid taken by the adjacent girder end):

$$R_u = T + \frac{3M_f}{b}$$

$$= 48.7\,\text{kip} + \frac{3 \times 37.17\,\text{kip-ft} \times 12\,\text{in/ft}}{14\,\text{in}} = 144.3\,\text{kip}$$

$$\phi R_n = 0.90 \times 6.25 F_{yf} t_f^2 = 0.90 \times 6.25 \times 50\,\text{ksi} \times (0.71\,\text{in})^2 = 141.8\,\text{kip} \quad D/C = 1.02 \quad \text{(J10.1)}$$

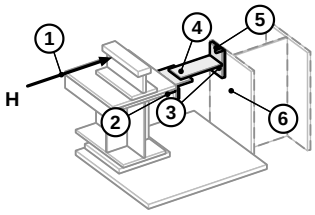
Bottom flange (LRFD A2c, crane at 15’-0” past grid 3, traction on bay 4, wheel over the grid taken by the adjacent girder end):

$$R_u = C\cos\alpha + \frac{3M_f}{b}$$

$$= 48.24\,\text{kip} + \frac{3 \times 33.11\,\text{kip-ft} \times 12\,\text{in/ft}}{14\,\text{in}} = 133.4\,\text{kip}$$

$$\phi R_n = 0.75 \times 0.80 t_w^2 \left[1 + 3\frac{l_b}{d}\left(\frac{t_w}{t_f}\right)^{1.5}\right]\sqrt{\frac{EF_{yw}t_f}{t_w}} = 197\,\text{kip} \quad D/C = 0.68 \quad (\text{J10.3},\ l_b = 1.04\,\text{in},\ d = 14\,\text{in})$$

5.4 Cap tieback



CAP ANGLE TIE

Cap tieback angle bolted to the channel cap

Detail: L4X4X1/2 × 6 1/2" cap angle with 2-5/8" A325 bolts in sleeved horizontal slots; tie PL 3/8" × 4 1/4" × 9 3/4" with 2-5/8" A325 slip-critical bolts; end PL 5/8" × 4 1/4" × 6", 5/16" fillets both faces; 2-5/8" A325 bolts through shims to the column flange (field-drilled)

Load path: (1) Side thrust H at the rail head loads the top flange → (2) Top flange and cap carry it to the column-side channel flange → (3) Cap angle bolts in horizontal slots (pull with prying; the girder slides along the runway) → (4) Angle and tie plate, lapped with slip-critical bolts → (5) End plate, bolts above and below the tie, and shims → (6) Column flange, or the stiffener at the tie level when needed

ISOMETRIC · NOT TO SCALE

One cap tieback angle (AISC DG7, top flange connected directly to the column) at each girder end, 3 1/2 in from the girder end to its centerline. The L4X4X1/2 angle is bolted through the column-side flange of the C15X50 channel with one leg level with the top of the cap; its bolts are pretensioned against steel sleeves (1 3/16 in OD) in 1 1/4 in × 2 in horizontal slots, so the girder slides 3/8 in each way along the runway with end rotation and thermal travel while the angle stays with the column. A 3/8 in × 4 1/4 in tie plate is lapped on the angle with two slip-critical bolts in short slots across the runway, spans the 7 3/4 in clear gap and is shop welded to a 5/8 in end plate, bolted through finger shims to the W14X90 column flange with one bolt above and one below the tie plate in field-drilled holes. The bolted end bearings restrain the bottom flange. Pull governs the bolts (tension with prying); push is checked for bearing and tie-plate buckling. Sheet S-04.

Input	Value	Input	Value
Angle / length	L4X4X1/2 / 6 1/2 in	Setback from girder end	3 1/2 in
Cap bolts	2 - 5/8 in A325, 2 1/4 in below top, 3 1/4 in pitch	Slots / travel	1 1/4 in × 2 in / 3/8 in each way
Tie plate	3/8 in × 4 1/4 in × 9 3/4 in	Lap bolts	2 - 5/8 in A325, Class B, 2 1/2 in pitch
End plate	5/8 in × 4 1/4 in × 6 in	Column bolts	2 - 5/8 in A325, 2 in each side
Shims	0 to 1/4 in	End plate fillets	5/16 in, both faces
Tie force F (pull / push)	6.58 kip	Fatigue range ΔF	2.35 kip
Column stiffener at the tie	not needed	Tie stiffness k after contact / k_{sec} with the play	262 kip/in / 34.7 kip/in

Section classification

Check	Member	Ratio	Limit	Classification
Column flange, tie bolt holes, F13.1 applicability (column flexural strength modifier) AISC 360-16 F13.1	Cap tieback	—	—	N/A

Loads: tie design force F = H + P_{br} = 4.09 kip + 2.50 kip = **6.58 kip**, pull or push. H: largest top-flange restraint force at a grid (LRFD A2b, grid 5). P_{br}: AISC 360-16 Appendix 6.3.1b with M_r = 134 kip-ft, plus 6.2.2 with P_r = 10.0 kip. Required brace stiffness 5.68 kip/in vs 34.7 kip/in provided (the secant stiffness with the tie's play; 262 kip/in after contact). Fatigue range ΔF = 2.35 kip.

Strength

Governing: Lap bolts, slip resistance in short slots (AISC 360-16 J3, J3.9, J3.2): 3.29 kip ≤ 8.36 kip, D/C = 0.39, OK.

Worked: Column bolts, tension with prying (AISC Steel Construction Manual Eq. 9-20; AISC 360-16 J3.7)

$\phi R_n = 0.75 F_{nt} A_b = 0.75 \times 90\,\text{ksi} \times 0.3068\,\text{in}^2 = 20.71\,\text{kip} \quad (\text{A325},\ d_b = 0.625\,\text{in})$

$T_u + q_{ep} + q_c = 3.434\,\text{kip} \ \le\ 20.71\,\text{kip} \quad (F = 6.584\,\text{kip}) \quad D/C = 0.17$

Worked: Cap bolts, tension with prying (AISC Steel Construction Manual Eq. 9-20; AISC 360-16 J3.6, J3)

$\phi R_n = 0.75 F_{nt} A_b = 0.75 \times 90\,\text{ksi} \times 0.3068\,\text{in}^2 = 20.71\,\text{kip} \quad (\text{A325},\ d_b = 0.625\,\text{in})$

$T_u + q = F/2 + q = 6.584\,\text{kip}/2 + 0\,\text{kip} = 3.292\,\text{kip} \ \le\ 20.71\,\text{kip} \quad D/C = 0.16$

Check · reference	Equation	Value	Limit	D/C	Result
Lap bolts, slip resistance in short slots AISC 360-16 J3, J3.9, J3.2	$R_n = \mu D_u h_f T_b n_s k_{sc};\ \phi = 0.85\ (\Omega = 1.76)$	3.29 kip	8.36 kip	0.39	OK
Angle heel, eccentric hanger path (H1-1) AISC 360-16 H1	$\frac{F}{\phi F_y A_g} + \frac{8}{9}\frac{F(t_p + t_a)/2}{\phi F_y Z} \le 1$ where F tie force (pull or push); t_a angle thickness	0.18	1.00	0.18	OK

17 other limit states checked, max D/C 0.32 (Appendix A).

Stiffness and fatigue

The tie stiffness after contact is k = 262 kip/in, from the series model below. The angle is not clamped, so the strength and buckling cases and the Appendix 6 brace stiffness check use its secant stiffness k_{sec} = 34.7 kip/in, the play taken up at the brace force (below, Play at the restraints); the service cases use k with the rigid-body take-up added. The bolt holes in the channel flange at each tie are Category B details (pretensioned bolts), screened in Appendix A.

Compliance (pull, in series)	Share
Plate	5.0%
Lap bolts	3.4%
Angle leg	35.6%
Channel flange	5.7%
Cap bolts	2.6%
End plate	13.9%
Column bolts	2.2%
Shims	0.0%
Column flange	31.7%
Total	**100%**

Play at the restraints

The stiffnesses above act after contact. At the top the angle is not clamped: it slides on its sleeves between the channel flange and the plate washers, so a reversal of the tie force first takes up $g_t = L_{sleeve} - t_a$ = 1/16 in; at the bottom the sliding bearing bears across its slot after g_b = 1/16 in (0.031 in each way when centered); the locating bearing is clamped by its pretensioned bolts. The analysis represents the play as follows. **Strength and buckling:** each restraint with play enters the strength cases, their stresses and the buckling eigenvalue (load-height lateral-torsional buckling and torsional stability), at its secant stiffness, the play taken up at the Appendix 6 brace force P_{br} = 2.5 kip: $k_{sec} = (1/k + g/P_{br})^{-1}$ = 34.7 kip/in for the tie, which the brace stiffness check also uses. **Uneven take-up:** each bay is restrained only at its ends, so the play moves it as a rigid body; ends that take up differently twist the bay uniformly by at most $\Delta\theta = (g_t + g_b)/h_r$ = 0.0086 rad (h_r = 14.5 in from the tie line to the bottom flange), a St Venant torque $GJ\Delta\theta/L$ = 0.13 kip-ft with shear stress 0.23 ksi and no warping stress ($\varphi'' = 0$). **Movement:** the rail-head movement check adds the rigid-body take-up $g_t + \Delta\theta(y_r - y_t)$ = 0.11 in. **Cyclic:** each reversal of side thrust slides the restraint across its play under pretensioned bolts that are not themselves cycled; the tie fatigue range is the full force reversal, and the play is reinspected with the rail ends at each periodic crane inspection.

Fatigue basis: section 4.4.

Detail and location	Cat.	F_{TH}	F_{SR} at n_{SR}	$f_{sr,max}$	$f_{sr,eq}$	$\Sigma n_i/N_i$	Miner $f_{sr,eq}/F_{SR}$	All-cycle $f_{sr,max}/F_{SR}$	Result
① **Tie plate at the end plate fillets, root** Cap tieback	C″ (root, R_{FIL})	none	17.3 ksi	4.02 ksi	—	—	—	**0.23**	**OK**

Screened details (5), closest to F_{TH}: Tie plate, net section at the slip-critical lap 13 ksi vs 16 ksi; Cap bolts, axial tension 5.19 ksi vs 7 ksi; Angle heel, channel and column flange roots, local bending 12.5 ksi vs 24 ksi; Tie plate at the end plate fillets, toe 4.02 ksi vs 10 ksi.

Movement

Governing: Angle slots, travel each way along the runway (Fisher & Van de Pas (EJ 2002); AISC DG7 §11.2): 0.30 in ≤ 0.38 in, D/C = 0.80, OK.

$$\theta h_b + \alpha L \Delta T + t_{set} = 0.00536 \times 12.37\,\text{in} + 0.108\,\text{in} + 0.125\,\text{in} = 0.0662\,\text{in} + 0.108\,\text{in} + 0.125\,\text{in} = 0.2992\,\text{in} \;\le\; u = 0.375\,\text{in} \quad D/C = 0.80$$

Worked: Steel sleeves, bolt installation load (AISC 360-16)

$$A_g = \frac{\pi}{4}(d_o^2 - d_h^2) = \frac{\pi}{4}\left[(1.188\,\text{in})^2 - (0.6875\,\text{in})^2\right] = 0.7363\,\text{in}^2 \quad (\text{sleeve})$$

$$1.5T_b = 1.5 \times 19\,\text{kip} = 28.5\,\text{kip} \;\le\; \phi F_y A_g = 1.00 \times 50\,\text{ksi} \times 0.7363\,\text{in}^2 = 36.82\,\text{kip} \quad D/C = 0.77 \quad (T_b\ \text{Table J3.1})$$

Check · reference	Equation	Value	Limit	D/C	Result
Angle slots, travel each way along the runway Fisher & Van de Pas (EJ 2002); AISC DG7 §11.2	$\theta h_b + \alpha L \Delta T + t_{set} \le u$ where L distance from the locating support; ΔT temperature change; h_b height of the cap bolts above the bearing; u slot travel each way; θ end rotation	0.30 in	0.38 in	0.80	OK
Cap tie sleeves and plate washers Controlling: Steel sleeves, bolt installation load AISC 360-16	$1.5T_b \le \phi F_y A_g;\ \phi = 1.0\ (\text{sleeve})$	28.5 kip	36.8 kip	0.77	OK
Tie plate, axial force with lap moment and support movement (H1-1) AISC 360-16 H1	$\frac{P_r}{P_c} + \frac{8}{9}\frac{B_1 Fe + 6EI\Delta_s/L^2}{\phi F_y Z_y} \le 1$ where F tie force (pull or push); e lever arm to the toe of the horizontal leg; L tie plate clear span; Δ_s support movement	0.36	1.00	0.36	OK

End rotation moves the girder top along the runway by $\theta \cdot h_b$ (h_b = 12.4 in); at sliding ends thermal travel adds to it. The angle slots take both. Vertical girder movement at the tie (0.013 in, factored) bends the tie plate about its weak axis.

6 DETAILING, FABRICATION AND APPLICABILITY

6.7 Items by others and forces delivered

Items this package needs from others, the acceptance criterion and the forces delivered to each.

Item	What is needed	Acceptance criterion	Force and location delivered	Responsible party
Existing frame, connections and foundations beyond the checked column	Check the building frame, roof and wall framing, the column-to-foundation connections, anchors and foundations for the crane forces combined with the building loads; the existing column member is checked in section 4.7 (braced frame assumed).	AISC 360-16 and ACI 318 strength under the ASCE 7-16 §2.3 combinations, crane loads as L (§4.9); IEBC alteration provisions.	Factored per-grid concurrent sets, section 3.5 and S-00: V at e = 33.0 in from the column centerline with Mx = V·e, and the weak-axis seat moment My at the seat El. 118'-0"; lateral at the tie line El. 119'-3 13/16", up to the tie design force 6.58 kip; longitudinal at the seat on the girder line. Unfactored reactions by load type: section 3.3.	Building engineer of record
Existing column, frame type and second-order effects	Confirm that the column line is braced (non-sway) across the runway, or give the frame's B2 from its analysis. The existing column check (section 4.7) is conditional on a braced frame: pinned base, braced top, B1 only.	AISC 360-16 Chapter C and Appendix 8 (B2 for a sway frame); repeat the H1 check with B2.	Column check combinations and moments, section 4.7.	Building engineer of record
Existing column, drift under the 10-year wind (TR-13 §5.3)	Frame displacement at the top of the crane girder under the 10-year wind, from the frame analysis.	AIST TR-13 §5.3: H/400 or 2 in, whichever is less; H = 232 in (column base to the top of the girder), limit 0.58 in.	—	Building engineer of record
Existing column, rail gauge change under gravity (TR-13 §5.3)	Outward displacement of each column at the rail under D and under the roof snow (snow reduced 50% if ≤ 30 psf, else 25%), from the frame analysis.	AIST TR-13 §5.3: gauge change +1 in / −1/2 in.	Column alone, runway dead load plus static crane: +0.073 in (7% of the limit). Frame spread: by others.	Building engineer of record
Runway seismic, seismic design category, TR-13 combinations 5a and 7a	SDS, the seismic design category, Ie and the existing lateral system's R with their source, or a record that seismic design of the runway is not required at the site (ASCE 7 §1.4.2 for SDC A). Confirm Cs ≤ 0.25 across and ≤ 0.29 along the runway, or have the runway's seismic forces checked.	ASCE 7-16 Chapter 12; AIST TR-13 §3.10 combinations 5a and 7a on the girder, bearings, ties and bracket. The members and connections checked here carry the runway's own seismic forces up to Cs = 0.25 across (the side thrust at each grid, factored 1.6 against 1.0E, over its dead and empty-crane reaction) and 0.29 along (the traction at a locating bearing over the longest bay and the empty crane): the crane forces act at the rail head, above the masses, so they also bound the tie and torsion effects.	Seismic weight added at the runway level 59.7 kip: 47.2 kip of new steel for the two runways (girders, rails, clips and joint bars, bearings, end stops, cap tiebacks and support steel, the takeoff in section 2.3), the crane 12.5 kip.	Building engineer of record (site seismic data)
Existing lateral system, alteration evaluation (IEBC)	Evaluate the existing lateral force-resisting system for the added runway mass and crane forces.	IEBC §503.4 / §805.3: demand-capacity ratio of existing lateral elements not more than 10% above that without the alteration (verify against the adopted edition).	Seismic weight as above.	Building engineer of record
Crane-level strut by others on the girder line (girder web, 27 in from the column centerline) at grids 1, 2, 3, 4, 5, k ≥ 22.7 kip/in, F = 10 kip (LRFD)	The existing-column check relies on a crane-level strut on the girder line (girder web, 27 in from the column centerline), connected at the seat level of every locating column (grids 1, 2, 3, 4, 5) and carried to the braced bay; the existing column check relies on it.	AISC 360-16 strength for the line force; on the girder line, an axial stiffness at each seat at least 10 times the column's own (the column check keeps 1/11 of the force).	Longitudinal at the seat on the girder line: crane stop 10.0 kip at grids 1 and 5, traction 3.20 kip at grids 1–5 (factored). Factored LRFD forces at each locating seat: traction 3.2 kip; crane stop 10 kip at grids 1, 5; strut design force (the line force to the braced bay) F = 10 kip. Axial stiffness at each seat at least 10 times the column's own stiffness there: k_s ≥ 22.7 kip/in, so the column keeps at most 1/11 of the force.	Building engineer of record
Crane-level strut on the girder line, collector, brace connections, braced-bay columns and foundation	Design the crane-level strut at each locating column on the girder line (the existing column check relies on it: the longitudinal force leaves the column at the seat with no column torsion or weak-axis bending), the collector to the braced bay, the brace end connections (gussets, clevises, bolts, welds), the braced-bay columns, anchors and foundation.	AISC 360-16 and ACI 318 strength; the strut stiff enough to hold the girder line at the seat.	Longitudinal at the seat on the girder line: crane stop 10.0 kip at grids 1 and 5, traction 3.20 kip at grids 1–5 (factored).	Building engineer of record
Crane data, Demo crane CR-5, 5 US tons	Signed confirmation of the data used: wheel loads 7.50 kip maximum / 3.50 kip empty per wheel, 3.70 kip minimum on the far runway, 2 wheels at 120 in; side thrust share s = 0.5 to this runway; driven-wheel load 7.50 kip; bumper force 10.0 kip and its height above the rail; bridge speed; the 0.9 trolley factor in the bumper energy.	The values used in sections 1.4 and 2.2; any increase requires a re-check.	—	Crane supplier
Rail and crane wheels	Confirm that the Demo R-6 crane rail suits the crane wheels (tread width, flange clearance, wheel load) and the rail joints and rail clips specified.	Crane supplier wheel and rail requirements.	—	Crane supplier

Item	What is needed	Acceptance criterion	Force and location delivered	Responsible party
Opposite runway	Build the opposite runway identical to this one, opposite hand (S-00, S-01): girders, bolted end bearings, haunches, column stiffeners, cap tiebacks and end stops; survey its column line as this one.	This calculation applies where the opposite columns, seat elevation and frame match those used here; otherwise a separate check.	As this runway (section 3.5).	Engineer of record for the runway; contractor survey
Existing column survey	Field-verify every existing column before fabrication: W14X90 d = 14.0 in, bf = 14.5 in, tf = 0.71 in, tw = 0.44 in, plumb, and nothing attached inside the flanges at the haunch and column stiffener levels; base metal per the mill certificates.	Surveyed dimensions within the ASTM A6 tolerances of the values used here; any difference to the engineer of record before fabrication.	—	Contractor (survey), reviewed by the engineer of record
Load test at 125% of the rated load (S-00)	Load test the crane and runway per ASME B30.2 and the crane supplier before service.	1.2 × empty-crane wheel load + 1.6 × 1.25 × (loaded − empty) = 1.2 × 3.50 kip + 1.6 × 1.25 × 4.00 kip = 12.2 kip per wheel ≤ the design wheel load 15.0 kip with impact: the runway checks cover the test.	—	Owner and crane supplier (perform); engineer of record (witness)

SCOPE OF WORK

FURNISH AND INSTALL ONE CRANE RUNWAY, TWO IDENTICAL GIRDER LINES (GRID B OPPOSITE HAND TO GRID A), EACH 125'-0" LONG IN 5 SIMPLE SPANS, FOR 5 TON (10 KIP) MAINTENANCE CRANE: W14X61 + C15X50 CAPPED RUNWAY GIRDERS, RAIL, RAIL ATTACHMENTS, END STOPS AND CONNECTIONS, SUPPORTED ON NEW BRACKETS WELDED TO THE EXISTING BUILDING COLUMNS. THE EXISTING COLUMN RECEIVING THE RUNWAY IS CHECKED FOR THE ADDED CRANE REACTIONS.

GENERAL NOTES

1. VERIFY ALL DIMENSIONS, ELEVATIONS AND EXISTING CONDITIONS BEFORE FABRICATION. DO NOT SCALE DRAWINGS. REPORT DISCREPANCIES TO THE ENGINEER.
2. PROVIDE BOLTS, HOLES AND FAYING SURFACES AS DETAILED. PRETENSION ALL RUNWAY CONNECTION BOLTS (AIST TR-13 §5.14); SNUG-TIGHT BOLTS ARE NOT PERMITTED ON THE RUNWAY.
3. PROVIDE E70XX WELD METAL. USE QUALIFIED WELDING PROCEDURES AND PERSONNEL. PROVIDE CONTINUOUS FILLETS WHERE NOTED. SMOOTH STARTS / STOPS. FIELD WELD ATTACHMENTS TO EXISTING STEEL AND THE RAIL CLIPS AS FLAGGED; VERIFY WELDABILITY AND PREPARE SURFACES.
4. PROVIDE INDEPENDENT BEARINGS AND TIES AT EACH GIRDER END. EACH GIRDER LOCATES AT ITS END ON THE LOWER-NUMBERED GRID AND SLIDES AT THE OTHER END. BOLT BEARINGS WITH STANDARD HOLES AT THE LOCATING END AND SLOTS, PLATE WASHERS AND BOLTS PRETENSIONED AGAINST STEEL SLEEVES AT THE SLIDING END (2/S-02). KEEP GIRDER AND CAP ENDS CLEAR; SEE 1/S-03. PROVIDE TEMPORARY RESTRAINT UNTIL PERMANENT TIES ARE COMPLETE.
5. INSPECT WELDS, BOLT INSTALLATION AND FAYING SURFACES. PERFORM THE INSPECTION AND NDT LISTED UNDER SPECIAL INSPECTIONS ON THIS SHEET AND IN THE CALCULATION REPORT.
6. VERIFY RAIL GAUGE, ALIGNMENT, LEVEL AND OPERATING CLEARANCES WITH THE CRANE SUPPLIER. COMPLETE INSPECTION AND COMMISSIONING BEFORE CRANE OPERATION.
7. DASHED BUILDING FRAMING, ANCHORS AND FOUNDATIONS ARE REFERENCE ONLY. SOLID WELDED BRACKETS: SEE 2/S-03. VERIFY EXISTING SECTIONS AND ATTACHMENTS. OPPOSITE RUNWAY SHOWN FOR CONTEXT.
8. THESE DRAWINGS AND THE CALCULATION REPORT OF THE SAME REVISION FORM ONE PACKAGE. WHERE THEY DIFFER, THE MORE STRINGENT REQUIREMENT GOVERNS UNTIL CLARIFIED BY THE ENGINEER OF RECORD.
9. ALL WORK SHALL CONFORM TO THE BUILDING CODE ADOPTED BY THE AUTHORITY HAVING JURISDICTION AND THE STANDARDS LISTED UNDER DESIGN CRITERIA.

EXISTING BUILDING NOTES

1. EXISTING CONSTRUCTION IS SHOWN DASHED AND IS BASED ON THE SURVEY AND SOURCES LISTED BELOW. FIELD VERIFY MEMBER SIZES, CONDITION, PLUMBNESS AND ELEVATIONS BEFORE FABRICATION AND REPORT DIFFERENCES TO THE ENGINEER OF RECORD.
2. DO NOT CUT, DRILL OR WELD EXISTING STEEL UNTIL ITS MATERIAL IS CONFIRMED (MILL CERTIFICATES OR CHEMICAL ANALYSIS, AISC 360-16 APPENDIX 5.2.3).
3. FIELD WELDS TO THE EXISTING COLUMNS: EXISTING STEEL ASTM A572 GR. 50 (MILL CERTIFICATES). WELD ONLY TO A WPS APPROVED BY THE ENGINEER OF RECORD FOR THAT STEEL; LOW-HYDROGEN PROCESSES ONLY. PREHEAT AND INTERPASS TEMPERATURE AT LEAST 150 °F (PROJECT REQUIREMENT); AWS D1.1/D1.1M:2020 TABLE 5.8, CATEGORY B, MINIMUM 50 °F FOR THE THICKEST PART AT THE WELD, 1 IN (HAUNCH TOP FLANGE; OVER 3/4 IN THROUGH 1-1/2 IN). WELD WITH THE CRANE OUT OF SERVICE AND THE COLUMN UNDER DEAD LOAD ONLY; SEQUENCE THE WELDS TO LIMIT HEAT INPUT.
4. SHORE OR UNLOAD EXISTING MEMBERS AS REQUIRED BY THE ERECTION PROCEDURE. DO NOT REMOVE EXISTING BRACING WITHOUT A TEMPORARY REPLACEMENT APPROVED BY THE ENGINEER.
5. EXISTING COLUMN: FICTITIOUS DEMONSTRATION: ORIGINAL BUILDING CALCULATION SHEETS 14-17, COLUMN LINES A AND B (THE SAME FRAMES).
6. ITEMS MARKED BY OTHERS IN THE CALCULATION REPORT (FRAME, CONNECTIONS TO EXISTING MEMBERS, ANCHORS AND FOUNDATIONS NOT CHECKED HERE) SHALL BE VERIFIED BY THE ENGINEER OF RECORD FOR THE REPORTED FORCES.

CRANE RUNWAY INSTALLATION

1. RAIL: DEMO R-6 CRANE RAIL; GAUGE (CRANE SPAN) 40'-0 1/2" ± 1/4", THE TOTAL OF THE SETTING TOLERANCES OF THE TWO RAILS. SET EACH RAIL C/L ON ITS GIRDER WEB C/L WITHIN 1/8" AND THE RAIL TOP WITHIN 1/8" OF THE THEORETICAL ELEVATION. THE DESIGN ALLOWS A RAIL-TO-WEB ECCENTRICITY OF 1/4" FOR SETTING AND WEAR; IT IS NOT A SETTING DIMENSION; IN SERVICE 1/4" MAX. (TR-13 §5.18). VERIFY THE GAUGE AND TOLERANCES WITH THE CRANE MANUFACTURER BEFORE SETTING RAILS.
2. RUNWAY FABRICATION AND ERECTION TOLERANCES (AIST TR-13 §5.18), UNLESS THE CRANE MANUFACTURER REQUIRES LESS: COLUMN BASE LINES ±1/8 IN; SEATS SURVEYED AND SHIMMED, HOLES DRILLED TO THE SET GIRDER (IN PLACE OF THE ±1/8 IN SEAT POSITION AND ELEVATION); GIRDER SWEEP 1/4 IN PER 50 FT; CAMBER +1/4 IN / −0 IN PER 50 FT; BEARING-ZONE FLATNESS ±1/32 IN; GIRDER END DEPTH ±1/16 IN; ADJACENT TOPS OF GIRDER FLANGES WITHIN 1/16 IN; RAIL GAUGE ±1/4 IN AT 68 °F; RAIL LINE AND GRADE 1/4 IN PER 50 FT, 3/8 IN TOTAL. SURVEY BOTH RAILS UNLOADED AFTER ERECTION, BEFORE THE LOAD TEST AND DURING COMMISSIONING: GAUGE, LINE, ELEVATION AND STRAIGHTNESS AT EVERY SUPPORT AND AT MIDSPAN; SUBMIT THE SURVEY.
3. BEARING SEATS (FIELD WELDED): SURVEY THE SEATS; SHIM TO THE THEORETICAL ELEVATION (STEEL SHIMS, FULL BEARING, 3 PLIES AND 1/2 IN MAX., DRILLED TO THE BOLT PATTERN); FIELD DRILL OR REAM THE SEAT HOLES AFTER THE GIRDER IS LINED AND LEVELED.
4. RUNWAY CLEARANCE TEST BEFORE THE LOAD TEST (AIST TR-13 §1.5.6.2): CRANE UNLOADED, HOOK FULLY UP, RUN THE CRANE THE FULL LENGTH OF THE RUNWAY THREE TIMES, WITH THE TROLLEY AT EACH END OF THE BRIDGE AND AT CENTER; CHECK THE FIG. 1.1 CLEARANCES, INCLUDING THE CLEARANCE TO THE ROOF TRUSSES DEFLECTED UNDER SNOW, AND TO THE EXISTING ROOF FRAMING, BRACING, LIGHTS, PIPING AND OTHER OBSTRUCTIONS; REPORT ANY SHORTFALL TO THE ENGINEER OF RECORD BEFORE COMMISSIONING.
5. INSTALL THE RUNWAY END STOPS (3/S-04) AT BOTH ENDS OF BOTH RUNWAYS BEFORE THE CRANE IS OPERATED OR LOAD TESTED.
6. LOAD TEST THE CRANE PER ASME B30.2 AND THE MANUFACTURER. THE TEST LOAD SHALL NOT EXCEED 125% OF THE RATED LOAD; COORDINATE ANY TEST LOAD ABOVE THE RATED LOAD WITH THE ENGINEER OF RECORD BEFORE TESTING.

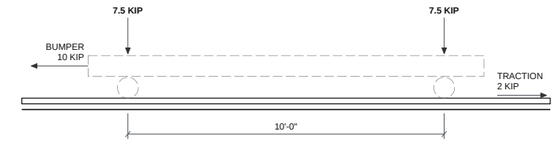
DESIGN CRITERIA · CODES AND STANDARDS

STANDARD	EDITION / SCOPE
BUILDING CODE	2021 IBC AS ADOPTED BY THE FICTITIOUS DEMONSTRATION JURISDICTION
EXISTING BUILDING CODE	IEBC OF THE SAME EDITION AS THE BUILDING CODE: ALTERATION; EXISTING STRUCTURE EVALUATED FOR THE ADDED CRANE LOADS, SEE EXISTING STRUCTURE EVALUATION
ASCE/SEI 7	2016 (§4.9 UNCHANGED IN 7-22): CRANE LOADS, COMBINATIONS
AISC 360	2016: STEEL DESIGN, LRFD
AIST TECHNICAL REPORT 13 (2021)	GUIDE FOR THE DESIGN AND CONSTRUCTION OF MILL BUILDINGS: RUNWAY LOADS AND CRITERIA
AISC DESIGN GUIDE 7	3RD ED. (2019) WITH 2023 ERRATA
AWS D1.1/D1.1M	2020: STRUCTURAL WELDING CODE - STEEL (CYCLICALLY LOADED); CLAUSE AND TABLE NUMBERS CITED ARE OF THIS EDITION (AISC 360-16 REFERENCES THE 2015 EDITION)
RCSC	2014 SPECIFICATION FOR STRUCTURAL JOINTS, AS REFERENCED BY AISC 360-16
ASME B30.2	2022: OVERHEAD AND GANTRY CRANES; INSPECTION, TESTING AND OPERATION

CRANE DATA · DEMO CRANE CR-5 · 5 US TONS

ITEM	VALUE
RATED CAPACITY	5 TON (10 KIP)
AIST CRANE TYPE / CONTROL	MAINTENANCE / PENDANT
SERVICE	FICTITIOUS CMAA C CRANE; AIST CLASS B BUILDING (400,000 CYCLES COUNTED)
WHEELS ON RUNWAY	2 AT 10'-0"
MAX STATIC WHEEL LOAD	7.5 KIP
VERTICAL IMPACT (ADOPTED)	25% (AIST / ASCE 7 MIN. 10%)
SIDE THRUST, THIS RUNWAY	2 KIP
LONGITUDINAL TRACTION	2 KIP
BUMPER (STOP) FORCE	10 KIP
LOAD SOURCE	FICTITIOUS SUPPLIER SCHEDULE DEMO-CS-05: 7.5 KIP MAX / 3.5 KIP EMPTY PER WHEEL ON THE NEAR RUNWAY, 3.7 KIP MINIMUM ON THE FAR RUNWAY; BRIDGE SPEED 100 FT/MIN; BUMPER STROKE 4 IN, EFFICIENCY 0.5; END-TRUCK BUMPER FACE 15 IN BEYOND THE END-WHEEL CENTERLINE.

WHEEL LOADS · DEMO CRANE CR-5 · 5 US TONS



WHEEL LOADS: MAXIMUM WITHOUT IMPACT, PER THE CRANE SUPPLIER; ADD 25% VERTICAL IMPACT. SIDE THRUST 2 KIP ON THIS RUNWAY AND TRACTION ACT AT THE RAIL HEAD. CRANE, END TRUCKS AND BUMPERS BY THE CRANE SUPPLIER.

DESIGN CRITERIA · RUNWAY

ITEM	VALUE
RUNWAY GIRDER	W14X61 + C15X50 (CAPPED), FY = 50 KSI
SPANS	25'-0" + 25'-0" + 25'-0" + 25'-0" + 25'-0" SIMPLE SPANS
DESIGN METHOD	LRFD; RUNWAY: AIST TR-13 §3.10; BUILDING CHECKS: ASCE 7 §2.3/§2.4
AIST BUILDING CLASS	CLASS B (TR-13 TABLE 1.1, ALL 400,000 LOAD CYCLES COUNTED). RUNWAY INSPECTION AT THE TR-13 APPENDIX C INTERVAL FOR CLASS B
VERTICAL / LATERAL DEFLECTION	L/1000 / L/400 OF EACH BAY (ONE CRANE, NO IMPACT)
RAIL TWIST / HEAD MOVEMENT	NO ROTATION LIMIT ENTERED (OWNER / CRANE SUPPLIER), ROTATION NOT CHECKED; RAIL-HEAD LATERAL MOVEMENT L/400 OF THE SHORTEST BAY, TWIST INCLUDED (OWNER / CRANE SUPPLIER CRITERIA, ONE STATIC CRANE)
FATIGUE SPECTRUM	400,000 STRESS CYCLES AT EACH GIRDER DETAIL (ONE PER CRANE PASSAGE) IN 3 DUTY BINS (60,000 AT RATED LIFT: 2 STRESS CYCLES PER FULL-LOAD REPETITION); WHEEL-LOCAL DETAILS, ONE CYCLE PER WHEEL GROUP: RAIL CLIP BASE WELDS 800,000, CAP WELDS UNDER THE WHEELS 800,000, RAIL JOINT BARS AND BOLTS 800,000
ELEVATIONS	T.O.R. 119'-9 5/8", T.O.S. 119'-2 7/8" (DATUM 100'-0")

LOAD COMBINATIONS · RUNWAY GIRDER (LRFD)

NO.	COMBINATION	NO.	COMBINATION
1A	1.4D+1.4CDM	9	0.9(D+CDS)+1.6CVS,MIN+1.6CSS
2A	1.2(D+CDM)+1.6L+CVM+CSS+CLS	10 (TRACTION)	0.9(D+CDS)+1.6CLS
2B	1.2(D+CDM)+1.6(CVM+CSS+CLS)+L	10 (BUMPER)	0.9(D+CDS)+CBS
2C	1.2(D+CDS)+1.6(CVS+CI+CLS)+L	A2B	1.2D+1.6(CDM+CVM+CSS+CLS+L)
7A	0.9D+CDM	A2C	1.2D+1.6(CDS+CVS+CI+CLS+L)
8	1.2(D+CDS)+CVS+CBS		

AIST TECHNICAL REPORT 13 WITH ASCE 7 §2.3. CD CRANE DEAD, CV VERTICAL, CSS SIDE THRUST, CLS LONGITUDINAL, CI IMPACT, CBS BUMPER; SUFFIX M MULTIPLE CRANES, S ONE CRANE, MIN MINIMUM LIFTED LOAD. D DEAD, L LIVE. IMPACT AND SIDE THRUST ARE IN SEPARATE COMBINATIONS (2B AND 2C; AISC DG7 3RD ED. WITH ERRATA 2/16/2023), ONE RULE FOR THE GIRDERS, SUPPORTS, AND THE EXISTING COLUMNS. ALSO EVALUATED: 1, 2, 6 AND 7, 3A, 4A (EQUAL TO OR BOUNDED BY THE CASES LISTED). 7A IS EVALUATED HERE WITHOUT THE SEISMIC LOAD E; 5A AND 7A WITH E ARE BY THE ENGINEER OF RECORD (EXISTING STRUCTURE EVALUATION, ITEM 6).

SUPPORT REACTIONS · UNFACTORED, PER SUPPORT

GRID (STATION)	D	CD	CV	CI	CSS	CLS·HR/L
1 (0'-0")	2.64 KIP	5.06 KIP	5.79 KIP	2.71 KIP	1.45 KIP	±0.14 KIP
2 (25'-0")	4.58 KIP	5.60 KIP	6.40 KIP	3.00 KIP	1.60 KIP	±0.14 KIP
3 (50'-0")	4.58 KIP	5.60 KIP	6.40 KIP	3.00 KIP	1.60 KIP	±0.14 KIP
4 (75'-0")	4.58 KIP	5.60 KIP	6.40 KIP	3.00 KIP	1.60 KIP	±0.14 KIP
5 (100'-0")	4.58 KIP	5.60 KIP	6.40 KIP	3.00 KIP	1.60 KIP	±0.14 KIP
6 (125'-0")	2.64 KIP	5.06 KIP	5.79 KIP	2.71 KIP	1.45 KIP	±0.14 KIP

CD CRANE EMPTY, CV LIFTED, CI IMPACT, CSS SIDE THRUST AT RAIL HEAD (ONE CRANE). RUNWAY LONGITUDINAL FORCE CLS = 2 KIP AT THE RAIL HEAD; CLS·HR/L IS ITS END COUPLE AT THE BEARINGS OF THE BAY THAT CARRIES IT, HR THE HEIGHT OF THE RAIL HEAD ABOVE THE BEARING AND L THE SPAN. ALL CRANE COMPONENTS ARE LIVE LOAD L (ASCE 7 §4.9). FACTORED: BRACKET DESIGN FORCES ON S-03.

FORCES DELIVERED TO THE BUILDING COLUMNS · FACTORED LRFD, CONCURRENT SETS PER GRID

GRID	SET	COMBINATION	V	MX = V·E	MY,SEAT	H TIE (DESIGN)	H SEAT	F
1	MAX V	LRFD A2C	24.4 KIP	67.2 KIP-FT	+7.33 KIP-FT	±2.93 KIP	∓0.43 KIP	−3.20 KIP
	MAX H	LRFD A2B	20.1 KIP	55.3 KIP-FT	+6.03 KIP-FT	±6.28 KIP	∓1.47 KIP	−3.20 KIP
	STOP	LRFD 8	15.3 KIP	42.0 KIP-FT	+4.58 KIP-FT	±2.72 KIP	∓0.23 KIP	−10.0 KIP
2	MAX V AND MAX F	LRFD A2C	29.0 KIP	79.9 KIP-FT	±20.4 KIP-FT	±2.96 KIP	∓0.47 KIP	−3.20 KIP
	MAX H	LRFD A2B	24.2 KIP	66.7 KIP-FT	+16.3 KIP-FT	±6.58 KIP	∓1.53 KIP	−3.20 KIP
	MAX F	LRFD A2C	28.6 KIP	78.7 KIP-FT	−20.4 KIP-FT	±2.95 KIP	∓0.45 KIP	+3.20 KIP
3	MAX V AND MAX F	LRFD A2C	29.0 KIP	79.9 KIP-FT	±20.4 KIP-FT	±2.96 KIP	∓0.47 KIP	−3.20 KIP
	MAX H	LRFD A2B	24.2 KIP	66.7 KIP-FT	+16.3 KIP-FT	±6.58 KIP	∓1.53 KIP	−3.20 KIP
	MAX F	LRFD A2C	28.6 KIP	78.7 KIP-FT	−20.4 KIP-FT	±2.95 KIP	∓0.45 KIP	+3.20 KIP
4	MAX V AND MAX F	LRFD A2C	29.0 KIP	79.9 KIP-FT	±20.4 KIP-FT	±2.96 KIP	∓0.47 KIP	−3.20 KIP
	MAX H	LRFD A2B	24.2 KIP	66.7 KIP-FT	+16.3 KIP-FT	±6.58 KIP	∓1.53 KIP	−3.20 KIP
	MAX F	LRFD A2C	28.6 KIP	78.7 KIP-FT	−20.4 KIP-FT	±2.95 KIP	∓0.45 KIP	+3.20 KIP
5	MAX V	LRFD A2C	29.0 KIP	79.9 KIP-FT	±20.4 KIP-FT	±2.96 KIP	∓0.47 KIP	−3.20 KIP
	MAX H	LRFD A2B	24.2 KIP	66.7 KIP-FT	+16.3 KIP-FT	±6.58 KIP	∓1.53 KIP	−3.20 KIP
	STOP	LRFD 8	8.47 KIP	23.3 KIP-FT	+3.07 KIP-FT	±2.61 KIP	∓0.12 KIP	+10.0 KIP
6	MAX V	LRFD A2C	24.4 KIP	67.2 KIP-FT	−7.33 KIP-FT	±2.92 KIP	∓0.42 KIP	—
	MAX H	LRFD A2B	20.1 KIP	55.3 KIP-FT	−6.03 KIP-FT	±6.25 KIP	∓1.44 KIP	—
STRUT	CRANE-LEVEL STRUT BY OTHERS, GIRDER LINE	TRACTION / STOP	—	—	—	—	—	±10.0 KIP

FACTORED LRFD. EACH ROW IS ONE CONCURRENT SET; DO NOT COMBINE ROWS. V: GIRDER REACTIONS ON THE SEAT, EL. 118'-0", AT E = 33.0 IN FROM THE COLUMN CENTERLINE (THE FAR EDGE OF THE BEARING PLATE, A CONSERVATIVE REACTION POSITION; THE GIRDER WEB IS AT 27.0 IN). MX = V·E (COLUMN STRONG AXIS). MY,SEAT: MOMENT OF THE UNBALANCED END REACTIONS ABOUT THE GRID (COLUMN WEAK AXIS). H TIE: TOP-FLANGE RESTRAINT FORCE PLUS THE APPENDIX 6 BRACE FORCE (2.50 KIP), AT THE TIE LINE, EL. 119'-3 13/16"; DESIGN VALUE 6.58 KIP. H SEAT: BOTTOM-FLANGE RESTRAINT FORCE AT THE BEARING BOLTS, ∓ OPPOSITE IN SIGN TO H TIE. F: LONGITUDINAL FORCE AT THE LOCATING BEARING, + TOWARD THE NEXT GRID. STOP FORCE 10.0 KIP AT GRIDS 1 AND 5 ONLY. TRACTION REACHES THE BAY UNDER THE CRANE, OR THROUGH THE FLOATING RAIL STRING AN END BAY AT ITS STOP. STRUT: THE CRANE-LEVEL STRUT BY OTHERS ON THE GIRDER LINE, 27.0 IN FROM THE COLUMN CENTERLINE, AT THE SEAT LEVEL OF GRIDS 1, 2, 3, 4, 5, AXIAL STIFFNESS K ≥ 22.7 KIP/IN AT EACH SEAT; DESIGN FORCE F = 10.0 KIP (TRACTION 3.20 KIP, STOP 10.0 KIP), CARRIED TO THE BRACED BAY; THE EXISTING COLUMN CHECK RELIES ON IT. V INCLUDES THE BEARING HARDWARE ON THE SEAT; THE HAUNCH AND OTHER SUPPORT STEEL (0.56 KIP PER GRID, DEAD) GOES STRAIGHT INTO THE COLUMN.

SHEET INDEX

SHEET	TITLE
S-00	COVER, GENERAL NOTES & DESIGN CRITERIA
S-00A	GENERAL NOTES & DESIGN CRITERIA (CONT.)
S-01	GENERAL ARRANGEMENT
S-02	CONNECTIONS & CAP CHANNEL
S-03	SUPPORTS & BRACKETS
S-04	CAP TIEBACK ANGLES & END STOPS

GENERAL NOTES CONTINUE ON SHEET S-00A.

ENGINEER STAMP / SEAL

NOT SEALED

DEMONSTRATION - NOT FOR CONSTRUCTION

REFERENCE FRAMING SHOWN DASHED

REV	REVISION DESCRIPTION	BY	DATE
0	ISSUED FOR REVIEW	MSB	2026-10-13

CRANECALC

STRUCTURAL DESIGN WORKSHEET

DIMENSIONS GOVERN / DO NOT SCALE

FICTITIOUS DEMONSTRATION

DRAWING RESPONSIBILITY

ORIGINATOR: MSB

CHECKER:

ENGINEER OF RECORD:

COVER, GENERAL NOTES & DESIGN CRITERIA

CEDAR WORKS · NEW 5-TON RUNWAY IN THE EXISTING 5-BAY BUILDING

PROJECT: DEMO-FOCUS-005

SITE: 1450 FOUNDRY ROAD, RIVERTON, OHIO (FICTITIOUS) · LENGTHS: FEET & INCHES

DRAWING / SHEET NUMBER

SHEET S-00 1 OF 6

PAGE SIZE: ARCH D / 36 X 24 IN

REVISION: 0 (SAME AS THE CALCULATION)

DATE: 2026-10-13 SCALE: NONE

DETAIL INDEX

DETAIL	TITLE
1/S-01	ISOMETRIC VIEW
2/S-01	RUNWAY PLAN
3/S-01	RUNWAY GIRDER ELEVATION - GRID A
4/S-01	TYPICAL RUNWAY SECTION
1/S-02	GIRDER BEARING / COLUMN BRACKET
2/S-02	GIRDER END BEARINGS / LOCATING AND SLIDING
3/S-02	SLIDING BEARING BOLT / SECTION
4/S-02	RAIL CLIP / SECTION
5/S-02	RAIL JOINT / ELEVATION
6/S-02	RAIL LAYOUT / PLAN
7/S-02	CAPPED GIRDER SECTION
8/S-02	CAP END DEVELOPMENT
1/S-03	ADJACENT GIRDER ENDS AT COLUMN
2/S-03	COLUMN BRACKET / TRANSVERSE SECTION
3/S-03	TAPERED HAUNCH / TOP FLANGE PLAN
4/S-03	TAPERED HAUNCH / SECTION AT STIFFENERS
5/S-03	COLUMN STIFFENERS / PLAN AT BRACKET
1/S-04	CAP TIEBACK ANGLE / SECTION AT SUPPORT
2/S-04	CAP TIEBACK ANGLES / PLAN AT SHARED COLUMN
3/S-04	END STOP / ELEVATION
4/S-04	END STOP / PLAN
5/S-04	END STOP / SECTION AT FACE

MATERIALS

ITEM	SPECIFICATION
RUNWAY GIRDER	ASTM A992, FY = 50 KSI
CAP CHANNEL	ASTM A572 GR. 50 CHANNEL, FY = 50 KSI
PLATES, BARS, ANGLES, TIES	ASTM A572 GR. 50, FY = 50 KSI, FU = 65 KSI
CRANE RAIL	ASTM A759 DEMO R-6 CRANE RAIL, FY = 60 KSI
BOLTS	ASTM F3125 GR. A325 (GROUP A), PRETENSIONED; SLIP-CRITICAL CLASS B FAYING SURFACES AT CLAMPED PLIES; CLASS A AT THE BEARING PLATES, SHIMS AND SEATS; BOLTS IN SLOTTED SLIDING PLIES PRETENSIONED AGAINST STEEL SLEEVES, THE SLOTTED PLY NOT CLAMPED (MATERIAL NOTE 1)
NUTS AND WASHERS	ASTM A563 GR. DH HEAVY HEX NUTS; ASTM F436 HARDENED WASHERS; PLATE WASHERS ASTM A572 GR. 50
BOLT SLEEVES	MACHINED FROM ASTM A108 COLD-FINISHED CARBON STEEL BAR (GRADE 1018 OR 1045) TO THE OD, BORE AND LENGTH DETAILED; FY = 50 KSI MIN. BY MILL TEST REPORT
WELD METAL	E70XX, AWS D1.1
RAIL CLIPS	FICTITIOUS SUPPLIER DATA RC-W6: WELD-ON FLOATING CRANE RAIL CLIP FOR A 6 IN RAIL, OR EQUAL, PER MANUFACTURER; SEE 4/S-02
COLUMN BRACKETS	BUILT-UP TAPERED I HAUNCH PLATES, ASTM A572 GR. 50, FY = 50 KSI, FU = 65 KSI; FIELD WELDED TO EXISTING COLUMNS: VERIFY WELDABILITY
COLUMN STIFFENERS	ASTM A572 GR. 50 PLATES, FY = 50 KSI; FIELD WELDED INSIDE THE EXISTING COLUMNS AT THE BRACKETS

MATERIAL NOTES

1. BOLTING: AIST TR-13 §5.14: RUNWAY BOLTED CONNECTIONS ARE PRETENSIONED HIGH-STRENGTH BOLTS, SLIP-CRITICAL WHERE THE LOADING IS CYCLIC (RCSC SPECIFICATION 2014 §4.2–§4.3, AS REFERENCED BY AISC 360-16 J3.1). BOLTS IN SLOTTED SLIDING PLIES ARE PRETENSIONED AGAINST STEEL SLEEVES SO THE SLOTTED PLY IS NOT CLAMPED; PLATE WASHERS COVER THE SLOTS. BOLTED BEARINGS: CLASS A FAYING SURFACES (SLIP COEFFICIENT 0.30) AT THE BEARING PLATE, SHIMS AND SEAT; CLEAN PAINT, LOOSE SCALE AND OIL OFF THE SEAT AT THE BEARING. RAIL JOINTS: TIGHT JOINTS, BOLTS PRETENSIONED AND RE-TIGHTENED PER 5/S-02; RAIL ANCHORAGE PER 6/S-02.
2. WELDING: E70XX LOW-HYDROGEN WELD METAL, AWS D1.1/D1.1M:2020, CYCLICALLY LOADED PROVISIONS (CLAUSE 4 PART C); QUALIFIED OR PREQUALIFIED WPS AND QUALIFIED WELDERS. CONTINUOUS FILLETS AS DIMENSIONED; NO INTERMITTENT WELDS ON CYCLIC LOAD PATHS. SMOOTH STARTS/STOPS AND REMOVE TEMPORARY WELDED ATTACHMENTS BY GRINDING. CAP: CONTINUOUS FILLET EACH W TOP-FLANGE EDGE, SIZE AND END DEVELOPMENT PER THE CAP SECTION AND DEVELOPMENT DETAILS. FULL CAP BEARING CONTACT REQUIRED. BEARING STIFFENERS CJP TO W TOP FLANGE, BOTTOM FITTED.

SPECIAL INSPECTIONS (IBC 1704, 1705 / AISC 360 CHAPTER N)

ITEM	REQUIREMENT	FREQ.
STATEMENT	STATEMENT OF SPECIAL INSPECTIONS BY THE ENGINEER OF RECORD (IBC 1704.3); APPROVED AGENCY RETAINED BY THE OWNER (IBC 1704.2)	-
FABRICATOR	AISC-CERTIFIED FABRICATOR, OR SHOP INSPECTION BY THE APPROVED AGENCY (IBC 1704.2.5)	-
MATERIAL	MILL CERTIFICATES: STEEL, BOLTS, NUTS, WASHERS, WELD FILLER	P
WELDING	AISC N5.4 TASKS BEFORE, DURING AND AFTER WELDING; QUALIFIED WPS AND WELDERS	O / P
NDT	UT 100% OF CJP GROOVE WELDS (GIRDER BEARING STIFFENERS TO THE W TOP FLANGE; HAUNCH FLANGES TO THE EXISTING COLUMN FLANGE (FIELD)); MT AT FILLET TERMINATIONS ON CYCLIC DETAILS; AWS D1.1:2020 CLAUSE 8 ACCEPTANCE CRITERIA FOR CYCLICALLY LOADED CONNECTIONS FOR THE RUNWAY GIRDER WELDS AND THE WELDS TO THE EXISTING COLUMN (AIST TR-13 §5.17)	P
BOLTING	AISC N5.6 TASKS; PRETENSION VERIFICATION PER RCSC; FAYING SURFACE CLASS; SLEEVE LENGTHS AND SLOT TRAVEL AT SLEEVED BOLTS	O / P
EXISTING STEEL	FIELD-VERIFY SIZES, CONDITION AND WELDABILITY BEFORE WORK; UT THE COLUMN FLANGE FOR LAMINATIONS UNDER THE HAUNCH FLANGE CJP WELDS BEFORE WELDING (LOADED THROUGH ITS THICKNESS)	P
FIELD WELDS TO EXISTING STEEL	BASE METAL CONFIRMED, WPS APPROVED, PREHEAT 150 °F MIN.; VISUAL AND MT OF EVERY FIELD WELD TO EXISTING STEEL, INCLUDING THE COLUMN STIFFENER WELDS	P
COLUMN STIFFENERS	FIT-UP AT THE BRACKET FORCE LEVELS WITH THE CORNER CLIPS CLEAR OF THE COLUMN FILLETS (SURVEY THE COLUMN AT BOTH LEVELS FIRST); VISUAL OF EVERY STIFFENER WELD, MT OF THE FLANGE FILLETS	P
FIELD WELDS	RAIL CLIP BASE WELDS: VISUAL OF EVERY WELD; MT AT THE RAIL CLIP WELD TERMINATIONS (CYCLIC ATTACHMENTS ON THE CAP CHANNEL, NDT ABOVE)	P
RAIL / RUNWAY	ALIGNMENT, GAUGE AND LEVEL SURVEY AFTER ERECTION	P
PROJECT REQUIREMENTS	VISUAL INSPECTION OF ALL WELDS AND BOLT INSTALLATIONS; VERIFY PRETENSION AND FAYING-SURFACE PREPARATION. MT AT ATTACHMENT TERMINATIONS AND REPAIRED WELDS (AIST TR-13 §5.17). RECORD BASELINE RAIL AND CONNECTION CONDITION BEFORE COMMISSIONING.	-

STEEL: O OBSERVE, P PERFORM (AISC 360 N5).

SUBMITTALS

1. SHOP AND ERECTION DRAWINGS FOR ALL NEW STEEL, SHOWING HOLES, SLOTS, WELDS AND FIELD WORK.
2. CRANE MANUFACTURER WHEEL LOADS, WHEEL SPACING, BUMPER FORCE AND RAIL REQUIREMENTS, SIGNED BY THE MANUFACTURER.
3. WELDING PROCEDURE SPECIFICATIONS, WELDER QUALIFICATIONS AND BOLT CERTIFICATIONS.
4. RAIL SURVEY AND CRANE LOAD-TEST REPORT.

DEFERRED SUBMITTALS / BY OTHERS

SUBMIT THE FOLLOWING TO THE ENGINEER OF RECORD FOR REVIEW AND TO THE BUILDING OFFICIAL FOR APPROVAL BEFORE INSTALLATION:

1. CRANE, END TRUCKS, CRANE-MOUNTED BUMPERS, CONDUCTOR BARS AND ELECTRIFICATION: CRANE SUPPLIER.

EXISTING STRUCTURE EVALUATION · WITH THE PERMIT SUBMITTAL

THE CALCULATION REPORT CHECKS THE EXISTING RUNWAY COLUMN FOR THE ADDED CRANE LOADS. THE ENGINEER OF RECORD SHALL EVALUATE THE FOLLOWING FOR THE FORCES IN THE CALCULATION REPORT AND INCLUDE THE EVALUATION IN THE PERMIT SUBMITTAL; THEY ARE NOT DEFERRED SUBMITTALS. EVALUATION: FICTITIOUS EXISTING STRUCTURE EVALUATION DEMO-ER-02: COLUMN LINE A AND B FRAMES, BRACING, CONNECTIONS AND FOUNDATIONS FOR THE ADDED CRANE LOADS.

1. EXISTING FRAME, CONNECTIONS & FOUNDATIONS BEYOND THE CHECKED COLUMN.
2. EXISTING COLUMN · FRAME TYPE AND SECOND-ORDER EFFECTS.
3. EXISTING COLUMN · DRIFT UNDER THE 10-YEAR WIND (TR-13 §5.3).
4. EXISTING COLUMN · RAIL GAUGE CHANGE UNDER GRAVITY (TR-13 §5.3).
5. CRANE-LEVEL STRUT BY OTHERS ON THE GIRDER LINE (GIRDER WEB, 27 IN FROM THE COLUMN CENTERLINE) AT GRIDS 1, 2, 3, 4, 5, K ≥ 22.66 KIP/IN, F = 10 KIP (LRFD).
6. RUNWAY SEISMIC · TR-13 COMBINATIONS 5A AND 7A.
7. EXISTING LATERAL SYSTEM · ALTERATION EVALUATION (IEBC).
8. CRANE LONGITUDINAL FORCE · BRACING PATH.

EXISTING STRUCTURE EVALUATION · WITH THE PERMIT SUBMITTAL (CONT.)

SYMBOLS AND LINE TYPES

———	NEW STEEL, PLATES AND WELDS (SOLID)
– – – –	EXISTING OR REFERENCE CONSTRUCTION, BY OTHERS (DASHED)
·····	HIDDEN EDGES
—·—·—	GRID LINES AND CENTERLINES
═══	CRANE RAIL
(A)	GRID DESIGNATION
1 / S-02	DETAIL NUMBER OVER THE SHEET WHERE IT IS DRAWN; REFERENCES READ 1/S-02
1/4	FILLET WELD, ARROW SIDE; SIZE LEFT OF THE SYMBOL
1/4 / 1/4	FILLET WELD, BOTH SIDES
1/4 / 1/4	FIELD WELD (FLAG); SHOP WELD U.N.O.
EL.	ELEVATION DATUM
(1'-2")	REFERENCE DIMENSION (SECTION DEPTH OR A SIZE SET BY OTHER DIMENSIONS); NOT A SETTING-OUT DIMENSION

ABBREVIATIONS

BRG. BEARING · C/L CENTERLINE · CJP COMPLETE JOINT PENETRATION · CLR. CLEAR · COL. COLUMN · COMB. LOAD COMBINATION · CONT. CONTINUOUS · D/C DEMAND / CAPACITY RATIO · EA. EACH · ED. EDITION · EL. ELEVATION · FIG. FIGURE · FL. FLANGE · FREQ. FREQUENCY · GR. GRADE · HORIZ. HORIZONTAL · MAX. MAXIMUM · MIN. MINIMUM · MT MAGNETIC PARTICLE TESTING · NDT NONDESTRUCTIVE TESTING · O.C. ON CENTER · OD OUTSIDE DIAMETER · PL PLATE · QTY QUANTITY · REF. REFERENCE (EXISTING OR BY OTHERS) · SC SLIP-CRITICAL · STD STANDARD HOLE · STIFF. STIFFENER · T.O. TOP OF · T.O.R. TOP OF RAIL · T.O.S. TOP OF STEEL · THRU THROUGH · TYP. TYPICAL · U.N.O. UNLESS NOTED OTHERWISE · UT ULTRASONIC TESTING · WPS WELDING PROCEDURE SPECIFICATION

ISSUE

DEMONSTRATION - NOT FOR CONSTRUCTION: FICTITIOUS DEMONSTRATION DATA. DRAWINGS AND CALCULATION REPORT REVISION 0, 2026-10-13.

DEMONSTRATION - NOT FOR CONSTRUCTION

REFERENCE FRAMING SHOWN DASHED

ENGINEER STAMP / SEAL

NOT SEALED

REV	REVISION DESCRIPTION	BY	DATE
0	ISSUED FOR REVIEW	MSB	2026-10-13

CRANECALC

STRUCTURAL DESIGN WORKSHEET

DIMENSIONS GOVERN / DO NOT SCALE

FICTITIOUS DEMONSTRATION

DRAWING RESPONSIBILITY

ORIGINATOR: MSB

CHECKER:

ENGINEER OF RECORD:

GENERAL NOTES & DESIGN CRITERIA (CONT.)

CEDAR WORKS · NEW 5-TON RUNWAY IN THE EXISTING 5-BAY BUILDING

PROJECT: DEMO-FOCUS-005

SITE: 1450 FOUNDRY ROAD, RIVERTON, OHIO (FICTITIOUS) · LENGTHS: FEET & INCHES

DRAWING / SHEET NUMBER

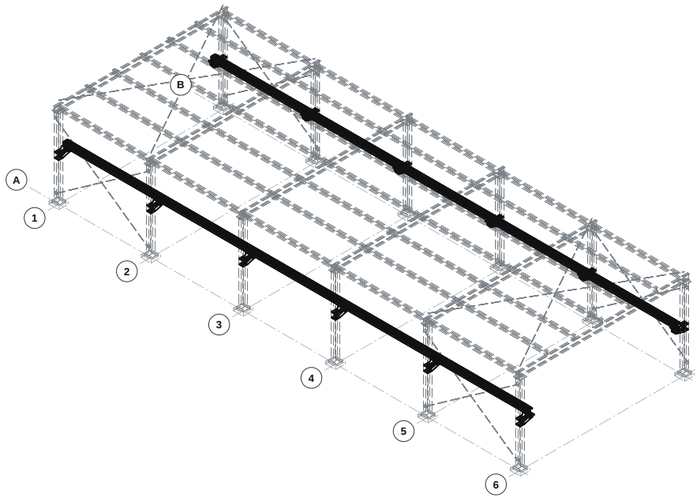
SHEET S-00A 2 OF 6

PAGE SIZE: ARCH D / 36 X 24 IN

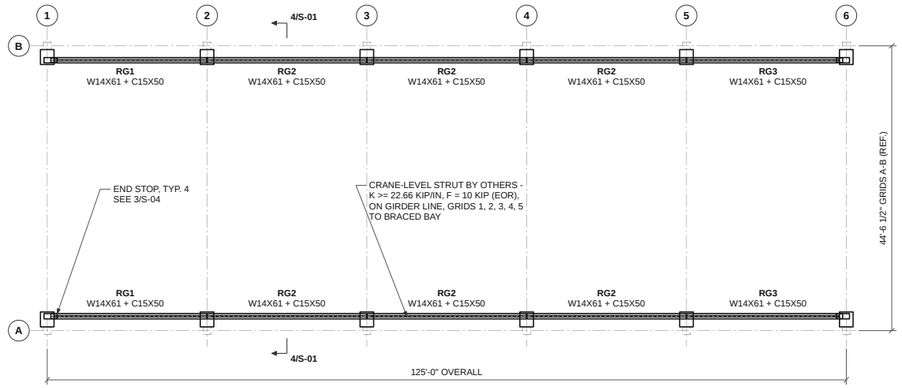
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DATE: 2026-10-13 SCALE: NONE

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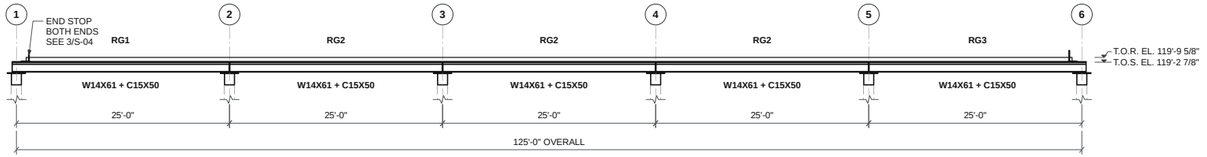


1 S-01 **ISOMETRIC VIEW**
NOT TO SCALE



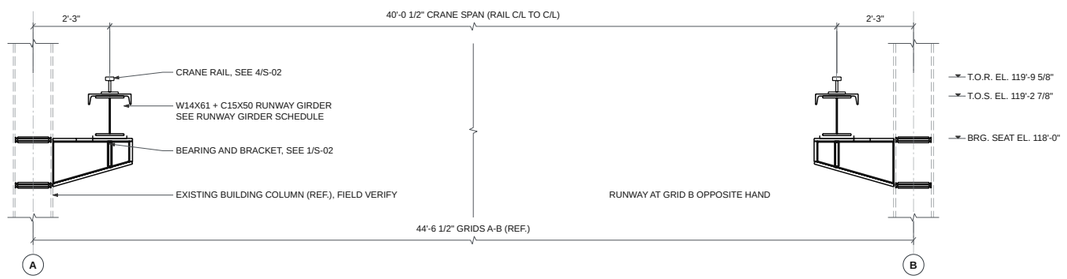
2 S-01 **RUNWAY PLAN**
SCALE: 3/32" = 1'-0"

SIMPLY SUPPORTED BAYS / CONNECTIONS: S-02 / SHARED SUPPORT: 1/S-03



DATUM EL. 100'-0" = REFERENCE FINISHED FLOOR. T.O.S. = TOP OF W STEEL; CAP ABOVE T.O.S. / SEE 7/S-02. ELEVATIONS FROM THE SURVEYED COLUMN SEAT.

3 S-01 **RUNWAY GIRDER ELEVATION - GRID A**
SCALE: 1/8" = 1'-0"



4 S-01 **TYPICAL RUNWAY SECTION**
SCALE: 1/2" = 1'-0"

RUNWAY GIRDER SCHEDULE

MARK	QTY	SECTION	LENGTH	GRIDS	CAMBER	ENDS
RG1	2	W14X61 + C15X50; W ASTM A992, CAP ASTM A572 GR. 50 CHANNEL (CAP: SEE 7/S-02)	25'-5 1/2"	1-2	NONE; NATURAL UP	LOCATES AT THE LOWER-NUMBERED GRID, SLIDES AT THE OTHER; SEE 2/S-02
RG2	6	W14X61 + C15X50; W ASTM A992, CAP ASTM A572 GR. 50 CHANNEL (CAP: SEE 7/S-02)	24'-11"	2-3, 3-4, 4-5	NONE; NATURAL UP	LOCATES AT THE LOWER-NUMBERED GRID, SLIDES AT THE OTHER; SEE 2/S-02
RG3	2	W14X61 + C15X50; W ASTM A992, CAP ASTM A572 GR. 50 CHANNEL (CAP: SEE 7/S-02)	25'-5 1/2"	5-6	NONE; NATURAL UP	LOCATES AT THE LOWER-NUMBERED GRID, SLIDES AT THE OTHER; SEE 2/S-02

SHEET NOTES

1. SEE S-00 FOR GENERAL NOTES, DESIGN CRITERIA, MATERIALS, SPECIAL INSPECTIONS AND SUPPORT REACTIONS.
2. GRIDS ARE SHOWN AS CENTERLINES (DASH-DOT); COLUMNS AND BUILDING FRAMING ARE EXISTING OR BY OTHERS AND ARE SHOWN DASHED FOR REFERENCE. FIELD VERIFY GRID DIMENSIONS AND COLUMN LOCATIONS BEFORE FABRICATION.
3. GRID B RUNWAY IS IDENTICAL AND OPPOSITE HAND TO GRID A U.N.O. QUANTITIES IN THE SCHEDULE ARE FOR BOTH RUNWAYS.
4. ERECT GIRDERS WITH ANY NATURAL MILL CAMBER UP.
5. GIRDER LENGTHS ARE OUT-TO-OUT OF STEEL WITH THE END GAP AT EACH SHARED SUPPORT; SEE 1/S-03.

ENGINEER STAMP / SEAL

NOT SEALED

DEMONSTRATION - NOT FOR CONSTRUCTION
REFERENCE FRAMING SHOWN DASHED

REV	REVISION DESCRIPTION	BY	DATE
0	ISSUED FOR REVIEW	MSB	2026-10-13

CRANECALC
STRUCTURAL DESIGN WORKSHEET

DIMENSIONS GOVERN / DO NOT SCALE
FICTITIOUS DEMONSTRATION

DRAWING RESPONSIBILITY

ORIGINATOR
MSB

CHECKER

ENGINEER OF RECORD

GENERAL ARRANGEMENT

CEDAR WORKS · NEW 5-TON RUNWAY IN THE EXISTING 5-BAY BUILDING

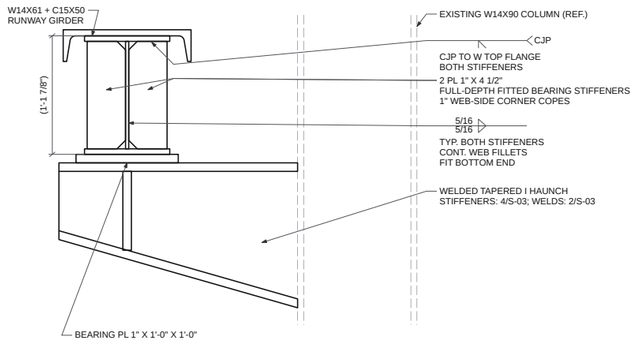
PROJECT: DEMO-FOCUS-005
SITE: 1450 FOUNDRY ROAD, RIVERTON, OHIO (FICTITIOUS) · LENGTHS: FEET & INCHES

DRAWING / SHEET NUMBER

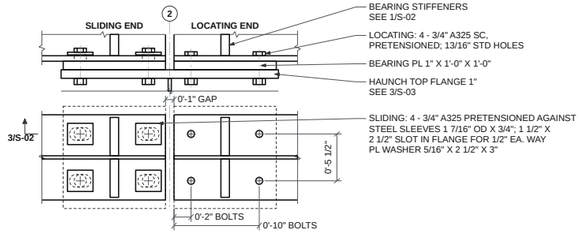
SHEET S-01 3 OF 6

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DATE: 2026-10-13 SCALE: AS SHOWN

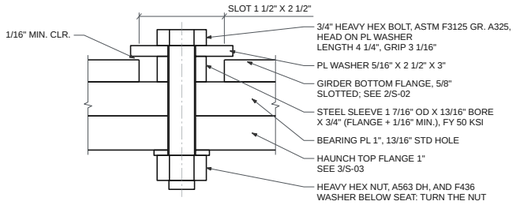
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1 S-02 GIRDER BEARING / COLUMN BRACKET
SCALE: 1 1/2" = 1'-0"



2 S-02 GIRDER END BEARINGS / LOCATING AND SLIDING
SCALE: 1 1/2" = 1'-0"



SET THE SLEEVE AND PLATE WASHER, INSERT THE BOLT FROM ABOVE AND PRETENSION IT AGAINST THE SLEEVE BY TURNING THE NUT, HEAD HELD. THE PLATE WASHER CLEARS THE FLANGE, SO THE FLANGE SLIDES 1/2" EACH WAY IN THE SLOT UNCLAMPED. CENTER THE SLEEVE IN THE SLOT AT ERECTION. TYP. EACH SLIDING BOLT.

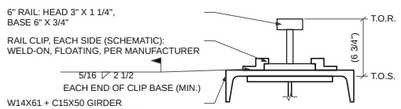
3 S-02 SLIDING BEARING BOLT / SECTION
SCALE: 6" = 1'-0"

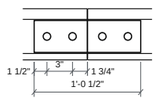
SHEET NOTES

1. SEE S-00 FOR GENERAL NOTES, DESIGN CRITERIA, MATERIALS, SPECIAL INSPECTIONS AND SUPPORT REACTIONS, AND S-01 FOR GIRDER MARKS, LENGTHS AND ELEVATIONS.
2. NEW CONSTRUCTION IS SHOWN SOLID; EXISTING AND REFERENCE CONSTRUCTION IS SHOWN DASHED. DIMENSIONS GOVERN; DO NOT SCALE THE DRAWINGS.
3. DETAIL REFERENCES READ DETAIL / SHEET. DETAILS ARE TYPICAL FOR BOTH RUNWAYS, OPPOSITE HAND AT GRID B, U.N.O.

CAP CHANNEL ATTACHMENT NOTES

1. PROVIDE 5/16" CONTINUOUS FILLETS AT BOTH W TOP-FLANGE EDGES. WELD METAL 70 KSI. NO WELD STRENGTH INCREASE FOR DIRECTION IS CREDITED.
2. STRAIGHTEN AND FIT CHANNEL BEFORE WELDING. VERIFY FULL CONTACT ALONG THE W TOP FLANGE. DO NOT FORCE A GAPPED CAP INTO SERVICE; REFER FIT-UP DISCREPANCIES TO THE ENGINEER.
3. CHANNEL AND W TOP FLANGE SHALL BE UNPERFORATED EXCEPT THE END STOP BOLT HOLES (4/S-04) AND THE CAP TIEBACK BOLT HOLES IN THE COLUMN-SIDE CHANNEL FLANGE (1/S-04). DO NOT ADD TEMPORARY ATTACHMENTS OR CHANGE RAIL CLIP WELDS WITHOUT FATIGUE REVIEW.
4. BEARING STIFFENERS: CJP TO W TOP FLANGE, DOUBLE BEVEL WELDED FROM BOTH FACES, NO BACKING, FIRST SIDE BACKGOUGED; FIT BOTTOM END. CONTINUOUS WEB WELDS AS DETAILED ON 1/S-02. INSPECT THE CJP WELDS PER THE SPECIAL INSPECTIONS ON S-00 (UT 100%).
5. CRANE SERVICE CLASS: CMAA C. CAP CHANNEL DETAILS ARE NOT APPLICABLE TO CMAA E/F SERVICE. REFERENCE AIST TECHNICAL REPORT 13 AND AISC DESIGN GUIDE 7.





5 S-02 RAIL JOINT / ELEVATION
SCALE: 1 1/2" = 1'-0"

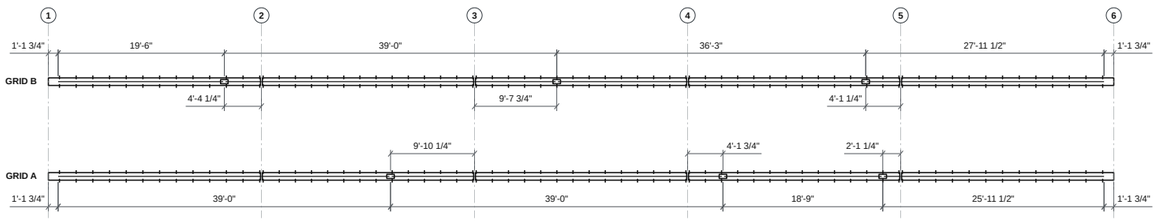
TIGHT JOINT (AIST TR-13 §5.16.1): 2 BARS 1 1/4" X 3 3/4" X 1'-0 1/2"; 4 - 7/8" A325 PRETENSIONED IN 15/16" STANDARD HOLES IN THE BARS AND THE RAIL WEB. RAIL ENDS BUTTED, GAP 1/16" MAX.; RE-TIGHTEN THE BOLTS AFTER 30 AND 90 DAYS OF OPERATION. SYMMETRIC

RAIL CLIPS: WELD-ON, PER MANUFACTURER (FICTITIOUS SUPPLIER DATA RC-W6: WELD-ON FLOATING CRANE RAIL CLIP FOR A 6 IN RAIL, OR EQUAL), RATED ≥ 10 KIP LATERAL AND ≥ 5 KIP UPLIFT PER CLIP (ALLOWABLE), IN OPPOSING PAIRS AT 2'-0" O.C. MAX. AND A PAIR EACH SIDE OF EVERY RAIL JOINT, FLOATING; WELD BASE TO CAP CHANNEL PER MANUFACTURER (MIN. 5/16" FILLET X 2 1/2", EACH END OF THE BASE).

RAIL ANCHORAGE: THE FLOATING RAIL IS ANCHORED BY THE END STOPS (3/S-04); NO ANCHOR CLIPS OR RAIL NOTCHES. RAIL JOINTS AND CLIP SPACING: 6/S-02.

RAIL STEEL: DRILL HOLES AND SAW-CUT ENDS; NO FLAME CUTTING OR PUNCHING. DO NOT WELD TO THE RAIL.

4 S-02 RAIL CLIP / SECTION
SCALE: 1 1/2" = 1'-0"

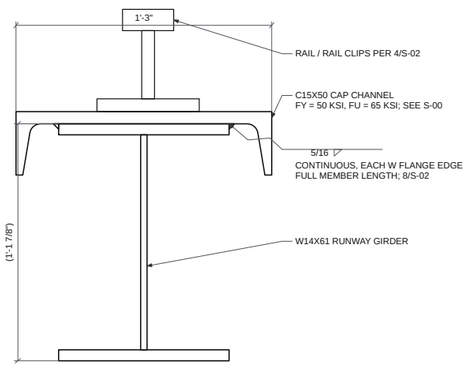


RAIL PIECES 39'-0" MAX. (MILL LENGTH); SAW-CUT THE SHORTER PIECES SHOWN, NO FLAME CUTTING. KEEP RAIL JOINTS 2'-0" MIN. FROM GIRDER JOINTS AND FROM THE OPPOSITE RAIL JOINTS, AND OPPOSITE JOINTS 2'-0" MIN. FROM ANY CRANE WHEEL SPACING (10'-0").

RAIL CLIP PAIRS (TICKS) AT 2'-0" O.C. MAX., A PAIR EACH SIDE OF EVERY JOINT; CLIPS: 4/S-02; JOINTS: 5/S-02. RUNWAYS DRAWN CLOSER THAN THE CRANE SPAN.

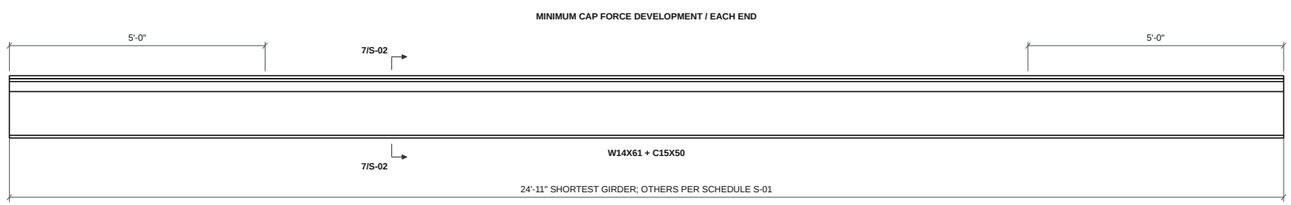
RAIL ANCHORAGE: FLOATING RAIL CLIPS, NO RAIL-SEAT FRICTION RELIED ON. TRACTION AND BRAKING (3.2 KIP FACTORED) REACH THE GIRDER UNDER THE WHEELS OR, AFTER RAIL CREEP, THROUGH AN END STOP (3/S-04); BOTH PATHS ARE DESIGNED. TIGHT RAIL JOINTS PER 5/S-02 (GAP AND RE-TIGHTENING THERE), STAGGERED AS SHOWN. THE RAIL STRING MOVES THERMALLY UP TO 17/32" INTO THE CLEARANCE GAP AT AN END STOP. AT EACH PERIODIC CRANE INSPECTION CHECK THE RAIL ENDS AT THE STOPS FOR CREEP.

6 S-02 RAIL LAYOUT / PLAN
SCALE: 1/8" = 1'-0" ALONG RUNWAY



CHANNEL WEB TO BEAR FULLY ON W TOP FLANGE
NO INTERMITTENT WELDS / NO UNAPPROVED FIT-UP GAPS

7 S-02 CAPPED GIRDER SECTION
SCALE: 3" = 1'-0"



REPEAT AT EACH SIMPLY SUPPORTED MEMBER

CONTINUOUS WELDS EXTEND THROUGH DEVELOPMENT REGIONS

8 S-02 CAP END DEVELOPMENT
SCALE: 3/4" = 1'-0"

DEMONSTRATION - NOT FOR CONSTRUCTION
REFERENCE FRAMING SHOWN DASHED

ENGINEER STAMP / SEAL
NOT SEALED

REV	REVISION DESCRIPTION	BY	DATE
0	ISSUED FOR REVIEW	MSB	2026-10-13

CRANECALC
STRUCTURAL DESIGN WORKSHEET
DIMENSIONS GOVERN / DO NOT SCALE
FICTITIOUS DEMONSTRATION

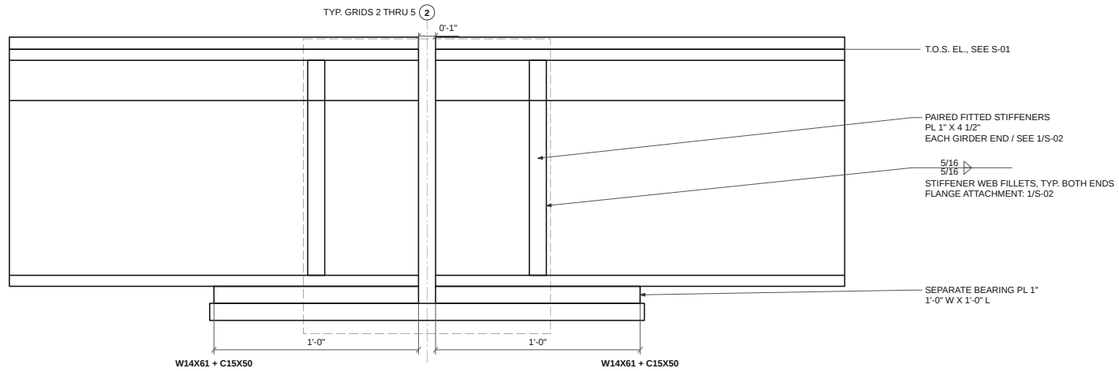
DRAWING RESPONSIBILITY
ORIGINATOR: MSB
CHECKER:
ENGINEER OF RECORD:

CONNECTIONS & CAP CHANNEL
CEDAR WORKS · NEW 5-TON RUNWAY IN THE EXISTING 5-BAY BUILDING

PROJECT: DEMO-FOCUS-005
SITE: 1450 FOUNDRY ROAD, RIVERTON, OHIO (FICTITIOUS) · LENGTHS: FEET & INCHES

DRAWING / SHEET NUMBER
SHEET S-02 4 OF 6
PAGE SIZE: ARCH D / 36 X 24 IN
REVISION: 0 (SAME AS THE CALCULATION)
DATE: 2026-10-13 SCALE: AS SHOWN

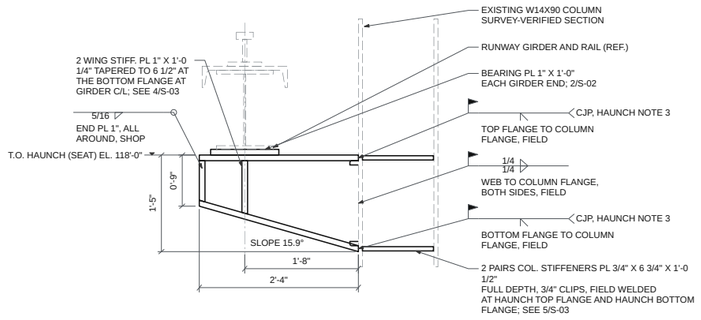
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WELDED COLUMN BRACKET BELOW / 2/S-03. DO NOT BRIDGE GIRDER OR CAP ENDS. END GRIDS: ONE GIRDER END, SIMILAR.

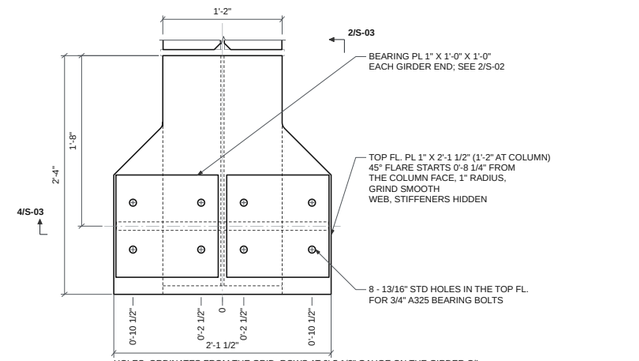
1 S-03 **ADJACENT GIRDER ENDS AT COLUMN**

SCALE: 3" = 1'-0"



2 S-03 **COLUMN BRACKET / TRANSVERSE SECTION**

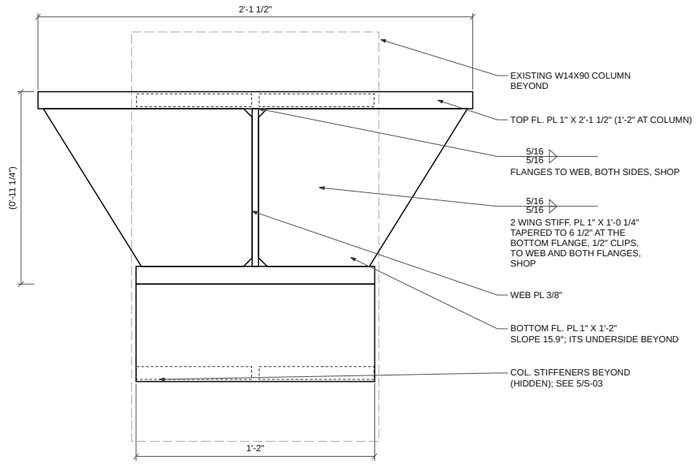
SCALE: 1" = 1'-0"



HOLES: ORDINATES FROM THE GRID; ROWS AT 0'-5 1/2" GAUGE ON THE GIRDER C/L.
EDGES: FLANGE END 0'-2 1/4", TIP 0'-4 1/4" (TO END PL); COLUMN FACE 1'-5 1/4".
AT A RUNWAY-END GRID: ONE BEARING CENTERED ON THE GRID (SUPPORT WEB), THE GIRDER RUNNING 0'-6" PAST IT; HOLES 0'-4" EACH SIDE OF THE GRID.

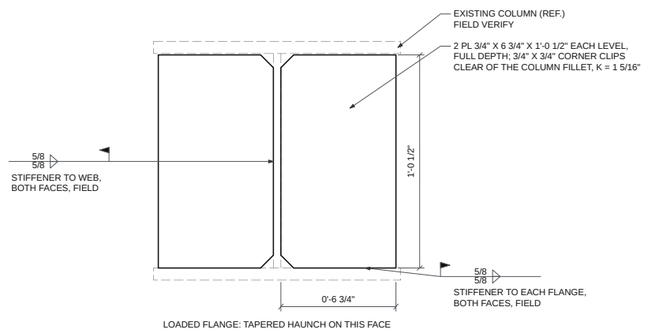
3 S-03 **TAPERED HAUNCH / TOP FLANGE PLAN**

SCALE: 1 1/2" = 1'-0"



4 S-03 **TAPERED HAUNCH / SECTION AT STIFFENERS**

SCALE: 3" = 1'-0"



LOADED FLANGE: TAPERED HAUNCH ON THIS FACE

UPPER PAIR C/L 0'-0 1/2" BELOW T.O. SEAT; LOWER PAIR C/L 1'-4 1/2" BELOW T.O. SEAT.

5 S-03 **COLUMN STIFFENERS / PLAN AT BRACKET**

SCALE: 3" = 1'-0"

SHEET NOTES

1. SEE S-00 FOR GENERAL NOTES, DESIGN CRITERIA, MATERIALS, SPECIAL INSPECTIONS AND SUPPORT REACTIONS, AND S-01 FOR GIRDER MARKS, LENGTHS AND ELEVATIONS.
2. NEW CONSTRUCTION IS SHOWN SOLID; EXISTING AND REFERENCE CONSTRUCTION IS SHOWN DASHED. DIMENSIONS GOVERN; DO NOT SCALE THE DRAWINGS.
3. DETAIL REFERENCES READ DETAIL / SHEET. DETAILS ARE TYPICAL FOR BOTH RUNWAYS, OPPOSITE HAND AT GRID B, U.N.O.

INDEPENDENT SUPPORT NOTES

1. WELDED GRAVITY BRACKET: 2/S-03. BOLTED END BEARINGS TRANSFER LONGITUDINAL FORCE AND UPLIFT (2/S-02); CAP TIEBACK ANGLES (1/S-04) TRANSFER LATERAL FORCE. FACTORED FORCES: BRACKET DESIGN FORCES ON THIS SHEET.
2. THE END OF EACH BAY AT ITS LOWER-NUMBERED GRID LOCATES LONGITUDINALLY. FULL CRANE TRACTION IS ASSIGNED TO EACH OCCUPIED BAY AND, VIA THE RAIL STRING, TO EACH END BAY AT ITS STOP, IN SEPARATE BOUNDING CASES; DO NOT ADD THESE CASES.
3. GIRDER ANALYSIS SPANS ARE GRID-TO-GRID; BEARINGS ARE INSET AS SHOWN. LOCATING ENDS: STANDARD HOLES, PRETENSIONED. SLIDING ENDS: SLOTS, BOLTS PRETENSIONED AGAINST STEEL SLEEVES (2/S-02).
4. REFERENCE AIST TECHNICAL REPORT 13 §5.8.1. GIRDER AND CAP ENDS REMAIN SEPARATE. RAIL JOINTS ARE DETAILED INDEPENDENTLY.

HAUNCH FABRICATION & EXISTING-STEEL WELDING NOTES

1. A BUILT-UP TAPERED I HAUNCH 2'-4" LONG, 1'-5" DEEP AT THE COLUMN TAPERING TO 0'-9" AT THE TIP (BOTTOM FLANGE AT 15.9°); WEB PL 3/8", TOP FL. PL 1" X 2'-1 1/2" (1'-2" AT COLUMN), BOTTOM FL. PL 1" X 1'-2", 2 WING STIFF. PL 1" X 1'-0 1/4" TAPERED TO 6 1/2" AT THE BOTTOM FLANGE AT THE GIRDER C/L, END PL 1". PLATES 50 KSI MIN. YIELD.
2. FIELD WELD THE HAUNCH TO THE EXISTING W14X90 COLUMN FLANGE: CJP GROOVE WELDS AT BOTH FLANGES AND 1/4" FILLETS BOTH SIDES OF THE WEB. SHOP: 5/16" FLANGE-TO-WEB FILLETS BOTH SIDES, CONTINUOUS; STIFFENERS FITTED TO BEAR ON THE TOP FLANGE AND WELDED TO THE WEB AND BOTH FLANGES WITH 5/16" FILLETS BOTH SIDES.
3. WELDS: E70XX LOW-HYDROGEN ELECTRODE, MATCHING FILLER. FLANGE CJP: SINGLE BEVEL ON THE HAUNCH FLANGE, 45°, 1/4" ROOT OPENING ON STEEL BACKING, WELDED THROUGH THE WEB ACCESS HOLES (1 1/2" X 3/4", 3/8" R, AISC J1.6); REMOVE THE BACKING, BACKGOUGE, REWELD AND ADD A 1/4" REINFORCING FILLET; UT PER S-00. CONTINUOUS WELDS AS SHOWN; NO INTERMITTENT WELDS. CUT THE TOP FLANGE FLARE WITH A 1" RADIUS AT THE RE-ENTRANT CORNERS AND GRIND SMOOTH.
4. WELDING TO THE EXISTING COLUMN: BASE METAL CONFIRMATION, APPROVED WPS, PREHEAT AND LOAD CONDITION PER THE EXISTING BUILDING NOTES ON S-00. REMOVE PAINT, COATINGS, MILL SCALE AND RUST FROM THE COLUMN FLANGE AT THE WELDS; CLEAN TO BRIGHT METAL.
5. KEEP GIRDER ENDS AND BEARINGS INDEPENDENT. DO NOT WELD THE SLIDING BEARING TO THE GIRDER. THE BEARING BOLTS LOCATE, GUIDE AND HOLD DOWN THE GIRDER THROUGH THE HAUNCH TOP FLANGE (2/S-02); DRILL TO MATCH (3/S-03). ONLY THE CAP TIEBACK ANGLES (1/S-04) CONNECT TO THE COLUMN SEPARATELY.
6. THE HAUNCH IS DESIGNED FOR EVERY CONCURRENT CASE OF THE BRACKET DESIGN FORCES BELOW, WITH ITS SELF-WEIGHT: FLEXURE AND SHEAR ON THE SECTIONS OF THE TAPER, TORSION FROM THE OFFSET BEARINGS, THE STIFFENERS, THE WELD GROUP ON THE COLUMN FACE, FATIGUE AND DEFLECTION. THE EXISTING COLUMN FLANGE AND WEB LOCAL CHECKS (J10) ARE IN THE CALCULATION, WITH THE COLUMN STIFFENERS AT BOTH HAUNCH FLANGE LEVELS.
7. COLUMN STIFFENERS: A PAIR OF PL 3/4" X 6 3/4" X 1'-0 1/2" PLATES (FULL DEPTH), ONE EACH SIDE OF THE COLUMN WEB, AT THE HAUNCH TOP FLANGE AND AT THE HAUNCH BOTTOM FLANGE (UPPER PAIR C/L 0'-0 1/2" BELOW T.O. SEAT; LOWER PAIR C/L 1'-4 1/2" BELOW T.O. SEAT), 3/4" CORNER CLIPS CLEAR OF THE ROLLED FILLET. FIELD WELD EACH PLATE TO BOTH FLANGES WITH 5/8" FILLETS AND TO THE WEB WITH 5/8" FILLETS, BOTH FACES, CONTINUOUS; NO WELD IN THE CLIPS.
8. EXISTING COLUMN: BEFORE FABRICATION, SURVEY THE COLUMN AT BOTH STIFFENER LEVELS FOR D, BF, TF, TW AND THE FILLET K, AND CONFIRM THAT NO EXISTING CONNECTION, BRACE, GIRT OR ATTACHMENT ON THE WEB, INSIDE EITHER FLANGE OR ON THE FAR FLANGE INTERFERES WITH THE PLATES OR THEIR WELDS; REPORT ANY TO THE ENGINEER OF RECORD. FIELD WELD THE STIFFENERS BEFORE THE BRACKET CARRIES LOAD.
9. THE STIFFENERS CARRY THE BRACKET'S CONCENTRATED FLANGE FORCES ABOVE THE UNSTIFFENED COLUMN STRENGTH, IN TENSION AND COMPRESSION (AISC 360 J10.8). THEY ADD NO PANEL-ZONE SHEAR STRENGTH; NO WEB DOUBLER PLATE IS PROVIDED.

BRACKET DESIGN FORCES / FACTORED LRFD ENVELOPE

GRID	CASE	COMB.	V	LEFT V	RIGHT V	MY,SEAT	F	H TOP (RESTRAINT)	H SEAT
1	MAX V & M	A2C	24.4	—	24.4	7.33	−3.20	±0.43	∓0.43
	MIN V	10-T	1.67	—	1.67	0.50	+3.20	±0.01	∓0.01
	MAX F	8	15.3	—	15.3	4.58	−10.0	±0.23	∓0.23
	MAX H	A2B	20.1	—	20.1	6.03	−3.20	±3.78	∓1.47
2	MAX V, M & F	A2C	29.0	2.41	26.6	20.4	−3.20	±0.47	∓0.47
	MAX V & M	A2C	29.0	26.6	2.41	−20.4	—	±0.45	∓0.45
	MIN V & END V	10-T	3.40	1.81	1.59	−0.19	+3.20	±0.02	∓0.02
	MAX F	A2C	28.6	26.4	2.19	−20.4	+3.20	±0.45	∓0.45
	MAX H	A2B	24.2	2.41	21.8	16.3	−3.20	±4.09	∓1.53
3	MAX V, M & F	A2C	29.0	2.41	26.6	20.4	−3.20	±0.47	∓0.47
	MAX V & M	A2C	29.0	26.6	2.41	−20.4	—	±0.45	∓0.45
	MIN V & END V	10-T	3.40	1.81	1.59	−0.19	+3.20	±0.02	∓0.02
	MAX F	A2C	28.6	26.4	2.19	−20.4	+3.20	±0.45	∓0.45
	MAX H	A2B	24.2	2.41	21.8	16.3	−3.20	±4.09	∓1.53
4	MAX V, M & F	A2C	29.0	2.41	26.6	20.4	−3.20	±0.47	∓0.47
	MAX V & M	A2C	29.0	26.6	2.41	−20.4	—	±0.45	∓0.45
	MIN V & END V	10-T	3.40	1.81	1.59	−0.19	+3.20	±0.02	∓0.02
	MAX F	A2C	28.6	26.4	2.19	−20.4	+3.20	±0.45	∓0.45
	MAX H	A2B	24.2	2.41	21.8	16.3	−3.20	±4.09	∓1.53
5	MAX V & M	A2C	29.0	2.41	26.6	20.4	−3.20	±0.47	∓0.47
	MAX V & M	A2C	29.0	26.6	2.41	−20.4	—	±0.45	∓0.45
	MIN V & END V	10-T	3.40	1.81	1.59	−0.19	+3.20	±0.02	∓0.02
	MAX F	8	8.47	2.41	6.06	3.07	+10.0	±0.12	∓0.12
	MAX H	A2B	24.2	2.41	21.8	16.3	−3.20	±4.09	∓1.53
6	MAX V & M	A2C	24.4	24.4	—	−7.33	—	±0.42	∓0.42
	MIN V	10-T	1.67	1.67	—	−0.50	—	±0.01	∓0.01
	MAX H	A2B	20.1	20.1	—	−6.03	—	±3.75	∓1.44

FORCES IN KIP, MOMENTS IN KIP-FT. LRFD FACTORED, AIST TR-13 COMBINATION NUMBERS PER S-00; ENVELOPE OF ALL CRANE POSITIONS. EACH ROW IS ONE CONCURRENT CASE. ITS FORCES ACTING TOGETHER; DO NOT COMBINE ROWS. V: GIRDER REACTIONS ON THE SEAT, DOWNWARD POSITIVE, BRACKET SELF-WEIGHT EXCLUDED; LEFT / RIGHT V: GIRDER END ON THE LOWER / HIGHER GRID SIDE. MY,SEAT = ΣV × A ABOUT THE GRID. A = ARM TO EACH REACTION, AT 0.8 OF THE BEARING LENGTH FROM ITS GIRDER END. MIN V: LEAST TOTAL V; END V: LEAST GIRDER END REACTION; NO UPLIFT. F: LONGITUDINAL FORCE AT THE LOCATING BEARING OF EACH BAY, AT ITS LOWER-NUMBERED GRID; NONE AT A SLIDING END. MY,SEAT AND F + TOWARD THE HIGHER-NUMBERED GRID. TRACTION ACTS EITHER WAY (BOTH SENSES LISTED WHERE IT GOVERNS), IN THE BAY UNDER THE CRANE OR VIA THE RAIL STRING AT AN END STOP; THE CRANE STOP FORCE ACTS ONLY AT STOP CONTACT, TOWARD THE STOP, AT GRIDS 1 AND 5. H TOP (RESTRAINT) / H SEAT: LATERAL FORCE OF ALL GIRDER ENDS AT THE GRID AT THE CAP TIEBACK ANGLES / THROUGH THE BEARING BOLTS INTO THE SEAT, ± TOGETHER; H SEAT IS SIGNED RELATIVE TO H TOP (RESTRAINT): − OPPOSITE (A COUPLE), + THE SAME WAY. H TOP IS THE RESTRAINT FORCE; THE TIE DESIGN FORCE (S-00 H TIE) ADDS THE APPENDIX 6 BRACE FORCE P BR = 2.50 KIP. COMB. 10-T: COMBINATION 10 (TRACTION) ON S-00.

ENGINEER STAMP / SEAL

NOT SEALED

DEMONSTRATION - NOT FOR CONSTRUCTION

REFERENCE FRAMING SHOWN DASHED

REV	REVISION DESCRIPTION	BY	DATE
0	ISSUED FOR REVIEW	MSB	2026-10-13

CRANECALC

STRUCTURAL DESIGN WORKSHEET

DIMENSIONS GOVERN / DO NOT SCALE

FICTITIOUS DEMONSTRATION

DRAWING RESPONSIBILITY

ORIGINATOR: MSB

CHECKER:

ENGINEER OF RECORD:

SUPPORTS & BRACKETS

CEDAR WORKS · NEW 5-TON RUNWAY IN THE EXISTING 5-BAY BUILDING

PROJECT: DEMO-FOCUS-005

SITE: 1450 FOUNDRY ROAD, RIVERTON, OHIO (FICTITIOUS) · LENGTHS: FEET & INCHES

DRAWING / SHEET NUMBER

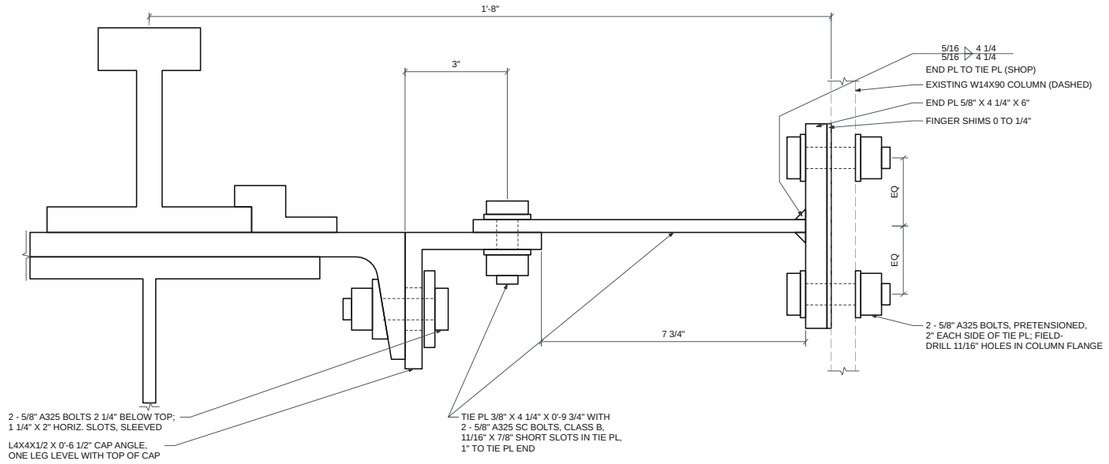
SHEET S-03 5 OF 6

PAGE SIZE: ARCH D / 36 X 24 IN

REVISION: 0 (SAME AS THE CALCULATION)

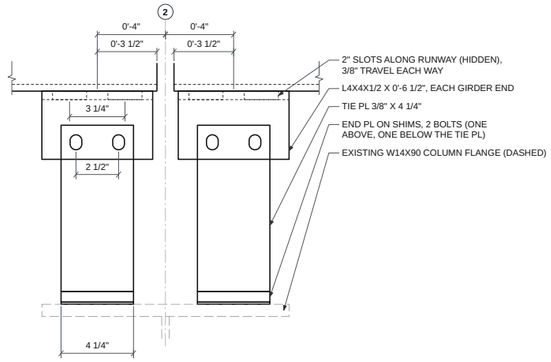
DATE: 2026-10-13 SCALE: AS SHOWN

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1 S-04 **CAP TIEBACK ANGLE / SECTION AT SUPPORT**

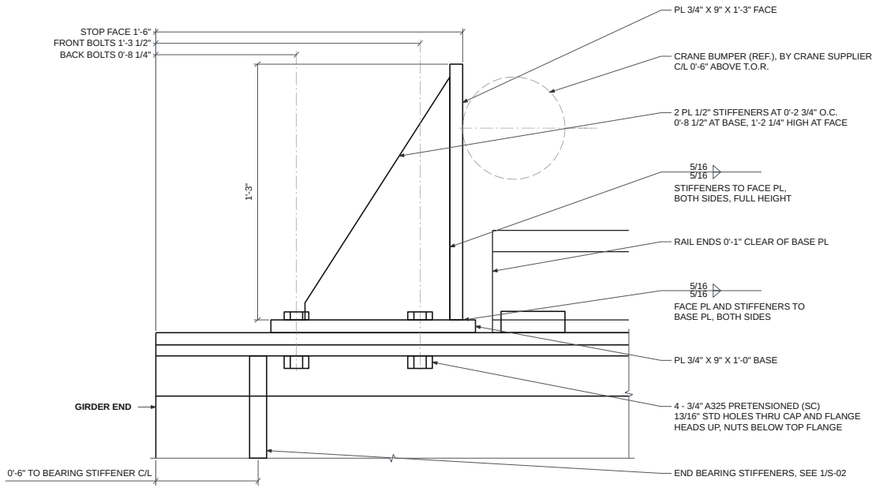
SCALE: 6" = 1'-0"



SEPARATE TIE AND END PLATE AT EACH GIRDER END.

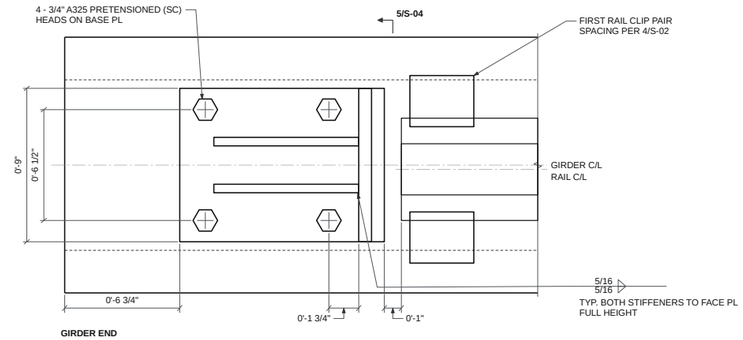
2 S-04 **CAP TIEBACK ANGLES / PLAN AT SHARED COLUMN**

SCALE: 3" = 1'-0"



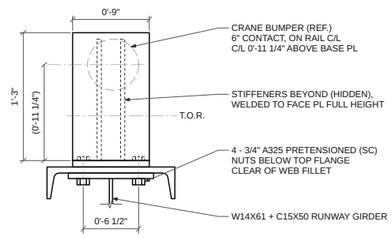
3 S-04 **END STOP / ELEVATION**

SCALE: 3" = 1'-0"



4 S-04 **END STOP / PLAN**

SCALE: 3" = 1'-0"



5 S-04 **END STOP / SECTION AT FACE**

SCALE: 1 1/2" = 1'-0"

SHEET NOTES

1. SEE S-00 FOR GENERAL NOTES, DESIGN CRITERIA, MATERIALS, SPECIAL INSPECTIONS AND SUPPORT REACTIONS, AND S-01 FOR GIRDER MARKS, LENGTHS AND ELEVATIONS.
2. NEW CONSTRUCTION IS SHOWN SOLID; EXISTING AND REFERENCE CONSTRUCTION IS SHOWN DASHED. DIMENSIONS GOVERN; DO NOT SCALE THE DRAWINGS.
3. DETAIL REFERENCES READ DETAIL / SHEET. DETAILS ARE TYPICAL FOR BOTH RUNWAYS, OPPOSITE HAND AT GRID B, U.N.O.

CAP TIEBACK ANGLE NOTES

1. CAP TIEBACK ANGLES (AISC DG7, TOP FLANGE CONNECTED DIRECTLY TO THE COLUMN): ONE AT EACH GIRDER END, 0'-3 1/2" FROM THE GIRDER END TO ITS CENTERLINE, ON THE COLUMN-SIDE CHANNEL FLANGE. THE BOTTOM FLANGE IS RESTRAINED BY THE BOLTED END BEARINGS.
2. CAP BOLTS: PRETENSION AGAINST STEEL SLEEVES 1 3/16" OD X 11/16" BORE X 9/16" LONG THROUGH 1 1/4" X 2" HORIZONTAL SLOTS IN THE ANGLE, UNDER 5/16" X 2 1/4" X 3 1/4" PLATE WASHERS. CENTER THE SLEEVES IN THE SLOTS AT ERECTION. THE ANGLE IS NOT CLAMPED: THE GIRDER SLIDES 3/8" EACH WAY.
3. DRILL STANDARD HOLES IN THE CAP CHANNEL FLANGE AT THE TIE ONLY; DO NOT WELD THE TIE TO THE CAP OR GIRDER. F436 1:6 BEVELED WASHERS UNDER THE CAP BOLT NUTS ON THE SLOPED INSIDE FACE OF THE CHANNEL FLANGE. INSTALL AND TENSION THE CAP BOLTS BEFORE THE TIE PLATE AND LAP BOLTS.
4. LAP BOLTS: PRETENSIONED, SLIP-CRITICAL, CLASS B FAYING SURFACES, IN SHORT SLOTS ACROSS THE RUNWAY IN THE TIE PLATE FOR FIT-UP; F436 WASHERS OVER THE SLOTS.
5. END PLATE: SHOP WELD TO THE TIE PLATE. FIELD-DRILL 11/16" HOLES IN THE EXISTING COLUMN FLANGE TO MATCH; NO FIELD WELDING TO THE COLUMN FOR THE TIE. FINGER SHIMS 0 TO 1/4" TAKE UP THE FIT-UP; PRETENSION THE COLUMN BOLTS.
6. AT A SHARED COLUMN THE TWO TIES SIT SIDE BY SIDE ON THE COLUMN FLANGE, EACH WITH ITS OWN END PLATE.

CAP TIEBACK CHECKS: SEE CALCULATION REPORT (MAX. D/C 0.77 STRENGTH, 0.80 SLOT TRAVEL, 0.23 FATIGUE).

END STOP DESIGN DATA

ITEM	VALUE
BUMPER FORCE, PER CRANE SUPPLIER	10 KIP
FACTORED (LRFD, AIST STOP COMBINATIONS)	10 KIP
BUMPER C/L ABOVE T.O.R. / CONTACT DIAMETER	0'-6" / 6"
FRONT BOLT TENSION / BOLT SHEAR	7.5 KIP / 2.5 KIP
BOLT C/L TO WELD TOE: FACE PL / STIFFENER	1 7/16" / 1 5/16", 1 5/16" MIN.
GOVERNING STOP CHECK	GIRDER TOP FLANGE · THICKNESS FOR NO PRYING: D/C 0.68 (THICKNESS RATIO)
QUANTITY	4: BOTH ENDS OF BOTH RUNWAYS
DATA SOURCE	FICTITIOUS SUPPLIER SCHEDULE DEMO-CS-05: BUMPER FORCE, BUMPER CENTERLINE HEIGHT AND CONTACT DIAMETER

END STOP NOTES

1. PROVIDE ONE STOP AT EACH END OF EACH RUNWAY, CENTERED ON THE GIRDER. THE STOP AT THE OPPOSITE END IS THE MIRROR IMAGE.
2. DRILL 13/16" STANDARD HOLES THROUGH THE CAP CHANNEL WEB AND THE TOP FLANGE ONLY AT THE LOCATIONS SHOWN, 0'-8 1/4" AND 1'-3 1/2" FROM THE GIRDER END. NO OTHER HOLES IN THE TOP FLANGE.
3. BOLTS: ASTM F3125 GR. A325 HEAVY HEX, PRETENSIONED, CLASS B FAYING SURFACES (SLIP-CRITICAL); HEADS ON THE BASE PLATE, A563 DH NUTS BELOW THE FLANGE, F436 WASHER UNDER THE TURNED ELEMENT. BOLT CENTERS ARE 1 5/16" MIN. CLEAR OF THE STOP WELD TOES FOR THE SOCKET (AISC MANUAL TABLE 7-15). VERIFY NUT CLEARANCE BELOW THE FLANGE AT THE BEARING STIFFENERS BEFORE DRILLING.
4. SHOP WELD THE FACE PLATE AND THE STIFFENERS TO THE BASE PLATE, AND EACH STIFFENER TO THE FACE PLATE OVER ITS FULL HEIGHT, WITH CONTINUOUS 5/16" FILLETS BOTH SIDES. GRIND THE FACE SMOOTH AT THE BUMPER CONTACT.
5. TERMINATE THE RAIL 0'-1" CLEAR OF THE STOP BASE PLATE EDGE (0'-1 3/4" CLEAR OF THE STOP FACE), 1'-7 3/4" FROM THE GIRDER END. THE FIRST RAIL CLIP PAIR IS AT THE RAIL END; CLIP SPACING PER 4/S-02. THE RAIL END BEARS ON THE STOP ONLY AFTER CREEP: THE STOPS ANCHOR THE FLOATING RAIL.
6. CONFIRM THE BUMPER FORCE, BUMPER HEIGHT AND CONTACT DIAMETER WITH THE CRANE SUPPLIER BEFORE FABRICATION. STOPS SHALL BE INSTALLED BEFORE THE CRANE IS OPERATED.
7. THE GIRDER AXIAL FORCE AND THE LOCATING END OF EACH END BAY INCLUDE THE BUMPER FORCE; THE END COUPLE IS INCLUDED IN THE GIRDER STOP COMBINATIONS (CALCULATION REPORT, RUNWAY END STOPS).

DEMONSTRATION - NOT FOR CONSTRUCTION

REFERENCE FRAMING SHOWN DASHED

ENGINEER STAMP / SEAL

NOT SEALED

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DRAWING RESPONSIBILITY

ORIGINATOR: MSB

CHECKER:

ENGINEER OF RECORD:

CAP TIEBACK ANGLES & END STOPS

CEDAR WORKS · NEW 5-TON RUNWAY IN THE EXISTING 5-BAY BUILDING

PROJECT: DEMO-FOCUS-005

SITE: 1450 FOUNDRY ROAD, RIVERTON, OHIO (FICTITIOUS) · LENGTHS: FEET & INCHES

DRAWING / SHEET NUMBER

SHEET S-04 6 OF 6

PAGE SIZE: ARCH D / 36 X 24 IN

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